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Review papers

A survey on river water quality modelling using artificial intelligence models: 2000–2020

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ABSTRACT

There has been an unsettling rise in the river contamination due to the climate change and anthropogenic activities. Last decades' research has immensely focussed on river basin water quality (WQ) prediction, risk assessment and pollutant classification techniques to design more potent management policies and advanced early warning system. The next challenge is dealing with water-related data as they are problematic to handle owing to their nonlinearity, nonstationary feature and vague properties due to the unpredictable natural changes, interdependent relationship, human interference and complexity. Artificial intelligence (AI) models have shown remarkable success and superiority to handle such data owing to their higher accuracy to deal with non-linear data, robustness, reliability, cost-effectiveness, problem-solving capability, decision-making capability, efficiency and effectiveness. AI models are the perfect tools for river WQ monitoring, management, sustainability and policymaking. This research reports the state of the art of various AI models implemented for river WQ simulation over the past two decades (2000–2020). Correspondingly, over 200 research articles are reviewed from the Web of Science journals. The survey covers the model structure, input variability, performance metrics, regional generalisation investigation and comprehensive assessments of AI models progress in river water quality research. The increasing contaminants, the lack of funding and the deficiency in data, numerous variables and unique data time series pattern based on the geological area have increased the need for river WQ monitoring and control even more. Hence, this is highly emphasising the involvement of AI models development which can deal with missing data, able to integrate the features of a black-box model and white-box models, benchmarked model and automated early warning system are few of many points need more research. Despite extensive research on WQ simulation using AI models, shortcomings remain according to the current survey, and several possible future research directions are proposed. Overall, this survey provides a new milestone in water resource engineering on the AI model implementation, innovation and transformation in surface WQ modelling with many formidable problems in different blossoming area and objectives to be achieved in the future.

1. Introduction

1.1. Research background

Humankind is blessed with a stunning piece of art, that is, the environment. The growing desire for comfort as time progresses has left us with toxic air, barren land and poisoned water. The propensity has brought the present condition with a need to examine, quantify and treat what was initially in its purest form. Consequently, the need for testing water was realised back in the 1960s, and water pollution acts were implemented to stop the deterioration of natural water quality (Barry, 1969). Water sources are divided into surface water sources (e.g., river, lake, reservoirs and coastal areas) and groundwater sources (e.g., infiltration galleries and springs). Rivers have been the most used water source on account of their accessibility and availability; this situation has led to the growth of most civilisation near river banks (Mustafa et al., 2017; Viessman et al., 1998).

Water quality (WQ) is divided into biological, hydro–morphological and physiochemical quality (Tchobanoglous and Schroeder, 1985). In addition to other variables, physiochemical WQ variables are widely used for river WQ modelling (Wan Mohtar et al., 2019). Only a few physical and chemical properties have been initially considered, but various chemicals that affect WQ have been discovered later. Physical and chemical variables produce synergistic effects on WQ along with several other environmental factors that may produce unpredictable WQ results (Wilhm and Dorris, 1968). There are predominantly few

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water quality variables that have been typically used to assess river health include dissolved oxygen (DO), biochemical oxygen demand (BOD), solids, nitrogen compounds, water temperature (WT), chlorophyll-a (Chl-a), electrical conductivity (EC), chemical oxygen demand (COD) and pH (Tchobanoglous and Schroeder, 1985).

Another frequently applied indicator is the water quality index (WQI), which is calculated by a complicated numerical formula using surplus time and may have inaccurate values. The process includes using multiple WQ variables for calculation and comparing them with the standard value for improved understanding of the non-scientific community, such as policymakers (Smith, 1990). WQI categorises WQ into classes, such as good, bad and worst, depending on the standard set by the study area governing bodies; the results are numerical but are presented in a simplified form (Fernández et al., 2004). Water quality research plays an important role in river WQ assessment, monitoring and management. Henceforward, the number of scientific approaches towards accessing, managing and understanding WQ data has increased considerably.

1.2. Problem statement

WQ assessment and its interpretation can be difficult to explain because they are influenced by several environmental factors and human activities (Kamel et al., 2013). This complexity has started a new geological period called Anthropocene, in which all earth system processes are altered by humans. Activities that influence WQ can be population explosion, urbanisation, industrialisation, irrigation and water-related engineering projects (Meybeck, 2003). A global-level study considered geospatial drivers, including catchment disturbance, pollution, water resource development and biotic factors, by using digital river networks and marine systems to obtain collective threat indices. This study showed that 80% of the world population lives in areas where water security threat may exceed 75 percentiles. Unsecure water indicates unhealthy and contaminated water and is the cause of water-related health risks (Vorosmarty et al., 2010).

WQ statistics have difficulty in managing the large amount of data (Chapman, 1992). Many studies have attempted to estimate the river WQ pollution. However, WQ sample gathering, testing and data handling are time-consuming and costly in terms of chemical, equipment and manpower. Traditional approaches bring difficulty in water-related projects and result in losses in economic value; these situations affect the decision making and productivity of the monitoring programmes (Ongley, 2000).

A productive and cost-effective methodology for WQ estimation with effective and robust advanced approaches are required to achieve the goal and overcome the previous limitations. WQ data are essential in defining the characteristics of the source, water regulations, addressing water management issues, investment decisions and dealing with pollution (Ongley, 1996). Water-related data are difficult to design as a result of their nonlinearity, nonstationary feature and vague properties due to the unpredictable natural changes, interdependent relationship and human interference (Votruba, 1988). As a result, these data have noise and are poor in quality. Therefore, reliable methodologies, such as artificial intelligence (AI) models and their advanced tools, are needed.

1.3. Research motivation

AI models, which are based on the learning process of the human brain, were proposed as forecasting and predictive tools by McCarthy (1956). AI models are powerful tools in many fields, such as robotics, hydrology, climatology and agriculture, including water resources. They are popular for their reliability, cost-effectiveness, problem-solving capability, decision-making capability, efficiency and effectiveness (Bandyopadhyay et al., 2007).

AI models are based on machine learning (ML) techniques and greatly impact human lives at present and will in the future. According to Mitchell (1997), "*the impact of machine learning is certain to grow over the coming years, as more and more data come online, we develop more effective algorithms and underlying theories for machine learning, the futility of hand-coding increasingly complex systems becomes clear, and human organizations themselves learn the value of capturing and using historical data*".

In river WQ-related studies, various issues must be addressed carefully for current researchers to discuss and understand what previous studies have accomplished. Many published review papers have reported the capacity of AI models in WQ studies. For instance, the first review reported prediction of WQ variables in water resource modelling (Maier and Dandy, 2000). The review studied the employment of artificial neural network (ANN) models until 1998, including model background, data utilisation, training and learning information, network architecture, optimisers and other model details. However, this study is outdated for the present scenario. Chau (2006) summarised the major AI models contribution in assessing and predicting coastal water quality researches. This study acknowledged the potential of different AI models, such as knowledge-based system, genetic algorithm (GA), ANN and fuzzy inference system (FIS), for WQ modelling. However, this study did not consider the river water researches. Solomatine and Ostfeld (2008) wrote an overview of data-driven models and evolutionary optimisation for river basin management. The review reported various gaps and limitations. The authors used two water sources, namely, Bagmati river basin in Nepal and Lake Kinneret (the Sea of Galilee) watershed, to assess the models that can provide the result on short-term forecasting and classification of hydrology-related data and flood-related data maps using WQ data and hydrological data. This study considered fewer models and lacks the advanced model information since it only studied researches published before 2008. Nicklow et al. (2010) reviewed the evolutionary algorithms (EAs) in planning and managing water supply, groundwater and WQ variable identification systems. Raghavendra and Deka (2014) reviewed the application of support vector machine (SVM) models on groundwater quality in hydrology. The study provided a brief idea of SVM models and their hybrid version for modelling hydrological data.

In 2010, a review research highlighted in detail the methods used to develop ANN models for river systems over the period of 1999–2007. The taxonomy of ANN model development selected inputs, data division and model structure were evaluated (Maier et al., 2010). The authors in another study elaborated the issue addressing the challenges and future aspect of EAs in solving water resource problems in the last 25 years. The study reviewed algorithms and their improvements in water distribution research and groundwater and river basin management (Maier et al., 2014). Che Osmi et al. (2016) conducted researches on river WQ data modelled with fuzzy-based methods. The study dealt with model architecture, membership function, operations and perspective of the integration of fuzzy with other models.

Considering all the previous review researches (Chau, 2006; Maier et al., 2010; Maier and Dandy, 2000; Solomatine and Ostfeld, 2008) which are dated back to (2000–2008), the advancement of AI research is very rapid and there is a need of review research such as the current survey which elaborates the current status and future prospect based on the previous progression. Three of the literature studies selected different water sources such as groundwater and coastal, which also makes this study unique considered this review strictly selected the studies on river WQ assessment. Additionally, out of eight past review studies, six of them is reporting only one model out of all AI models (e.g., EA (Maier et al., 2014; Nicklow et al., 2010), ANN (Maier et al., 2010; Maier and Dandy, 2000), SVM (Raghavendra and Deka, 2014) and fuzzy based models (Che Osmi et al., 2016)). Whereas, this survey is considered all the popular AI models. Thus, this review is up to date, takes in account the last 20 years of research progress in the field of AI researches and especially in the river WQ research which also the need of the hour and its novelty.

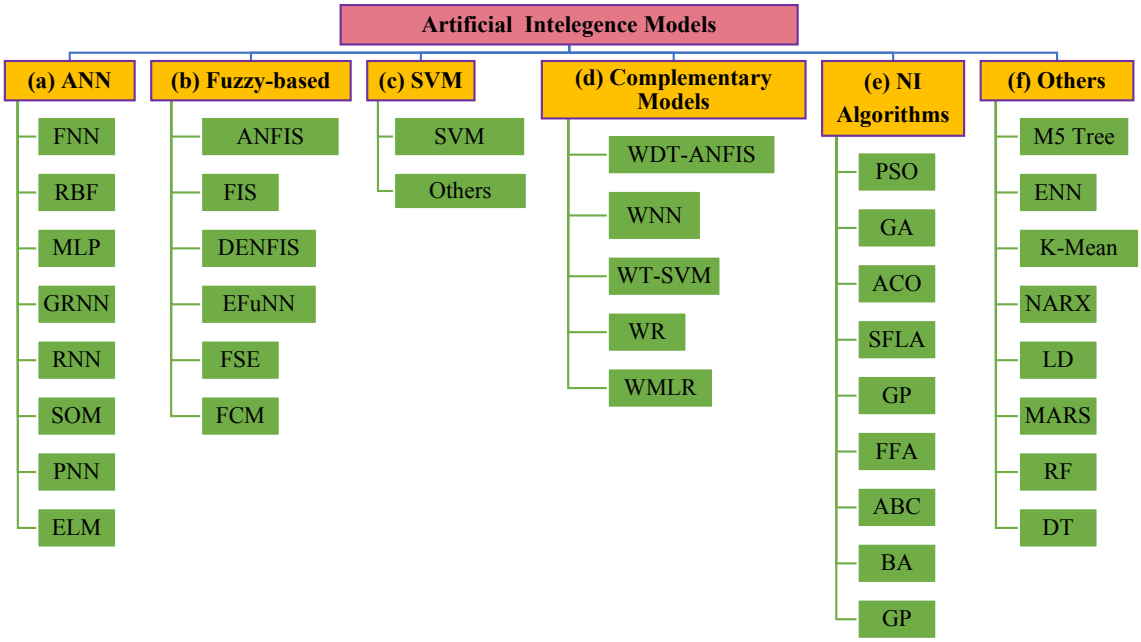


Fig. 1. Various applied version of artificial intelligence models for river water quality modeling: The major divisions are (a) ANN related models (b) Fuzzy-based model are hybrid models which uses fuzzy rules (c) SVM and Others represent various integrated and coupled techniques with basic model (d) Complementary models are the hybrid of wavelet transformation and other models (e) Natural algorithms are the memetic models (f) Other is the categories of the models which doesn't suitable with previously described classes.

In recent decades, AI models have gained the trust of the scientific community as a predictive and classification tool. Considerable researches have progressed on river WQ prediction, monitoring and management using various AI models. With each progressive year, new models have been developed. Thus, a comprehensive milestone that facilitates the contemporary researchers to understand the features of these models and other variables needed for modelling, as a result, the forthcoming studies can evaluate and design the next-generation models. Many reported studies are in a scattered form, and various models with their distinctive capability have been illustrated by researchers. Thus, the current review research tries to organise all the research findings into one survey. Reviewing various works of scientists worldwide reveals that this review is exclusive in its kind, and such an in-depth study of river WQ modelling and assessment has not been done previously to the best knowledge of the authors.

The current study deals with AI models used for river WQ modelling. Numerous models have been designed and applied over the years. Fig. 1 shows the classes and subclasses of these models for the better understanding. AI models have been divided into six subdivisions, namely, ANN model, kernel-based model, fuzzy-based models, complementary models, hybrid models (natural algorithms) that combine architectures of more than one model or techniques and other metaheuristic models that come from different model classifications. The first three classes are the conventional AI models and their improved versions frequently classified as the same by past researchers. However, complementary models take an account of the possible application of wavelet with the combination of other models thus, they have been categorized in a separate group, which allows the readers to better understand the comparative analysis of its performance and effectiveness to deal with river WQ data when coupled with AI models. The fifth category was created considering the studies which dealt with the nature-inspired algorithm for river WQ modelling, as these algorithms behaviour are based on a similar concept which makes them suitable for the same class. The last category was created only for the convenience of the readers and study analysis since the number of papers available is limited and will create no advantage in the present review. Thus, it classifies all other metaheuristic models which don't fall under any previous categories. However, in future, these other models can be further divided as per their architecture or concept when there is a considerable increase in the research articles which is also a present gap and future aspects.

1.4. Research objectives

The research on river quality modelling has increased considerably over time. The contributions of the present survey are summarised as follows. (i) This review is summarised 209 research works based on AI models and their advanced applications, growth over the past two decades (2000–2020), output data consideration, most preferred study area, performance indices, prediction time step and comprehensive assessment. (ii) This review is highlighted the capacity of the conventional models which are applied to the research field of river WQ are taken over by hybrid and complementary AI models as the result of unprecedented attention from scholars. The assessment of the researches will open a new path for researchers in the field of AI and soft computing with forthcoming ideas to fill the gaps presented. (iii) The survey is emphasised the breaches in the implementation of AI models in river WQ research and models that have not been tested with river WQ data. (iv) Finally, the survey reveals the advantages and weaknesses of each model, need for additional exploration of various river areas, issues with data collection and recommendation for future enhancements. Consequently, this review paper presents a snapshot of the historical development of AI models over the years. The in-depth description of the AI model architecture and mathematical theory has not been detailed in this review. However, an appropriate references have been cited in each introductory part of the AI models class for the readers desiring further profound knowledge.

The rest of the review article is divided into nine sections. Sections 2 to 7 details the major classes of AI models as previously described in Fig. 1 along with the subsection that briefly describes each model with related citation. The related bibliographic discussions are cited and reviewed, and the sections are assessed. Section 8 presents a general assessment of the reviewed papers. Section 9 and 10 contain the conclusion and recommendation for future research possibilities of the

impact of river WQ changes and modelling.

2. Artificial neural network (ANN)

2.1. Conceptual background

ANN models are popular AI models due to their robustness and capability to deal with nonlinear data and solve complex problems. ANN model is the brainchild of McCulloch (1943). ANN has been continuously progressing, and its efficiency can be increased with the help of various data optimisation process training algorithms and by selecting appropriate input techniques. ANN consists of three layers, namely, input, hidden and output layers. The number of layer increases when the problem is complex. Thus, the computational efforts also increase. ANN trains with the data provided using various training or learning algorithms. The most popular ANN learning algorithms are generalised regression neural network (GRNN) (Antanasijević et al., 2014, 2013), radial basis function neural network (RBFNN) (Chen et al., 1991), backpropagation (BP), levenberg–marquardt (LM), bayesian regularization (BR), gradient descent, steepest descent gauss-newton, quick propagation and adaptive learning (Maier and Dandy, 1999; Rumelhart et al., 1988; Samarasinghe, 2006). A new version of ANN model called extreme learning machine (ELM) is recently gaining popularity due to its simple-structured single hidden layer feed-forward neural network (FFNN). It works as a linear-in-the parameter model and easily reaches global optima (Huang and Chen, 2007; Huang et al., 2015). Another time series model that can handle nonlinear data is a dynamic neural network called nonlinear autoregressive with eXogenous (NARX) network. It is a subclass of the recurrent neural network (RNN). It uses a gradient–descending learning algorithm that makes it effective in learning; this feature results in its fast network convergence with improved generalisation and memory retention (Diaconescu, 2008; Tsungnan Lin et al., 1996). The typical architecture of ANN is depicted in Fig. 2 (Hameed et al., 2017).

ANN can work efficiently because it can use an input significance approach that can determine the potential input for output variables (Diop et al., 2018). ANN can use a model-based approach that constructs individual models for different inputs to determine the important one (Salih et al., 2019). Thus, its time consumption is high. By contrast, the model-free approach provides developers with choices or uses mutual information. Table 1 presents all the papers that reported the application of ANN models.

2.2. Bibliographic review of ANN models

2.2.1. Comparative studies of ANN with other AI models

Numerous studies have been conducted in the field of river WQ using ANN models. Jeong et al. (2001) applied recurrent ANN to predict chlorophyll-a (Chl-a) using various inputs, such as meteorological, hydrological and environmental variables. The model can produce a good result with 3 days of time lag input data. Four of the studies found that ANN performs better than the regression models such as MLR and nonlinear regression models. The results concluded that ANN is an efficient, accurate and economic tool for river WQ modelling (Anmala and Venkateshwarlu, 2019; Huang and Foo, 2002; Mitrović et al., 2019; Olyaie and Banejad, 2011). Four studies based on river WQ modelling determined that BPNN and FFNN models which use feed-forward and back-propagation algorithm are capable in predicting long-term trends, and perform accurately and their effectiveness is high because their ability to grasp the nonlinear relationship between the input and output variables in compare to models like linear partial least square (PLS) regression (Basant et al., 2010; Dogan et al., 2009; Najah et al., 2011; Yang et al., 2008). Khalil et al. (2011) used ANN, ensemble ANN (ENN) and ENN coupled with canonical correspondence analysis (CCA). CCA was used to manage the functional relationship and standardisation of variables. CCA–ENN was established as the improved model and can estimate the WQ mean value for sites. Niroobakhsh (2012) performed a comparative study to assess the prediction capability of two ANN models, namely, multi-layer perceptron (MLP) and RBF. The results showed that RBF can handle an immense number of data and can thus

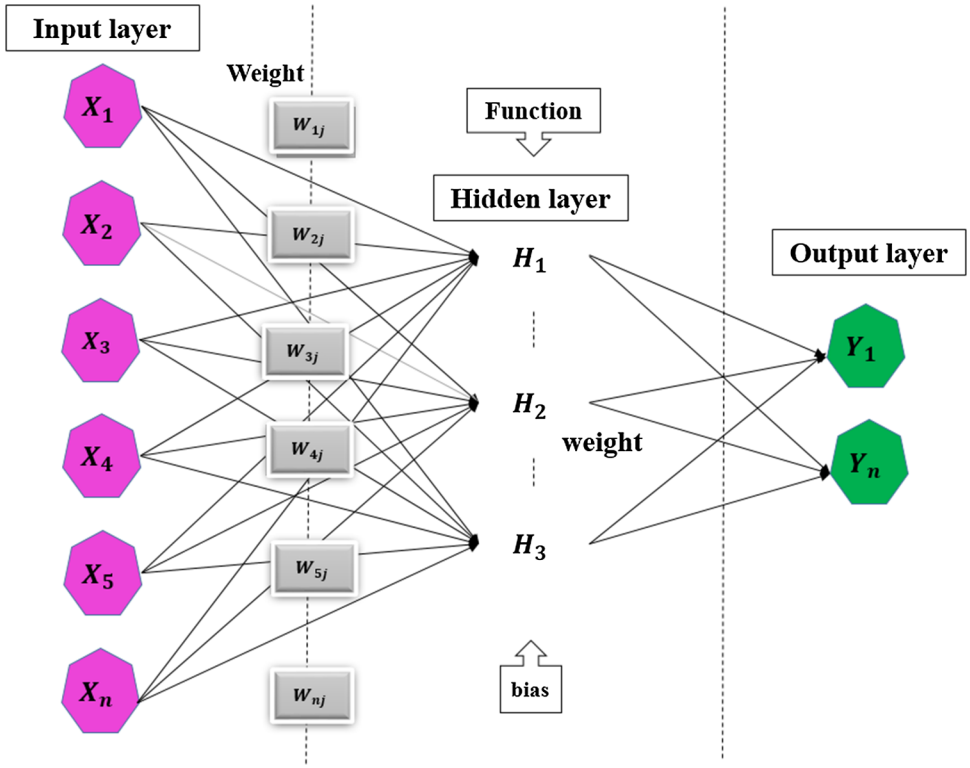


Fig. 2. Typical ANN model architecture with three layers i.e. input, hidden and output layer for river WQ modeling where X = inputs, Y = output, bias is justified by the training process and w = weight. Most common transfer functions are Sigmoidal, Gaussian and Transign.

Table 1
The summary of the research reported the implementation of ANN and its learning approaches.

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
1.	(Jeong et al., 2001)	Nakdong River, Korea	RNN	I_{rad}, P_{pre}, Q, E_{vap}, WT, SD, Tu, pH, DO, NH_4, NO_3-N, PO_4, Si, Rot., Clad, Cop. / Chl-a	MSE	Seasonal (1994–1998)	The study showed the river ecosystem is complex due to discrete seasonal variations of phytoplankton subtleties. RNN is able to manage such diversity with inputs like meteorological, hydrological, physical, chemical and biological. The prediction of Chl-a showed high accuracy and the relationship with input was informative.
2.	(Huang and Foo, 2002)	Apalachicola, River, Florida	FF-BPANN, MRA	Sa, Q_{in}, P_{pre}, Freshwater Q_{in} / Sa	RMSE, R	Hourly, Daily (July 1993)	The study aimed to predict the salinity change in river water due to salinity intrusion which may affect aquatic life using FF-BP-ANN. The ANN proved to correlate various variables like inflow, rainfall and freshwater inflow with high accuracy and cost-effectiveness. The results were acceptable for BP-ANN and faster but not for multiple regression analysis.
3.	(Suen and Eheart, 2003)	Sangamon River, USA	BPNN, RBFNN, MRA	$R_{pre.}$, WT/ NO_3	RMSE	Daily (1993–2000)	BPNN, RBFNN, and MRA were studied along with the incorporation of the Boolean output method. The results show ANN produces a better result with the Boolean method as it simplifies the number of neuron requirement and increases the computational speed.
4.	(Bowden et al., 2005)	Murray River, S. Australia	Hybrid GA, GRNN, ANN, SOM	Sa, Q_{in}, SD / Sa	RMSE	Not given (1992–1998)	The authors used partial mutual information criteria and SOM to remove unnecessary data and hybrid GA and GRNN were used for selecting influential input for the ANN model. The salinity of the river was predicted in advance of 14 days with proper input selection and outperformed previous models.
5.	(Diamantopoulou et al., 2008)	Strymon River, Greece	ANN	NO_3-N, EC, DO, WT, pH, Na, Ca, K, Mg, $P_{Tot.}$, Cl, NO_3, HCO_3, Q / NO_3, EC, DO	R, MAE, RMSE, $M_{\%}$	Monthly (19980–1990)	The study aims to develop the ANN model to fill missing data of WQ variables. ANN was able to predict the monthly value of outputs with great precision and thus, it can be used for predicting as well as filling missing data.
6.	(Yang et al., 2008)	Liming River, China	ANN	Q_1, Q_2, Q_3Q_4, UV, pH /COD	MSE	Minutes (2004–2005)	The authors designed an online water quality monitoring and management system combined with ANN for estimating COD. This model was able to help in the estimation of the quantity of water released from the reservoir to maintain required WQ by dilution.

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Table 1 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
7.	(Sahoo et al., 2009)	Truckee River, USA	BPNN, MRA, CDNA	Rs, Tmx / WT	R, RMSE, ME	Daily (January 1999-September 2002	The study considers that WT plays a governing role in the determination of various patterns in water bodies and thus, used BPNN, MRA and CDNA models for prediction. It was found that water and air temperature are highly dependable when there is no lag where in case of a reverse condition almost independent. BPNN was found to be the superior model than others considered.
8.	(Singh et al., 2009)	Gomti River, India	BPNN	pH, DO, BOD, Al, H, TS, COD, NH_3-N, NO_3-N, Cl, PO_4, K, Na, / DO, BOD	DC, RMSE	Monthly (January 1996-December 1999, January 2002- December 2005)	The study suggests that BPNN is an effective tool to predict variables like DO and BOD in a long term basis as per the case study results on Gomti River, India. The results show that pH, TS and COD have an influence on the BOD model whereas DO has a great impact due to COD, NH_3-N, NO_3-N, and PO_4 changes.
9.	(Amiri and Nakane, 2009)	21 River of Chugoku district, Japan	ANN, MLR	Urban, agriculture, grassland, waterbody, human population density / $N_{Tot.}$	R, p-value	Monthly (2001)	A wide range of data from 21 Rivers of Chugoku districts, Japan using GIS was extracted. Yet, there was a lack of data for other rivers in the area which led to the randomization of data as a hypothetical river basin. The sensitivity test from input data was insufficient but results were capable to predict Total nitrogen. This paper is an example that how to deal with a lack of data and predict the output using ANN.
10.	(Dogan et al., 2009)	Melen River, Turkey	ANN	COD, WT, DO, Q, Chl-a, NH_4-N, NO_3-N, NO_2 / BOD	AARE, MSE, DC	Daily. Weekly (2001–2002)	The author's goal was to estimate the ability of the ANN model to improve BOD prediction accuracy. A sensitivity test was done to improve the prediction quality of ANN in which COD was found to be the most influential parameter. ANN performance was up to the mark and was able to correlate other variables with BOD.
11.	(Chen et al., 2009)	Changle River, China	BPNN	Q, WT, T_t, R_{pre}, N, P/ N, P, DO	DC	Seasonal, Monthly (July 2006 – December 2007)	Researchers aimed to identify the agricultural non-point source pollution by designing a predictive model. They were able to identify the seasonal variation. BPNN was found to be reliable, easy and fast in response.
12.	(Basant et al., 2010)	Gomti River, India	PLS2, FFBP-ANN	pH, Al, H, TS, COD, NH_4-N, NO_3-N, Cl, PO_4, K, Na, DO, BOD /DO, BOD	RMSE, Bias, SEP, DC, NSE, A_f	Monthly (10 Years)	The authors used predictive linear and non-linear models for BOD and DO prediction. It was found both PLS2 and FFBP-ANN performed well yet ANN models can grasp is little more for the non-linear nature of WQ.

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Table 1 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
13.	(Zhang et al., 2010a,b)	Beiyun River, China	BPNN	DO, WT, pH, COD, COD_{cr}, BOD, NH_3-N/ DO	RMSE, ARE	Monthly (January 2007-December 2009)	The study applied the BPNN model to predict DO for the river and found that DO have a direct relationship with WT, BOD, NH_3-N. A more accurate model was constructed by establishing relativity between the variables.
14.	(Khalil et al., 2011)	Nile Delta, Egypt	ANN, Ensemble ANN, (CCA) and ENN	(33 WQ data were taken but used 4) BOD, TVS, TDS, $N_{Tot.}$/ BOD, TVS, TDS, $N_{Tot.}$	NASH, RMSE, relative RMSE, mean, Bias, relative Bias-	Daily, yearly (Not given)	Three models were implemented to estimate Nile delta water quality. Out of 33 WQ variables four where selected whose absolute factor correlation coefficient was found greater than or equal to 0.7. The evaluation shows EANN with CCA is a superior model than ANN and EANN models. BIAS index suggested that EANN is better than ANN. Overall, the conclusion derived that modified ANN models are better than conventional ANN.
15.	(Zali, et al., 2011)	Kinta River, Malaysia	ANN	COD, DO, BOD, pH, TSS, NH_3-N/ WQI	DC, RMSE, SSE	Monthly (2002–2006	The study aims to reduce less substantial WQ variables and estimate DO, TSS and NH_3-N using the ANN model. BOD and COD were found to have the least influence on WQI forecasting and if included in the study, it may increase the cost and time of the overall estimation. The author suggested for intermittent monitoring the program which may improve the efficiency of ANN by great fold.
16.	(NASIR, 2011)	Juru River, Malaysia	ANN	DO, BOD, COD, NH_4-N, TSS, TSS / WQI	DC, R, RMSE	Not given (6 January 2003–5 November 2007)	The study used the ANN model to predict WQI and eliminate less sensitive variables from the test. The initial input consisted of 30 WQ variables but after the sensitivity test, 6 were selected for modelling and suggested that ANN is an excellent method for WQI prediction and management.
17.	(Banejad and Ehsan, 2011)	Morad River, Iran	ANN, MLP, BPNN	EC, TDS, TSS, Tu, Na, HCO_3, PO_4, N-NH_4, DO, BOD, NO_3-N / BOD, DO	R, RMSE, MAE	Monthly (September 2008-September 2009)	The study suggests that ANN models are capable to analyze the pattern of WQ variables and able to form an excellent prediction. ANN was able to do manage non-linear data more efficiently, both in time and space than others.
18.	(Najah et al., 2011)	Johor River, Malaysia	ANN, MLP	TSS, TSS, TDS, EC, Tu / TDS, EC, Tu	MAE, SSE, AMPE	Not given (1998–2002)	In this study, six ANN models were proposed with randomly defined data that are able to predict the output accurately. The models were found to be consistent and vigorous in their performance i.e. predicting DO, EC and Tu.

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Table 1 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
19.	(Asadollahfardi et al., 2011)	Talkheh Rud River. Iran	RNN, MLP	Q, TDS / TDS	MAPE, R, MAE	Monthly (1991–2003)	The study considered the importance of salinity in the selected area and found MLP is a faster predictive model but lacks accuracy in compare to RNN performance in TDS prediction.
20.	(Gazzaz et al., 2012)	Kinta River, Malaysia	FFMLP	pH, BOD, DO, K, FC, EC, WT, Ca, Mg, Zn, COD, TDS, Tu, Fe, Na, Cl, As, PO_4-P, NO_3-N, E.coli, Total coliform, TSS, NH_4 -N / WQI	DC	Monthly (Feb-Nov 1997–2007)	23 WQ variables were analyzed using a parallel, fully connected FFMLP network which was able to find the relationship between them and predict data more efficiently and faster rate. The model has been established to be a reliable and comprehensive tool for WQI prediction.
21.	(Niroobakhsh, 2012)	Jajrood River, Iran	MLP, RBF	TDS, EC, Q / TDS	DC, RMSE	Monthly (40 Years)	This study modelled two ANN models i.e. MLP and RBF to measure TDS for over 40 years. The models produced good results over time and results were effective as well as accurate. They can be used for other water quality management sites. Among both RBF had a closer outcome in respect of MLP.
22.	(Chebud et al., 2012)	Kissimmee River, South Florida	ANN	Chl-a, Tu, $P_{Tot.}$ / Chl-a, Tu, $P_{Tot.}$	DC, RMSE	Seasonal (1998–1999,2009–2010)	ANN model was applied to reduce uncertainty and nonlinear water quality data. It was able to give an excellent relationship between inputs dry and wet season. The study investigated the potential of limited in the case unreachable site or data availability.
23.	(Liu et al., 2012)	Jiaozhou River, China	BPNN	COD, COD_{cr}, DO, NH_3, VHB, M_{oil} / COD, COD_{cr}, DO, NH_3, VHB, M_{oil}	NA	Not given (2005–2006)	The research objective was to minimize cost and time of data collection so used a single model of AI. BPNN performed well and accurately for WQ prediction.
24.	(Heydari et al., 2013)	Delaware River, Pennsylvania, USA	ANN, MLP	DO, WT, pH, Q, EC/ DO, EC	R, RMSE, MAE	Monthly (July 1995-March 2006)	The research aims to predict DO and EC data of the Delaware river using ANN and MLP, and assess the efficiency of the model. It was suggested that ANN models are fully capable of predicting river water quality and proved to be less expensive.
25.	(Rahim and Ahmad, 2013)	Perak River, Malaysia	ANN, ANN-CCA	Zn, TSS, TS, Tu, NO_3, BOD, COD /BOD, COD	r-value, MSE,	Monthly (March 2000-November 2004)	The author's motivation was to use the least number of variables and yet the prediction accuracy could be good so that it can be used when there are fewer data and less time. In this study, ANN and ANN-CCA were used which produced excellent results. Out of 28 input variables only five were chosen by applying CCA, who had a max influence on COD and BOD.

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Table 1 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
26.	(Jiang et al., 2013)	Lanzhou, Yellow River, China	ANN and ANN-MCS	PI / PI	DC	Seasonal (Not given)	The authors have used ANN-MCS and MCS based model for risk assessment of pollution in the river, in reference to COD values. ANN-MCS have proven to be less time consuming, needs moderate computational cost and needed a smaller number of simulations (99% less) to give higher result than conventional ANN model.
27.	(Antanasijević et al., 2013)	Danube River, Serbia	RNN, BPNN, GRNN	Q, pH, WT, EC / DO	RMSE, MAE. d	Monthly (2004–2009)	In this study three ANN models i.e. GRNN, BPNN and RNN were tested for predicting DO. RNN was proved to be a superior model as it can be effectively used for an emergency case or in case of missing data.
28.	(Grbić et al., 2013)	Kielstau (Treene River) Germany	ANN, MLR	DIN, NO_3-N, auto Chl-a, Cl, SO_4, $P_{Tot.}$ / Chl-a	DC, NSE, RMSE	Daily (27th July 2009- 31st December 2010)	The objective of the study was to predict Chl-a as an indicator using ANN and MLR models. A sensitivity test was performed to rule out the least significant inputs. ANN efficiently solved the nonlinear and the complex problem on hand whereas MLR was only able to relate the independent and reliant variables.
29.	(Chu et al., 2013)	Liao River, China	HNN, BPNN	N, $P_{Tot.}$, DO, BOD, COD, COD_{cr}, NH_3-N, Cr, As, Cu, Zn, Pb / WQI	NA	Daily (July 2010)	The study integrated factor analysis and HNN along with a comparative model that is BPNN was used. FHNN was found to be a better model since it overcame the problem of overfitting along with good accuracy in the result.
30.	(Najah et al., 2013)	Johor River, Malaysia	LRM, RBFNN, MLP	EC, TDS, Tu, TS/ TDS	CE, DC, MAPE	Not given (1998–2002)	The study modelled LRM, RBF-NN, and MLP. MLP outpaced LRM but RBF-NN was found to be better, faster and less complex than others. Thus, the authors suggested that RBF-NN is an enhanced model than others considered.
31.	(Iglesias et al., 2014)	Nalon River, N. Spain	ANN	EC, pH, WT, NH_4-N, DO / Tu	DC	Daily (2010)	The author's objective was to use ANN as a means to reduce the time and price of WQ testing and yet able to monitor changes. Change in WT had the highest impact on turbidity and ANN performed well with input that has synergistic relation with the output.

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Table 1 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
32.	(Antanasijević et al., 2014)	Danube River, Hungary	GRNN-MCS	WT, pH, Na, SO_4, Cl, H, HCO_3, N-NO_3, EC / DO	MAE, RMSE	Monthly (2002–2000)	The paper deals with the selection of optimum data normalization (min–max, median, z-score, sigmoid and tanh) and input selection techniques (VIF, Correlation analysis and GA) and then GRNN was used for prediction DO. Min-Max and correlation have given the most accurate results. The MCS sensitivity test for DO shown that it has the highest influence with WT, pH, HCO_3, SO_4 and N-NO_3 in descending order. The authors conclude that ANN could be the most effective tool for DO forecasting.
33.	(Heddam, 2014a)	Klamath River, USA	GRNN, MLR	WT, DO, pH, EC, SD / DO	DC, RMSE, d, MAE	Hourly (26th April 2006-30th December 2009)	The study objective was to do a comparative study for assessing the efficiency of GRNN and MLR for river water modelling. GRNN performed is a lot better than the other model.
34..	(Sarkar and Pandey, 2015)	Yamuna River, India	FFBP-ANN	Q, pH, WT, BOD, DO/DO	RMSE, and DC	Monthly (1990–96)	The authors designed three ANN models with various combinations of input variables. The results favoured more use of ANN especially if data is on a daily basis and consistent. It was also suggested that such a model should be more encouraged in developing countries like India.
35.	(Nemati et al., 2015)	Tai Po River, Hong Kong	MLR, ANFIS, ANN	Cl, pH, EC, WT, NO_2, $P_{Tot.}$, NH_4-N, NO_3-N / DO	R, RMSE, NSE	Monthly (1991–2011)	MLR, ANFIS, and ANN models were applied to create a predictive model for DO in the Tai Po river using WQ variables. It was found that ANN and MLR outperformed ANFIS in this case. The sensitivity to DO was most with Cl then WT, NO_3-N, NO_2-N, NH_4-N and P_{Tot}.
36.	(Gazzaz et al., 2015)	Kinta River, Malaysia	ANN	WT, pH, Tu, TSS, TDS, EC BOD, DO, K, FC, Ca, Mg, Zn, COD, Fe, Na, Cl, As, PO_4-P, NO_3-N, Ecoli, Total coliform, NH_4-N/ Land use	DC	Seasonal (1997–2006)	The author's objective is to do modelling using land-use data and WQVs. The ANN model was able to uncover and develop relative relation with land use and water quality data. The land-use data was used as a replacement of water quality data as a predictor.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
37.	(Kim and Seo, 2015)	Nakdong River, S. Korea	ANN	WT, DO, EC, pH, Tu, Tr, AL, $N_{Tot.}$, $P_{Tot.}$, NO_3, PO_4, Si / Chl- α	RMSE	Bi-weekly (Nov 2009-Nov 2011)	The researchers evaluated algal dynamics of pre and post effect of four major river restoration projects in South Korea yet cannot portray precisely the direct relationship between descriptors and Chl-a since they didn't consider many essential descriptors in the study. As per sensitivity analysis, Chl-a is affected highly by wind velocity then EC, N and lowest by dam discharge. The model showed good results in the prediction of Chl-α concentration proving that algae is not much sensitive to changes and only be affected by long term water quality change of the river.
38.	(Sinha and (Saha), 2015)	Hooghly River, India	ANN	pH, DO, FC, BOD, Total Coliform/ WQI	NSE, RMSE, DC, AAD	Monthly, Seasonal (2002–2008)	The author's objective is to better understand the relationship between various WQ variables and the site of collection using ANN. ANN was successful to categorize the level of pollution at each point. The WQI was calculated using the DELPHI and CCME method in which DELPHI had a higher predictive capability.
39.	(Piotrowski et al., 2015)	BialaTarnowska and Suprasl River, Poland	ANN, ANFIS, WNN, MLP, kNN	Daily mean WT, Max WT, Min WT, AT, Q, S / WT	MSE	Daily (1983–2000)	ANN and other models like ANIFS, WNN, MLP, and kNN were implemented for comparative analysis to predict WT. ANN was found to perform a little better than others due to its ability to correlate with the WQ variables.
40.	(Chang et al., 2015)	Dahan River, Taiwan	ANN: BPNN, ANFIS, NARX	pH, EC, DO, BOD, COD, TSS, TC, $R_{pre.}$, Q, WT, TOC, $P_{Tot.}$, $N_{Tot.}$, NO_3-N, NO_2/ NH_3-N,	RMSE, MAE, CE,	Monthly/ Quarterly (June 2002 to June 2012)	The study advised a systematic analysis scheme using ANNs (BPNN, ANFIS, NARX). It was found that the water quality problem is present in the study area and ammonia has a significant effect on it. Ammonia was most influenced by WT, Q and rainfall. NARX was able to detect the time series, peak values of ammonia which is a useful tool for suggesting the right time for treatment. Here BPNN and the systematic analysis scheme modelled ammonia based on hydrological data more easily and efficiently without much help from water quality data.
41.	(Salami and Ehteshami, 2015)	Rivers of Ireland	ANN	DO, DO%, BOD, Cl, Al, H, WT, pH, NO_3, NH_4 / DO	MSE, MAE, R	Not given (2012)	The authors used ANN modelling and LM as an optimization tool for WQ variables forecasting. The aim was to reduce the price and effort need for data collection and testing exhausting AI model.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
42.	(Chang et al., 2016)	Dahan River, Taiwan	NARX, BPNN, MLR	pH, EC, DO, BOD, COD, TSS, TC, NH_3-N, WT/ $P_{Tot.}$	RMSE, MAE, DC	Monthly (July 2002- June 2012)	The systematical model scheme was applied to determine phosphorus concentration using NARX and BPNN. The model used the gamma test for input. NARX outperformed BPNN.
43.	(Gholamreza et al., 2016)	Zayanderud River, Iran	RBF, TDNN	pH, Ca, Na, Cl, SO_4, HCO_3, Mg, K, H / TDS	DC, IA, RMSE, MBE	Monthly (2001–2010)	The study targeted RBF and TDNN for TDS prediction using various input variables. The sensitivity analysis found Ca and SO_4 more sensitive to TDS. RBF performance was found to better than TDNN.
44.	(Heddam, 2016a)	Charles River, USA	MLP, MLR	WT, DO, pH, EC / Pc_1	DC, d, NSE, RMSE, MAE, MSRE	0.25 Hourly (13 May-04 November 2015)	MLP models were designed and compared with MLR for predicting phycocyanin as an indicator. MLP outperformed MLR and showed excellent results. It was established that the MLP model can be used for predicting Pc_1 with few input variables with high efficiency.
45.	(Heddam, 2016b)	Klamath and Link River, USA	RBFNN, MLP	WT, pH, EC, SD / DO	RMSE, MAE, DC	Hourly (1 June 2014–31 May 2015)	The author aims to design and predict DO with RBFNN and MLP using 4 WQ variables. Various models were generated for various inputs which lead to good performance. Both the models were able to predict DO least 7hrs in advance.
46.	(Csábrági et al., 2016)	Danube River, Hungary	MLR, BPNN, GRNN	pH, R_F, WT, EC / DO	RMSE, MAE, DC	Monthly (1998–2003)	The study revealed that GRNN is the best model and BPNN is better than MLR in predicting DO. The sensitivity test showed that DO is most influenced by pH and least by runoff
47.	(Tarke et al., 2016)	Godavari, Maharashtra, India	ANN	Tu, H, pH, Na, Ca, TSS, SO_4, Mg, EC/ TDS	RMSE, MAE, DC	Monthly (2001–2012)	The authors were successfully able to apply ANN on river water quality parameters. Co-relation analysis was done to establish the relationship between input and output where EC was found that most related input variables for TDS prediction.
48.	(Naseh et al., 2016)	Al-Arab River, Iraq	ANN	pH, H, $MgSO_4$, $CaSO_4$, Cl, SO_4, Tu, EC, TDS/ TDS	R, MSE	Monthly (January 2007-December 2012)	ANN was applied for TDS prediction for better planning and management of the study area. The study was evident upon the use of ANN for complex and non-linear data.
49.	(Heddam, 2016c)	Klamath River, USA	OP-ELM, MLP	DO/ DO	R, d, NSE, RMSE, MAE, MAPE, Bias	Hourly (1st January 2004-31st December 2013)	The authors have used optimally pruned ELM for predicting DO and compare with MLP performance. The study used only DO for input and out variables and successfully modelled it with 7 days' head prediction results. Even though both models were good OP-ELM was better than MLP.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
50.	(Heddam and Kisi, 2017)	Clackamas, Willamette, Umpqua, Tualatin, Delaware, Cumberland basin, USA	OP-ELM, MLPN, MLR	WT, EC, Tu, pH / DO	R, NSE, RMSE, MAE	Hourly (2008–2009)	The researchers have used ELM with sigmoid activation function, radial, online sequential and optimally pruned which was compared with MLP and MLR to predict DO. Among these OP-ELM showed the best accuracy with WQ variables and R-ELM achieved best without them. In both scenarios, MLR performed worst whereas MLP performed satisfactorily.
51.	(Gebler et al., 2017)	Low land rivers, Poland	MLP	Ma, pH,NH_4-N, NO_3-N,N, $N_{Tot.}$ $P_{React.}$, $P_{Tot.}$, DO, BOD / Ma	DC, NSE, RMSE	Monthly (Not given)	The research demonstrated the use of ANN for modelling Macrophytes metrics at a satisfactory level when compared to past research works. It also established the relationship between Ma and eutrophication as of the influence on nutrient concentration. This opened up the possibility to use aquatic plants as bio-indicator for environmental monitoring, but yet it took into account that other variables may influence the results.
52.	(Ahmed, 2017)	Surma River, Bangladesh	FFNN, RBFNN	BOD, COD/ DO	R, MSE, EC	Monthly (2010–2012)	DO was predicted in the Surma river using two ANN models i.e. FFNN and RBFNN. It was found that FFNN produced better results than RBFNN but the overall performance of both models was satisfactory to predict DO. These models can be used for water management and treatment system.
53.	(Hameed et al., 2017)	Langat River and Klang River, Malaysia	BPNN, RBFNN	DO, BOD, COD, NH_3-N, TSS, pH / BOD	RMSE, DC, NSE, RE %	Monthly (January 2001 October 2010)	ANN models were used to predict BOD in a tropical ecosystem. RBFNN was proved to be a superior model than BPNN in handling WQ variables, mimic the pattern and perform well with nonlinear data.
54.	(Csábrági et al., 2017)	Danube River, Hungary	MLR, MLP, RBFNN, GRNN	pH, WT, EC, R_F / DO	RMSE, MAE, DC, WI	Monthly (1998–2002)	During the study, the authors found that the nonlinear model performance was better than linear. GRNN and RBFNN were found to do well in different cases in respect of other models considered. The sensitivity analysis showed that pH has more influence over DO change than EC, temperature, and runoff. The author feels the necessity to design MLR models without respective parameters. They found that the use of random initialization identical model settings in MLP gives prominent results.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
55.	(Villas-Boas et al., 2017)	Piabanha River, Brazil	AANN	Al, NH_4, COD, DO, EC, FC, Fe, NO_3, pH, WT, SO_4, TC, Tu	MSE, FPE, Aic	Seasonal (2009–2012)	The study implemented AANN for WQ prediction along with nonlinear PCA which helped in finding the most relevant parameter due to which out of 30 WQ variables only 13 were selected.
56.	(Dezfooli et al., 2018)	Karoon River, Iran	PNN, kNN, SVM	FC, pH, DO, BOD, WT, $P_{Tot.}$, NO_3, Tu, TS / WQI	ER, EV, Acc	Monthly (2007–2012)	The research used SVM, k-NN, and PNN for WQ prediction. PNN produced cost-effective and good performance than others. The models identified FC as being the most effective WQ predictor.
57.	(Sharaf El Din et al., 2017)	Saint John River, Canada	BPNN, SVM	Tu, TSS, BOD, COD, DO/ WQI	RMSE, DC	Daily (27 June 2015, 10 April and 12 May 2016)	The authors combine the benefits of satellite imaging and using the AI model to get a better performing model. BPNN was successful and accurate in its predictive ability as well as in data fetching from images.
58.	(Sharaf El Din and Zhang, 2017)	Saint John River, Canada	BPNN	TSS, TS, TDS, COD, BOD, DO, EC, WT/ WQI	DC	Random (27 June 2015–23 August 2016)	The study aims to minimize the data size and shorten the assessment process using the AI model and satellite sensing. The result of the integrated technique was good and simplified than before.
59.	(Asadollahfardi et al., 2018)	Zayande River, Iran	MLP, ARIMA	Ca, Na, Cl, SO_4, HCO_3, H, TDS / TDS	VE, MBE, RMSE, DC	Monthly (2001–2010)	The author studied MLP and ARIMA to predict TDS, After optimization with LM, the MLP model performed well and found to be a more consistent model than box Jenkins time series.
60.	(Abba et al., 2018)	Yamuna River, Agra India	ANN, MLR, ANFIS	DO, pH, BOD, WT / pH	DC and RMSE	Monthly (1999–2005)	In this study sensitivity analysis showed that pH has more influence over DO change than BOD and temperature. ANN performance in pH prediction proved to be superior to THE MLR and ANFIS models.
61.	(Nhantumbo et al., 2018)	Dal river, skellefte, Vendel and Klaralven River Sweden	ANN, PPBM	pH, Al, WT / Na, Mg, Ca, K, HCO_3, SO_4, Cl and NO_3	RMSE, CE	Monthly (2001–2011)	The researchers highlighted that PPBM is much preferred in the case of studies in the developing country since some time there is a lack of sufficient data where in this scenario ANN doesn't work well. Even though ANN provides better estimates but PPBM can be calibrated if required. This model can produce a much better result if calibration data is good. The models didn't work well for nitrate estimation. The paper recommends the use of ANN combined with PPBM.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
62.	(Gebler et al., 2018)	River, Poland	ANN	pH, EC, $P_{Tot.}$, DO, P_{React}, NO_3– N, NH_4 -N, N, $N_{Tot.}$, BOD, HQA, HMS, MIR, IBMR, N, D / Ma	RMSE, standardized RMSE, DC	Seasonal (June-August)	The study aims to find the correlation between ecological indices, water pollution, hydro morphological deterioration in using various biological indices; physicochemical parameter with ANN. ANN results very good considering the complexity of the data used in the study; simultaneously it also proved that using so many indices improved the performance and prediction rate of macrophytes. MIR itself is not a sufficient indicator thus, it was recommended to be used along with ecological and physicochemical data.
63.	(Šiljić Tomić et al., 2018a)	Danube River, Serbia	BPNN	WT, pH, EC, CO_2, HCO_3, Al, BOD, COD, TSS, NO_3, Cl, SO_4, PO_4, $P_{tot.}$, H, DO / DO	RMSE, MAE, MAPE, DC	Monthly (2002–2011)	In the study, the response surface method was used and input was enhanced using BP-ANN. Box Behnken design was used for optimization and results showed that models gave satisfactory output and better than previous ANN models.
64.	(Šiljić Tomić et al., 2018b)	Danube River, Serbia	ANN, PNN	WT, Al, $P_{tot.}$, TSS, pH, Ca, H, EC, Mg, CO_2, NO_3, Cl, HCO_3, PO_4, SO_4 / DO	DC, FCPSE	Monthly (2002–2013)	DO was predicted using PNN and ANN models using various WQ variables. It was concluded by the author that less influential inputs can cause bias by multi-outliner records. PNN successfully predicted DO and established relationships with other WQ variables.
65.	(Alves et al., 2018)	Medioparanapanema, Brazil	ANN	WT, pH, N, DO, BOD, $P_{Tot.}$, TF, Tu, TSS/ WQI	DC, MAPE,	Monthly (March –December 2015)	The authors introduced portable UV vis spectroscopy along with ANN to predict the WQI. This integrated system can reduce cost, time and makes the analysis more comprehensive.
66.	(Voza and Vuković, 2018)	Morava River, Serbia	CA, DA, ANN	WT, TSS, DO, H, pH, EC, NO_3, N, PO4, Si, Na, K, Ca, Mg, Cl, SO_4, BOD / WQI	R, SC, DC	Bi-monthly (January 2005-December 2015)	The study used CA, DA and ANN to analyze the WQ and its variation over time. The models were able to categorize into period and pollutant change. The results showed the promise of identification of monitoring periods and according to management measures.
67.	(Abba et al., 2018)	Yamuna River, India	MLR, ANN, ANFIS	WT, BOD, pH / DO	DC, RMSE	Monthly (1999–2005)	The study compared MLR, ANN and ANFIS models for prediction DO outcomes. The results showed that ANN performed better than ANFIS but much superior to MLR.
68.	(Yi et al., 2018)	Nakdong River, South Korea	ELM, BPNN, LR, ANFIS	AT, $R_{pre.}$, S, N, P, N/ P ratio, Chl-a / Chl-a	DC, RMSE	Weekly (January 2013-December 2016)	The study compared ELM with BPNN, LR and ANFIS to find the best predictive model for Chl-a prediction which was used as an algal bloom indicator. ELM was modelled with an independent variable which proved to be the best model of all.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
69.	(Peña-Guzmán et al., 2019)	Salitre River, Colombia	ANN	BOD, COD, TSS, $N_{Tot.}$, $P_{Tot.}$, MBAS, pH, F&O, FC, DO / WQ	NA	Daily, Monthly (2009–2013)	The Salire WQ network was restructured for optimizing the controlling points using ANN. It was established during the study that the water quality can be categorized based on seasonal use by the industrial and commercial sector which will improve the WQ management of the study area. ANN was found to be also cost-effective. ANN results showed that monthly variation for dry and rainy season affects the WQ.
70.	(Mitrović et al., 2019)	Danube River, Serbia	ANN, MCS-ANN	WT, pH, TSS, H, Al, EC, BOD, COD, DO, CO_2, Ca, Mg, HCO_3, NO_3, PO_4, Cl, SO_4, P_{Tot} / WQ	MAE, RMSE, NSE, MAPE, DC	Monthly, Semi-monthly (2002–2011)	The study selected 18 WQ variables for constructing Monte Carlo simulation-based ANN model. The results proved to be better in the case of a single output model than multiple WQ variables. Multi-target fusion models have good results than single-target models or unused one which was successful due to MCS. The model is useful for multi targets with high accuracy, less effort and time.
71.	(Anmala and Venkateshwarlu, 2019)	Green River, Kentucky, USA	ANN	P_{pre}, WT, W, F, A, PO_4, $P_{Tot.}$, FC, / FC, Tu, pH, EC, Sa, TDS, DO, TSS, PO_4, $P_{Tot.}$, SO_4, NH_4 -N, NO_3N	DC, R	Monthly (May 2002-October 2002)	The study constructed various models with different input and output using a nonlinear regression model and ANN. Precipitation was found to be one of the influential variable especially FC. ANN proved to be better than all other models considered in the study.
72.	(Kim et al., 2019)	Nakdong River, S. Korea	ANN	Rs, P_{pre}, W_1, Q, S_h, WT, DO, EC, pH, Tu, Tr, AL, $N_{Tot.}$, $P_{Tot.}$, NO_3-N, Si / Chl-α	RMSE, MAE, R	Bi-weekly (November 2009-November 2011	The investigation revealed the pre and post effect of five major river restoration periods where the algal boom was measured in terms of Chl-a as output and inputs as meteorological, hydrological and other WQ variables using ANN models. The sensitivity test and ANN model were most affected by human activities and short term changes were responsible for the algal subtleties.
73.	(Li et al., 2019)	Qiantang River, China	RNN-DS, BPNN, SVR	COD, pH, $P_{Tot.}$, DO / COD, pH, $P_{Tot.}$, DO	RMSE, MAE, MAPE, NSE	Hourly, Daily (5^{Th} October 2007–5 May 2018)	The authors have integrated Dempster/Shafer theory with RNN to reduce the uncertainty of selection and handling long-term data. The comparison with SVR and BPNN proved that RNNs-DS is a more accurate and high performing model.

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	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
74.	(Zhu et al., 2019c)	Cedar, Fanno and Irondequoit creek River, US	MLP, MLR, ELM	WT, AT, Q DOY / WT	R, d RMSE, MAE	Daily (2001–2017)	ELM was modelled with MLP and MLR. ELM and MLP model performance was higher than the MLR model. ELM with sigmoidal and radial basis activation was the best performer. The use of discharge as input was found to have the least influence on WT.
75.	(Isiyaka et al., 2019)	Kinta River, Malaysia	ANN, HAC, PCA	DO, BOD, COD, TSS, pH, Tu, TDS, NH_3-N, Fe, As, Ca/ DO	DC, RMSE	Monthly (2006–2013)	The study applied clustering techniques with ANN to reduce cost, maintenance, staffing and time. HAC analysis was used for clustering the monitoring sites into two and PCA for pollution source clustering.
76.	(Jin et al., 2019)	Ashi River, China	Improved GA, BPNN	NH_3-N, Tu, EC/ NH_3-N, Tu, EC	RMSE, MAE, MRE, DC	Hourly (8 June 2014–26 October 2014)	The study was coupled with improved GA and BPNN by sliding window prediction pattern to investigate the variation and optimize initial weight and prevents from selecting a local optimal result. This improves prediction and able to design an early warning system which can detect WQ abnormalities.

perform effectively in total dissolved solids (TDS) prediction. Antanasijević et al. (2013) studied the predictive competencies of GRNN, BPNN and RNN models and used a regression MLR model. Amongst the three ANN models, RNN is the most efficient and is also better than MLR. RNN shows least over-training and improved DO prediction capability.

Grbić et al. (2013) demonstrated ANN and MLR model performance for Chl-a prediction and found that the ANN model is more effective in handling nonstationary variables when compared to MLR, adaptive neuro-fuzzy interference system (ANFIS) and ANN models. ANN and MLR models use BP and LM algorithms. ANN model is the most efficient and accurate model for DO prediction amongst the considered models. Sinha and Saha (2015) used ANN to establish a relationship between WQI and source network. Clustering analysis (CA) was conducted to identify the similarities. Feed-forward, BP and LM algorithms were used for the best performance of the ANN model. In another two studies, ANN model was used as a tool to predict the amount of TDS in the river; these algorithms are accurate and reliable for such prediction and current scenario (Naseh et al., 2016; Tarke et al., 2016). Ahmed (2017) determined that feed-forward neural network (FFNN) and RBFNN produce satisfactory results for DO prediction, but FFNN is slightly better in overall performance. Another research addressed the issue by reducing the tenacious application of WQ variables and satellite data modelling by using AI model. BPNN was used for mapping, and the results are surprisingly accurate and simpler than before. Another three studies used the same process to reduce the cost and timing of the project whilst ensuring easy interpretation of data easy for the decision body for improved management planning (Liu et al., 2012; Sharaf El Din et al., 2017; Sharaf El Din and Zhang, 2017). Antanasijević et al. (2019) applied ward neural network (wNN) system along with SOM as a tool to pre-process the dataset to extract the important features of the data and create a simple process. wNN model was used to split the data and had two parallel hidden layers for identifying a different part of WQ data patterns. Thus, this model is unique and increases the accuracy compared with GRNN results.

Suen and Eheart (2003) considered two ANN models, namely, BPNN and RBFNN, which were modelled and compared to predict nitrate contamination. The outcome from RBFNN is quicker than that from BPNN because the number of hidden layers is determined by the fuzzy min–max clustering algorithm, where BPNN applies a trial-and-error approach. This model is robust and highly accurate, but both present similar false-negative frequency. Similarly, the results of a study done by Najah et al. (2013) presented that the performance of RBFNN is more accurate and faster than that of the linear regression model (LRM) and multi-layer perceptron (MLP) in WQ assessment. RBFNN and FFNN were used in a study to find the DO with the help of input data as BOD and COD as they can influence the DO variation. Piotrowski et al. (2015) performed an extensive model assessment using ANN, fuzzy, wavelet and instance-based model, that is, k-nearest neighbours (kNN). The WT modelling results showed that each model is unique depending on the training algorithm, optimisation and data sources. ANN performs better, but the authors suggested that any new model must be assessed using various criteria to establish its performance efficiency. Heddam (2016a) studied RBFNN and MLP techniques for DO modelling and both models were able to successfully predict the DO at 72 hrs ahead of time; however, the reliability of the forecasting reduced with increase in time step ahead. Another model GRNN is gaining popularity because of its good performance. Thus, three different studies compared it with MLR, MLP, RBFNN, BPNN and RNN for WQ modelling. The results show that GRNN performance is accurate, efficient and faster as it doesn't need iterative training and overtraining chances are less (Csábrági et al., 2017, 2016; Heddam, 2014a). Dezfooli et al. (2018) studied SVM, kNN and probabilistic neural network (PNN) as WQ models and established the effect of each input variable using the trial-and-error method. PNN is the best model and performs well when few variables are omitted. kNN works well with all variables included, and SVM performs better with fewest variables. Asadollahfardi et al. (2018) applied auto-regressive integrated moving average (ARIMA) box Jenkins time series and MLP model for TDS estimation. MLP used 10 hidden layers, LM as an optimisation algorithm with Tansig transfer function which resulted

in a better and more reliable model than the others. Abba et al. (2018) simultaneously studied MLR, ANN, and ANFIS models to predict DO. The results showed that the performance of ANN is superior to those of ANFIS and MLR models as they suffered from the drawbacks of over-estimation and under-estimation, respectively.

2.2.2. Studies incorporated with various input–output selection techniques

Bowden et al. (2005) enhanced the GRNN model performance by using mutual information to characterise the important input and output WQ variables, SOM to reduce the dimensionality of inputs and GA to establish a relationship between them. Zali et al. (2011) identified the relationship of WQ variables using sensitivity analysis technique on inputs considered in the research with the classification of pollution sources. ANN successfully correlates and finds the pollution problems in the study area, that is, agricultural runoff and animal farm wastage. Similarly, NASIR (2011) used sensitivity input analysis, which is a model-based technique. It was used to identify the effective input–output relationship and eliminate the rest for ensuring the effectiveness of the ANN model prediction for predicting WQI. Gazzaz et al. (2012) reduced the massive quantity of data using principal factor analysis (PFA) to identify variables affecting WQ in the study area. MLP with ANN model was used to construct the relationship, and the result is fast and accurate because it can use achieved data to create a model that can be used for direct calculation of the river WQ from raw data. A combined model was introduced by Chu et al. (2013), where Hopfield neural network (HNN) was used as a classifier modelling tool and factor analysis (FA) were used together for classification and dimensionality reduction, respectively. The combination model can overcome overfitting caused by WQ inputs and performs better than other evaluation models. Rahim and Ahmad (2013) used CCA to reduce input variables with least or no predictive utility in the creation of the ANN model for BOD prediction which increased the computational cost and modelling accuracy. WQ estimation comes with uncertainty and very complex model design and was addressed by Jiang et al. (2013) using Monte Carlo simulation (MCS) with ANN to reduce time and cost without compromising the accuracy. MCS helps produce improved model WQ variables, including input and output along with possibility and numerical distribution, but needs high computation labour. Similar to previous research, Antanasijević et al. (2014) used MCS to address uncertainty. They also used two methods for input selection, namely, a model-free approach with correlation analysis and model-based approach with GA, to obtain the best result when incorporated with GRNN. These combined techniques are effective and produce enhanced DO prediction results. Šiljić Tomić et al. (2018a) focused on interpolation and extrapolation of the ANN model by using PNN with emphasis on the right data selection for the learning process to create the number of possible models and select the optimum competent model. The results of extrapolation are unaffected by less substantial inputs but may be biased yet, DO modelling results were good. The same researchers used box Behnken design for optimisation and multi-filter approach for input correlation, which was then modelled with ANN and found the coupled technique is effective in DO prediction (Šiljić Tomić et al., 2018b). Voza and Vuković (2018) studied two analysis techniques, namely, clustering and discriminant analysis (DA), to reduce the initial dataset and isolate the main source of pollutants. This screening increases the efficiency of the ANN model, which in turn can provide a coherent and reliable WQ management strategy.

Chen et al. (2009) used correlation analysis and quadratic polynomial stepwise regression analyses for selecting appropriate input to obtain improved results during BPNN modelling. The results are good, fast, easy to manage and can be useful for river and agricultural water management. Chang et al. (2015) conducted modelling using dynamic ANN coupled with gamma test to establish a spatiotemporal relationship and select the combination of best-fitted inputs by evaluating the noise. The model was accompanied by NARX, ANN and ANFIS models. Gamma test helped in the input selection and improved prediction performance. The model can find the water pollutant source and produce the most accurate results. Villas-Boas et al. (2017) evaluated the performance of auto-associative NN coupled with nonlinear principal component analysis (PCA) to address the issue of budget difficulties and inadequate WQ network design. The nonlinear PCA can understand the WQ relationship and perform and address the issues. Another paper reported the use of BPNN modelling along with correlation analysis for normalisation, similar to previous works. PCA was added in place of polynomial analysis to reduce the correlation of neurons. The results can cope with the missing data and decrease the monitoring index and can thus, be used for WQ management tools (Zhang et al., 2010a,b). Li et al. (2019) used D-S evidence theory (Dempster/Shafer) to deal with WQ data uncertainty by data fusion and improve data performance by combining it with RNN. The results are promising since they are comparable to those of support vector regression (SVR), BPNN and conventional RNN models. Isiyaka et al. (2019) investigated WQ by using the ANN model and multivariate statistical tools. Hierarchical agglomerative cluster (HAC) analysis helped in clustering the monitoring site in two and PCA was applied pollution source identification such as anthropogenic, surface runoff and rock weathering in which rock weather was the highest contributor of pollution. A sensitivity test was applied to recognise the most sensitive variable affected by pollution. All the techniques coupled with ANN model increased the overall performance.

2.2.3. Modelling with distinctive indicators

Kim and Seo (2015) used Chl-a as an indicator to assess algal growth by integrating ANN, and sensitivity analysis to find the most sensitive input to the required output. The authors also used clustering and partitioning to balance training, which led to highly accurate results using the ANN model. Heddam (2016b) used Phycocyanin as an indicator for phytoplankton and algal growth. MLP, ANN and MLR models were used for modelling, and LM was used for optimisation. MLP model performs well during training, validation, testing and prediction. Two research studies selected TDS as the output variables since TDS can be used as an indicator of salinity (salt deposits) in the study area, as often present due to the geographical location. On of the study applied Time delay NN (TDNN) and RBF models whose prediction accuracy was high and other modelled the WQ variables with MLP and RNN in which MLP obtained results faster than RNN, but RNN applicability and accurate was more than MLP (Asadollahfardi et al., 2011; Gholamreza et al., 2016). WQ can be assessed using indicators, such as macrophytes which were used by Gebler et al. (2017) as an indicator for the testing of the extent of river contamination. This study explored the possibility of using plants as bio-indicators for WQ monitoring and management. Similar output variables, namely, macrophytes, were taken as indicators in 2018, but the relative relation was established by considering water pollutants, hydro–morphological changes and physical, chemical and biological variables. The authors suggested that the relationship must be observed in more than one aspect, such as environmental and ecological, to increase model accuracy (Gebler et al., 2018). Kim et al. (2019) used Chl-a as an indicator to predict algal bloom using the ANN model in their study. The results are helpful for environmental management and assessing other factors affecting the algal changing aspects. During the sensitivity analysis, nitrogen shows the most impact on algal bloom variation in the study area.

2.2.4. Models addressing the issue of missing data and cost-effective approaches

Diamantopoulou et al. (2008) examined the ANN model and estimated the missing data by using the available relevant WQ variables. A similar outcome was achieved by Amiri and Nakane (2009), where data were taken using geographic information system (GIS); the lack of meticulous data was dealt by randomisation of the data to produce accurate results using ANN model. Singh et al. (2009) tested the BPNN model for DO and BOD computation. The model is able to capture

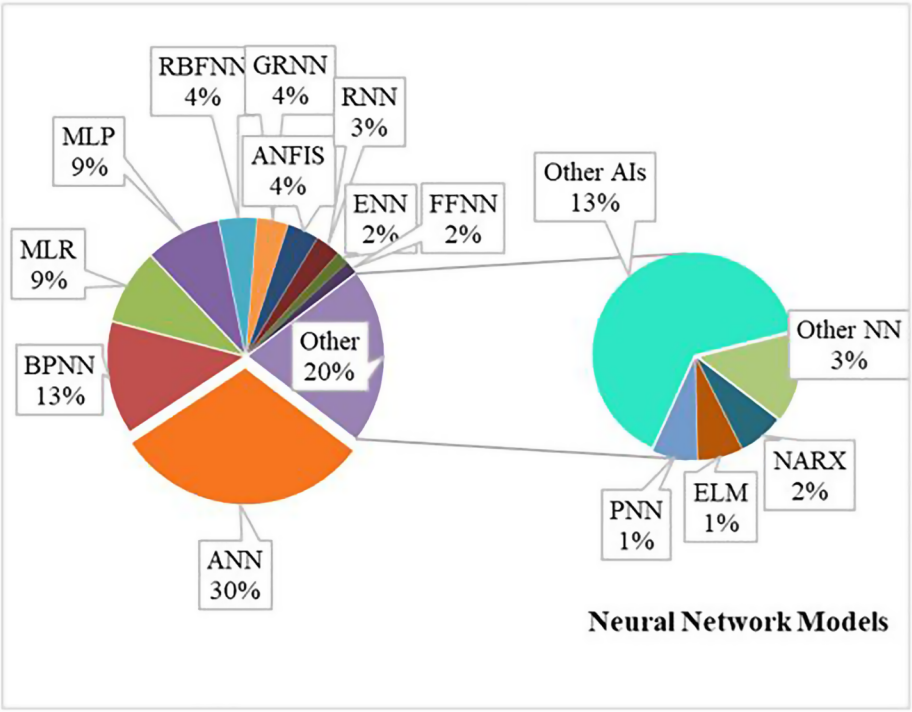


Fig. 3. Neural Network related algorithms**:** The figure represents Neural Network sub-class Models along with other comparative models (a) Models except ANFIS and MLR are ANN models (b) ANN Models which are least explored.

longterm trends of WQ variation, both in time and space. Thus, the study concluded that it can be used as a predictive tool for assessing river health. Sahoo et al. (2009) successfully predicted WT even with a lack of detailed data. The time delay approach was used to adjust nonlinear data for overcoming the predicament, and mutual information limited the time delay up to 3 days. GA was used for optimisation, and data were fed to a chaotic nonlinear dynamic algorithm (CNDA) and BPNN model. The results showed that the BPNN model produces the best results. A related study of Heydari et al. (2013) solved the problem of missing data and created the ANN model using available data of other WQ variables. Then the backpropagation MLP model was used for the network architecture and produced accurate results. Nhantumbo et al. (2018) addressed the same issue of cost and lack of details of variables using the ANN model and comparing with the physical process-based model (PPBM) with LM optimiser. The physical process-based model is satisfactory when calibrated and can be used along with the ANN model to obtain the best possible result. The data unavailability can be sometimes due to an inactive monitoring station. To address this point, Mitrović et al. (2019) used ANN model for virtual WQ monitoring and MCS algorithm for data sensitivity analysis. MCS with the ANN model is successful for complex multi-output targets.

Salami and Ehteshami (2015) selected inputs, such as WT, pH and EC, because these experiments are easy and low cost for DO prediction using FFNN along with LM and transign for the learning process. The study concluded that ANN model is a good tool for missing data, finding the relationship between data variables and overcoming problems from data collection and experimentation. Peña-Guzmán et al. (2019) used ANN model for handling WQ prediction for a large population, which is difficult, costly and time consuming with the traditional methods. The results are satisfactory and concluded that ANN model can be used for similar conditions.

Chebud et al. (2012) used the ANN model with GIS to quantify and establish a relationship between the inputs and minimise project cost. GIS was used for landscape analysis and dynamic monitoring and produces good quality images by removing haziness and cloud effect. The model can reduce uncertainty and manage the complexity of the relationship between WQ variables. Iglesias et al. (2014) minimised the cost and time of physiochemical testing by integrating ANN. The study used turbidity as a WQ indicator to reduce excessive modelling time, and the results are satisfactory. Sarkar and Pandey (2015) proposed ANN model for predicting DO and suggested that ANN can be used as an active tool for river WQ monitoring for developing countries. The only consideration will be consistent data and the same controlling factors for training and testing datasets. Gazzaz et al. (2015) correlated and used WQI as an indicator and applied ANN modelling. During the research, WQI was influenced by mining, rubber industries, forestry, urbanisation and agricultural activities. Chang et al. (2016) compared MLR, static-type BPNN, dynamic-type NARX and subclass of RNN models. Gamma test was used to estimate the importance and interdependence of the input variables and the noise. NARX model can predict the spatiotemporal features in the inputs, outperform other models and can perform effectively under missing data and costly WQ variables. Alves et al. (2018) proposed replacing the older method of WQ analysis with UV spectrophotometry with ANN modelling tool to decrease the time, need of costly infrastructures and other expenses of the project. The study introduced a simple and fast way to monitor WQ.

Four research study applied ELM and modified ELM models i.e. optimally pruned extreme learning machine (OP-ELM), radial basis ELM, online-sequential ELM and compared their performance with MLP, MLR, LR, BPNN and ANFIS models. OP-ELM model demonstrates higher predictable performance when compared to MLP and could predict WQ in advance from several hours to 7 days with improved accuracy. ELM models outperform others as the result of its use of a single hidden layer feed-forward network, can learn faster than other ANN models and has lesser number of epochs. It is superior in prediction accuracy, speed and in generalisation. Thus, the model is easy to apply. ELM can be applied using different activation functions, such as sigmoidal, sine, RBF, hard-line and triangular function, in which sigmoidal and RBF has the highest success among all. Thus, the designer can opt for the best design as per the feasibility of the project (Heddam, 2016c; Heddam and Kisi, 2017; Yi et al., 2018; Zhu et al., 2019c).

2.3. Model assessment

Assessment of the various literature surveys on ANN modelling for river WQ reveals the following matters. The popularity of the ANN model is due to the flexibility in its implementation in river WQ. The ANN model can process complex, nonlinear, nonstationary and vague data of river WQ without assumptions between input and output variables. Thus, ANN models are categorised as a 'black box model,' which uses the numerical approach without understanding the mechanism (Juditsky et al., 1995). Thus, this model can be applied easily. ANN models have been proven excellent predictive tools by the literature considered in this study for predicting the change in river WQ in advance, which is helpful for decision-makers and policymakers around the world, as shown in Fig. 3. The pie chart presents various ANN models, compares to the applied models in the 76 review papers and concludes that ANN performance is the highest; the traditional ANN model has been applied in 30% of the studies and remains the most popular. With the development in the ANN model architecture, the new generation of ANN models, such as GRNN (4%), PNN (1%), ELM (1%) and NARX (1%) models, continuously try to take the place of the previous ones, such as ANN (30%), BPNN (13%) and MLP (9%) models. These new models are gaining popularity in the last few years. Few hybrid models, such as RBFNN (4%) and RNN (3) models, have been studied along with the traditional models and has shown promising performance. Fig. 3 illustrates a few models, such as MLR (9%), ANFIS (4%) and other AI models (13%), used for comparison against ANN models.

ANN models perform better when coupled with various dimensionality reduction tools (e.g., PCA, CCA, SOM, PFA and FA) and clustering tools (e.g., CA, DA and GA) (Chen et al., 2009; Khalil et al., 2011; Sinha and (Saha), 2015; Voza and Vuković, 2018). The efficiency and accuracy of the results can be increased because of the enhanced understanding of function relation, identifying the similarities, decreasing the huge input data into a manageable amount and standardising the data.

Similarly, time and cost can be decreased, but quality can be increased using satellite and integration tool coupled with the ANN model (such as SOM, GIS, satellite data modelling and remote sensing). These tools are also able to get the most important features from data resulting in increased efficiency and performance (Ahmed (2017); Antanasijević et al., 2019; Diamantopoulou et al., 2008; Sharaf El Din et al., 2017; Sharaf El Din and Zhang, 2017).

The literature reveals that many comparative studies have been done using conventional ANN and other modified versions of the ANN model. Different models of ANN show good results in varying scenarios and advantages over one another. Out of 74 studies, 30% of the researchers have successfully used conventional ANN, and most of them have found that ANN is better than most models, such as ANFIS, MLR, wavelet and kNN models. This popularity is subjected to ANN model flexibility, computationally economical, reliable, consistent predictability and capability to handle nonlinearity (few examples supporting the efficiency of traditional ANN model (Anmala and Venkateshwarlu, 2019; Banejad and Ehsan, 2011; Grbić et al., 2013; Huang and Foo, 2002)).

The second most popular model is BPNN, which uses the BP algorithm along with LM algorithm. LM is the most famous training algorithm and finds optimum results in a minimisation problem. This feature gives the model high stability and efficiency during the learning process (Gorashi and Abdullah, 2012). Even though, RNN model has been implemented less compared with other popular ANN models but has the aptitude to achieve the least over-training with good prediction capability (Antanasijević et al., 2013; Asadollahfardi et al., 2011). GRNN model is fast in data processing as it does not require an iterative training process. Thus, this model can deal with large data. This condition is common in WQ assessment and predictions which makes it more suitable for handling WQ data (Cigizoglu and Alp, 2006).

Enhancement tools of multiple input variables can be applied with the ANN model to increase the efficiency of the model. Selection of the significant input attributes is essential because excessive input may cause overtraining, introduces extra local minima and compromises the exact meaning of the input–output relationship. For input selection, a model-free approach and model-based approach can be used (Antanasijević et al., 2014). Mutual information and multi-filter approach for variable characterisation can be helpful considering they do not depend on model performance and use nonlinear statistical dependence to measure and determine inputs and thus, are called model-free approach (Bowden et al., 2005). Methods, such as sensitivity and quadratic polynomial stepwise regression analyses, can be used in river WQ models to identify the most influential inputs over the outputs and eliminate the rest (Chen et al., 2009; NASIR, 2011). Another method of extrapolation can be used in data selection considering its ability to neglects the variables with least influence (Šiljić Tomić et al., 2018a).

Various ANN models in river WQ modelling use indicator outputs to assess other pollutants. For instance, Kim and Seo (2015) used chlorophyll as an indicator for algal growth in the river, Heddam (2016a,b,c) used phycocyanin as the phytoplankton and algal growth indicator and Gebler et al. (2017) used macrophytes for pollutant indication. Such studies are useful when collecting specific water samples is difficult due to unavailability of required fund, manpower or risk in reaching the specific area for data collection. The indicators can help fill the gap in those studies and achieve the goal of river health assessment.

ANN models reduce the cost of the project and can deal with missing data and insufficient meticulous data using randomisation, time delay, mutual information and virtual WQ modelling (Amiri and Nakane, 2009; Mitrović et al., 2019; Sahoo et al., 2009). They address the issue faced by many projects where the overall modelling, monitoring and assessment suffer due to insufficient fund and inadequate data, specifically in developing countries where ANN models can be an economic solution.

Several new models of ANN with good potential in the area or river WQ, such as ELM coupled with optimally prune method, can predict WQ in advance by many days. Similarly, ELM can process data in high speed, has high accuracy and good generalisation capability by achieving the least training error and the smallest norm of output weight and performs fast due to less number of epochs (Heddam, 2016c; Zhu et al., 2019c). This model has universal approximation capacity and improves overfitting problem by random feature mapping approach; the simple design and the learning without hidden layer tuning make ELM applicable to WQ modelling (Huang et al., 2015). In many cases including saline water, ELM and conventional ANN has successfully predicted WQ several hours ahead of time (Alizadeh et al., 2018).

Another model known as NARX can deal with missing data, noise and nonlinear inputs (Chang et al., 2016). wNN with SOM can find import data, simplify a process by using data splitting and apply parallel hidden layer technique, which makes it unique (Antanasijević et al., 2019). Least explored models can be implemented in the river WQ modelling combined with other AI models to achieve superior model performance. Furthermore, the challenges regarding the uniqueness of each study area, seasonal variation and other abnormalities common in WQ data must be assessed.

Data is normally split into two; training set and validation set. Generally, ANN models are not able to extrapolate beyond the range of training data which leads to poor prediction values and thus, it becomes hard to get good results with small data set. To solve the problem holdout method, synthetic testing data, trial method and cross-validation methods have been applied to improve the generalization of the model (Maier and Dandy, 2001; Master, 1993). The most popular technique is cross-validation technique which splits the data into three; training testing and validation which helps in regulate when the training has to be dismissed (Maier and Dandy, 2000). But unfortunately very less thought has been given over the issue in the recent

research considering WQ data may present itself with the issue of unavailability of uniform long term data as an example of these two research where data availability was one year monthly data (Amiri and Nakane, 2009; Chang et al., 2001).

3. Fuzzy logic-based modelling strategy

3.1. Conceptual background

Fuzzy logic was developed to deal with uncertainty, imprecision and vagueness. Zadeh (1965) introduced a fuzzy set as the class with defined criteria of membership. The fuzzy set was established with the idea of inclusion, blending, balance and relation. A fuzzy set consists of a membership function, a discourse and a membership value. Each fuzzy set is defined by a fuzzy membership function which allocates a membership value to every conceivable input with two types of rules: antecedents (IF) and consequents (THEN). The membership value is between 0 and 1 where 0 represents absolute non-membership and 1 represents full membership. The consideration of the rules is depended upon the type of interference systems such as Mamdani, Sugeno, Tsumoto and Kang (Mamdani and Assilian, 1975; Sugeno and Kang, 1988; Takagi and Sugeno, 1985). The number of fuzzy rules needed for inputs is depended on the complexity of the real system and hence the number of rules increases with the complexity. FIS is able to better understand the relationship between the variables and then interpreted them in simple language, it has been considered as “grey box” instead of “black box” model (Lindskog, 1997).

The architecture of adaptive network-based fuzzy inference system presented by Jang (1993) uses the hybrid learning process with input–output mapping and has functional blocks as a rule base, a database, a decision-making unit, a fuzzification interface and a defuzzification interface. ANFIS model has the advantage over ANN models in adaptive filtering, adaptive signal processing, such as identification, inverse coding and noise cancelling. Fuzzy rules are determined by three identification methods, namely, grid partition (GP2), subtractive clustering (SC) and fuzzy c-mean (FCM) clustering (Jang, 1996). The typical architecture of the five-layer ANFIS has been shown in Fig. 4 (Yaseen et al., 2018a).

Zhu et al. (2019b) concluded that the ANFIS model is a flexible AI model that can identify complexity and nonlinearity between variables. Another popular type of FIS model is dynamic evolving neural–fuzzy inference system (DENFIS), which can learn online and offline. DENFIS advances through incremental, hybrid learning and adds new inputs, new classes and fuzzy rules; thus, it is superior to MLP, ANFIS and evolving fuzzy neural network (EFuNN) (Wright, 1991a,b). FIS has been successfully applied for river WQ for prediction, assessment and classification. All these FIS applications have been reported in studies related to river WQ. Tables 2A and 2B classifies FIS model applications into (A) predictive and (B) classification models.

3.2. Bibliographic review of fuzzy-based models

3.2.1. Predictive models

FIS has been applied mostly as a classification tool, However, in any case, it has been applied to compare with existing models such as PNN, BN and grey relation method. It followed the modelling pattern of fuzzification, membership selection, rule determination and defuzzification. FIS was used with water pollution index to incorporate and administer information (e.g. individual knowledge and experience) and qualitative environmental variables. FIS shows the good result when coupled with MCS or cascade-type FIS is applied which increases accuracy and speed (Li et al., 2016a,b; Mahapatra et al., 2011; Nikoo et al., 2011; Nirmala et al., 2015).

One of the most popular implemented models of fuzzy is ANFIS which proved to be a better model than PNN, FIS and MLP in five different studies. ANFIS performance is good, accurate and quicker than MLP and PNN models as these model cannot produce reliable solution due to the nonlinear characteristic of the WQ variables and FIS takes longer in data interpretation. Moreover, this model is economic than laboratory analysis and can deal with a large amount of data. Data normalisation, correlation and regression analyses, and modelling conversion (inverse transformation) were applied to find important input variables which increased the performance many folds (Ahmed et al., 2017; Houari et al., 2019; Mahmoodabadi and Rezaei Arshad, 2018; Najah et al., 2014; Yan et al., 2010).

Deng et al. (2015) compared gaussian cloud transformation–fuzzy

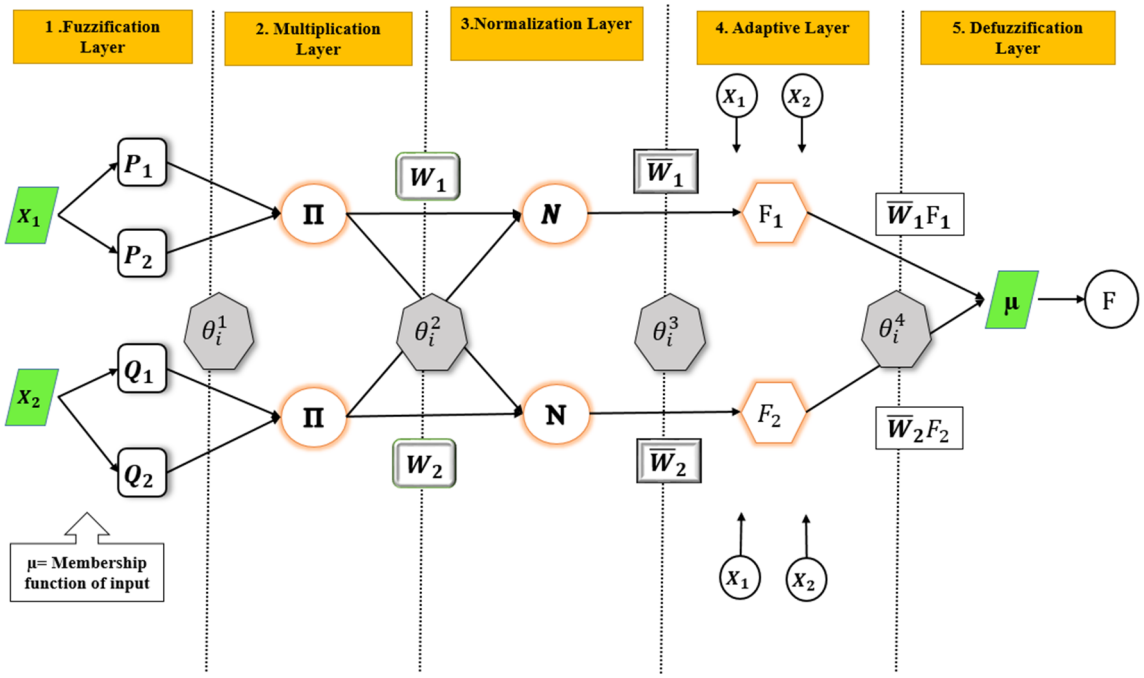


Fig. 4. Five layer ANFIS design where, θ_i^1 = Node Output, i = node, F = final output, X_1 and X_2 = Input variable, P_1, P_2, Q_1, Q_2 = Fuzzy rules (Yaseen, Ghareb, et al., 2018).

Table 2A
Summary of the researches demonstrated the utility of the fuzzy logic approach: Predictive Models.

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
77.	(Yan et al., 2010)	Major Rivers, China	ANFIS, ANN	DO, COD, NH_3-N/ DO	R, NSE, RMSE	Weekly (40th −18th week 2009)	The research classified the WQ of all the major rivers of China using ANFIS. ANFIS used with Gaussian function showed superior performance and had a high prediction result for rivers WQ. ANFIS outperformed ANN and generated continuous yield making the valuation more coherent.
78.	(Mahapatra et al., 2011)	Mahanadi, Ganga, Cauvery, Sabarmati, Godavari, Brahmaputra Rivers, India	ANN, CFS	pH, BOD, DO, FC, EC, WT, NH_4-N/ WQI	NA	Not given	The authors have chosen 6 rivers of India to predict water quality using the cascaded fuzzy system. It was found to be more effective and have a high approximation capacity than the conventional fuzzy system. They have fuzzified the input, and then defuzzified the output and the result performance was checked. If required, the rule base can be increased to improve the resulting output in CFS. The WQI results were compared to Indian, Malaysia and USA WQ criteria. The results were very close to Malaysia standards which proved the model predicted well.
79.	(Nikoo et al., 2011)	Jajrood River, Iran	FIS, BN, PNN	DO, FC, COD pH, NO_3, P_{Tot}, WT, Tu, TS/ WQI	ARE	Not given (2008)	In the study, three models had been considered FIS, BN and PNN which were optimized using Monte Carlo simulation to create a hybrid probabilistic water quality index. The results showed that these model were capable enough for assessment of probabilistic water quality.
80.	(X. Wang et al., 2013)	Haihe River, China	FSM	COD, BOD, NH_3-N/ WQI	NA	Weekly (2011)	The author applied the FSM model for the evaluation of river water quality. The river WQ was classified as per the water quality standard rank.
81.	(Orouji et al., 2013)	Sefidrood River, Iran	ANFIS, GP	pH, TDS, K, EC, SO_4, Ca, Mg, Na, Cl, Q / WQI	RMSE, DC, NSE	Monthly (20 years)	Two data-driven models ANFIS and GP were tested for WQ prediction in the river system. GP result accuracy was improved than ANFIS when different input models were constructed and proved to be more effective.
82.	(Najah et al., 2014)	Johor River, Malaysia	ANFIS, MLP	WT, pH, NH_4-N, NO_3-N /DO	DC, MAPE, R	Monthly (1998–2007)	The researchers have done a comparative study between ANFIS and MLP for predicting DO using various WQ variables. It was found the most influential WQ variable for DO was NH_3-N and NO_3-N where pH and WT had the least effect. ANFIS was proved to be a better model than MLP and also able to predict when the DO value was extreme.
83.	(Heddam, 2014b)	Klamath River, USA	DENFIS, MLR, MLP	pH, WT, EC, SD/ DO	RMSE, MAE, d, DC	Hourly (1st January to 30th December 2009)	The author had used DENFIS which worked on the evolutionary clustering method and equated with MLR and MLP to design an improved model for predicting DO. The model of DENFIS with the online-based system was found to be the superior model than others considered. It was suggested that this model could be used as a substitute system for DO prediction.
84.	(Deng et al., 2015)	Yangtze River, China	FTSM	DO, COD, WT, EC/ DO, COD, WT, EC	MSE, MAPE, NSE, R	DO, COD: Weekly (2004–2013) WT, EC: Monthly (1986–2014)	The authors have proposed model sedately for all four variables i.e. DO, COD, WT, EC using the Fuzzy time series model and cloud model theory. The paper was successful to achieve soft partitioning of undefined numerical data, decreased noise, improved data set and amplified prediction accuracy. The proposed model performance was better than other existing time series model like ARMA, SVM, ANN-GT etc.
85.	(Nirmala et al., 2015)	Chunnambar River, India	FIS	DO, BOD, pH, WT/ WQI	NA	Monthly (March 2013- February 2014)	The pollution content of the river was studied using FIS to better control and monitoring of river pollution. The results were very promising and close to the Indian surface water quality criteria.

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Table 2A (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
86.	(Li et al., 2016a,b)	Qu River, China	FWPI, FIS, FCA and GRM	DO, PI, BOD, NH_3-N, TP / WQI	R	Yearly (2006–2010)	The author designed the FWPI method derived from the fuzzy method. The results suggest the method can accurately distinguish the problems of water quality assessment and management. The new model was consistent and convenient to produce an analysis which suggests possible ways to reduce water pollution. The drawback of the model that it may not able to handle a large number of parameters quite well.
87.	(Ahmed et al., 2017)	Surma River, Bangladesh	ANFIS	pH, AL, H, TS, TDS, K, PO_4, NO_3, DO/ BOD	MAE, NSE, MAE, R	Monthly (2010–2012)	The study aims to examine the extent of the ANFIS application towards BOD prediction. The result showed that ANFIS performance was satisfactory and can be utilized in the future.
88.	(Heddam, 2017)	Klamath River, USA	EFuNN, MLR	WT, pH, EC, SD, DO/ DO	RMSE, MAE, DC	Hourly (1st January 2014–31 December 2014)	The study was done to test the efficiency of EFuNN and compared it with MLR. EFuNN was found to be a better and effective model than the other for DO prediction.
89.	(Mahmoodabadi and Rezaei Arshad, 2018)	Karoun River, Iran	ANFIS	pH, EC, TDS, SO_4, HCO_3, Cl, Ca, Mg, Na / Q	RMSE, DC, MAE	Monthly (1968–2015)	The researcher's objective was to predict pollution discharge using chemical WQ variables with ANFIS. Mann-Kendall test was used for the analysis of fluctuation in WQ which indicated increase over time whereas discharge was decreased. The study rivals that a steep decrease in discharge readily increases salinity. The results from ANFIS were very reasonable.
90.	(Tiwari et al., 2018)	Satluj River, India	FCM- ANFIS, SC_1 - ANFIS	pH, EC, Cl, DO, BOD, TDS, TSS, NO_3 N, NH_4-N, $P_{Tot.}$, FC / WQI	RMSE, MSE, DC	Monthly (1996–2012)	In this study two clustering techniques i.e. fuzzy C-mean and subtractive clustering with ANFIS was used. SC_1-ANFIS predictability was found to be better than FCM. During sensitivity test ammonia, chloride and faecal coliform showed more influence towards WQ changes.
91.	(Yaseen et al., 2018a,b,c,d)	Selangor River, Malaysia	ANFIS-FCM, ANFIS-GP_2, ANFIS-SC_1	DO, TSS, Tu, Ca, BOD, COD, WT, pH / WQI	MAE, RMSE, DC	Monthly (2000–2011)	Three hybrid ANFIS models were modelled with GP, SC_1 and FCM using WQ variables. GP was found to be the least performing model of all. ANFIS-SC_1 and ANFIS-FCM used fewer variables which resulted in less computational time and further increase inaccuracy. However, all the models showed good prediction capacity and robust nature.
92.	(Zhu et al., 2019b)	Dischmabach, Mentue, Rhone River Switzerland and Drava River, Croatia	MLP, ANFIS-GP, ANFIS-SC_1, ANFIS-FCM	WT, AT, Q, / WT	RMSE, MAE, R, d,	Daily River wise respectively (2004–2012, 2002–2012, 1984–2013, 1991–2016)	The authors choose one Croatian and three Swiss rivers to model the WT using MLP and ANFIS models with a combination of grid partition, subtractive clustering and fuzzy c-mean clustering. Each model performed better at different scenarios of river considered. The specific input was proved to be increasing the modelling accuracy considering the seasonal variations in this case.
93.	(Houari et al., 2019)	Tafna River, Algeria	ANFIS	Q, Sa / Sa	DC, RMSE, NSE	Daily (1977–79, 1982–1991)	The investigation aims to analyze the prediction capability of ANFIS. The result very good for daily time and it was concluded that the model is effective in other scenarios like hydrological and environmental variables.
94.	(Chen et al., 2019)	Fuyang River, China	SFM, PMFSFM	DO, NH_3-N, N, NO_3, NO_2, PO_4, pH, $P_{Tot.}$, COD/ DO	AA, Variance, H-Value	Not given	The author introduced fuzzy rules with SFM to eliminate the inability to deal with the complexity of WQ variables and led the new PMFSFM model. The results showed much higher accuracy and efficiency in the case of PMFSFM. It was also concluded that unbalanced data can take more time.

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Table 2A (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
95.	(Elkiran et al., 2019)	Yamuna, India	ANFIS, BPNN, SVM, ARIMA, SAE, WAE ENN	DO, BOD, COD, Q, pH NH_3, WT/ DO	DC, RMSE	Monthly (1999–2012)	The study developed a single and multiple predictive models using AI models and three ensemble techniques which were able to enhance the overall performance. The combined performance of ENN and ANFIS was superior to another single (BPNN, ARIMA, SVM) and combined models.

time series (GCT-FTS), ARIMA, RBFNN, nonlinear auto-regression (NAR), SVM, order series method (OSM) and ANN–GT (Gaussian transformation) models. The study aimed to develop multifactor WQ time series, where Gaussian transformation was applied to historical data, and multifactor FTS model was used on two different time period datasets. The new model can perform soft partitioning of uncertain data, decrease noise and construct improved quality training data. Thus, adaptive expectation model was applied to enhance the performance. Tiwari et al. (2018) incorporated FCM and SC techniques, which are powerful tools for multidimensional WQ data analysis, and were clustered with ANFIS to verify their predicting and classifying capabilities. SC is more accurate than FCM technique; thus, it was used for sensitivity analysis for better understanding the impact of the input variables on the output. Yaseen et al. (2018c) used SC, FCM and GP2 method to find the most effective predictive model. The result shows that SC is the most effective model then FCM and GP2, respectively. The first two models are less complex, robust and consume less time in WQI prediction. Zhu et al. (2019b) performed similar modelling combination of ANFIS with three identification methods i.e. SC, FCM, GP and different functions like genfis 1, 2 and 3, and MLP model. MLP performs better, but inaccuracy is less in FCM–ANFIS and GP–ANFIS models as the models can well manage input data and thus, performs better.

An integrated model for river WQ was introduced by Heddam (2014b) as DENFIS. It is based on evolving clustering method (ECM) was used for DO prediction, and the results were compared with those of MLR and MLP models. Online-based DENFIS outperforms other models with high accuracy. Heddam (2017) applied an evolving fuzzy neural network (EFuNN) model and compared its performance with that of the MLR model. The result is highly accurate when EFuNN model is used; thus, the model is a good alternative for DO prediction. Chen et al. (2019) experimented on a hybrid model, called possibility measured-based fuzzy support function machine (PMFSFM), to overcome the shortcoming of the individual methods. The hybrid model is accurate, considers the fuzziness of WQ variables and is less time consuming under few dimensions.

3.2.2. Classification models

WQ classification has been the primary and initial goal of many researchers. Chang et al. (2001) used fuzzy synthetic evaluation (FSE) approach in achieving this goal using the emerging technique of AI techniques. FSE can deal with the uncertainties of WQ and lack of the boundary of WQ and water to be utilised. It is also more relevant than other techniques. In this previous study, simple fuzzy classification, fuzzy information intensity and defuzzification were applied to the WQ data to interpret the data with high accuracy. Ning and Chang (2004) developed a fuzzy multi-objective evaluation (FMO-E) model to deal with a long-standing arbitrary situation. WQ analysis might increase the level of monitoring for reasonable utilisation of water at each location. GIS was used as the data retriever and an optimiser. The model can handle multiple objectives with a pragmatic approach, which increases the success of screening, selecting and sequencing the optimal expansion substitutes. In another study, interactive FMOE was utilised by Lee and Chang (2005) for efficient WQ management considering financially viable factors. The model can balance the economic and environmental factors to achieve high WQ model performance, which in turn increases the options for decision-makers.

Chen et al. (2005) applied the fuzzy contingent valuation (FCV) method to improve the river WQ by providing supportive information for cost-effective analysis, which can improve the management plan. Similarly, Nasiri et al. (2007) applied fuzzy multiple-attribute (FMA) decision support expert. This method is suitable for comprehensive data investigation and easy to assess and can evaluate WQ effectively. The model helps classify the WQ and improve the administration plan. Singh et al. (2007) explored the use of FMO–LP (Linear programming) for WQ evaluation for different disposal location. The model is flexible and enables water-related policies to be managed easily and cost-

Table 2B
The summary of the researches demonstrated on the utility of the fuzzy logic approach.

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
96.	(Chang et al., 2001)	Tseng-Wen River, Taiwan	FSE	DO, BOD, NH_3-N, pH, TSS / WQI	NA	Seasonal (1997)	In the study, authors have used three FSE techniques i.e. simple fuzzy classification, fuzzy information intensity and defuzzification for the classification of the seasonally based samples. The results were promising for WQ monitoring and managing.
97.	(Liou et al., 2003)	Keelung River, Taiwan	FST, FCM	DO, BOD, NH_3-N, pH, TSS / WQI	NA	Monthly (1991–2000)	Two-stage fuzzy set theory was applied to the piecewise linear distribution for calculating river pollution index and the fuzzy c-mean (FCM) for river quality index. The index showed a more systematic and simple method for data assessment.
98.	(Ning and Chang, 2004)	Kao-Ping River, Taiwan	FMOE	DO, BOD, $P_{Tot.}$, NH_3-N, NO_3-N, NO_2-N/ WQI	NA	Monthly, Seasonal (May-October)	The study addresses the challenges of the continuous and complex task of river monitoring using fuzzy multi-objective evaluation. Authors also integrated GIS for data archival, data and spatial analysis.
99.	(Lee and Chang, 2005)	Tou-Chen, China	IFMOLP	BOD / BOD	NA	Not given	This research deals with the fiscal and environmental issues related to river WQ management and tried to resolve it by using IFMOLP. IFMOLP was effectively able to optimize and appraise imprecise and arbitrary WQ data.
100.	(Sârbu and Pop, 2005)	Danube, Romania	Fuzzy-PCA	pH, COD, DO, Ca, Mg, Cl, SO_4, HCO_3, NH_4-N, NO_3, NO_2 PO_4, NH_4, Al, H, Fe/ WQI	NA	Monthly (1985–1996)	The investigation used conventional PCA and other Fuzzy PCA. The new model was a more comprehensible, more accurate and able to manage missing data and poor linear comparison.
101.	(Chen et al., 2005)	Dan-Shi River, Taiwan	FCV	BOD, DO, TSS, E.coli / BOD, DO	NA	Not given (1970–1980)	The study uses the FCV method by utilizing its flexibility in the assessment and classification. The new method was able to map the subjective assessment.
102.	(Ocampo-Duque et al., 2006)	Ebro River, Spain	FIS	DO, EC, pH, TSS, BOD, TOC, microbial, Ions, Toxic substances/ WQI	R	Not given (2002–2004)	FIS model was applied as a decision-making tool and found that it's a very appropriate technique to deal with water quality classification with a simple framework.
103.	(Nasiri et al., 2007)	Tha Chin River, Thailand	FST, FMAA	DO, BOD, TC, NH_3-N, NH_4-N, $P_{Tot.}$ / WQI	NA	Not given (1999)	The authors have proposed a fuzzy multi-attribute analysis support expert system using a fuzzy set theory for WQI and management. The new system was found more than capable to handle the fuzziness in WQ variables.
104.	(Singh et al., 2007)	Yamuna River, India	IFMOLP	Q, BOD, DO, Cost of treatment, (Drain) Q / DO	NA	Not given (1995–1996)	The study uses the fuzzy set theory in which multi-objective function and the fuzzy number was applied as IFMOLP. The model was able to set the WQ criteria and calculate the optimal treatment efficiency for the decision-makers.
105.	(Wang et al., 2008)	Naoli River, china	FSE	DO, COD, N, NO_3-N, NO_2-N, $P_{Tot.}$, TSS, NH_4-N/ WQI	R	Not given	The researchers effectively integrated FSE and parameter correlation analysis as well as did a sensitivity analysis to simplify the calculation and yet got better accuracy.
106.	(Karamouz et al., 2009)	Karoon and Dez River, Iran	FCA, GA	TSS, WT, DO, pH, TDS, EC, Cl, SO_4, NO_3, COD, Fe, Zn, Pb, Cd, Cr, Cu, Mn, BOD, NH_4, FC, Chl-a / WQI	NA	Monthly, weekly (Not given)	The authors used a fuzzy clustering genetic algorithm-based optimization model, Kriging method and analytical hierarchy process. The model used was good for zoning and designing of water quality monitoring network.
107.	(Mouton et al., 2009)	Zwalm River, Belgium	FST	Q, W_1, Z, $P_{Tot.}$, NH_4-N, NO_3-N, EC, DO / Q, W_1, Z, NH_4-N, NO_3-N, $P_{Tot.}$, EC, DO	NA	Not given (2000–2004)	The study integrated with the nearest hill-climbing algorithm with a data-driven model. The study was proven to be flexible enough to allow habitat selection. The FST model allows for better visualization and interpretation.
108.	(Alias et al., 2009)	Layang, Segget, Tebrau, Tukang Batu, Kempas, Buloh River, Malaysia	FAHP	BOD, COD, N, TSS, DO, pH, WT, Fe, Q_1, U, Economy / WQI	NA	Not given	The investigation objective was to deal with vague data for a number of combined rivers using FAHP which showed a representation of risk and level of assurance. The model was able to give better efficient enough for discoing-makers.

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Table 2B (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
109.	(Raman et al., 2009)	Sememyih River, Malaysia	FIS	DO, BOD, COD, NH_4-N, TSS, pH/ WQI	R	Not given (1997–2004)	The author used fuzzy logic to design and classify fuzzy WQI and the model was able to deal with non-stationary data. The results were helpful for monitoring and managing river WQ.
110.	(Huang et al., 2010)	Qiantang River, China	FCA	NH_3-N, f, COD, Cr, V-ArOH, DO, BOD, Cd, Hg, $P_{Tot.}$, CN/ WQI	NA	Monthly (2004)	FCA was used for water pollutant evaluation for monitoring purposes. The results showed seasonally and point pollution sources with a unique physicochemical variation.
111.	(Lu et al., 2010)	Wei River, China	FSE	pH, EC, BOD, DO, COD, f, Se, As, Hg / WQI	NA	Not given (Oct-Nov 2006)	The objective of the study was to assess the river health using a fuzzy synthetic evaluation. The results were able to do the classification and found that BOD, COD, and Hg are above the permissible level better when the entropy method was integrated.
112.	(Babaei et al., 2011)	Karoon River, Iraq	FST	TDS, Tu, pH, DO, TC, TSS, NH_3/ Fuzzy WQI	NA	Monthly (1999–2000)	The fuzzy model was implemented to assist in a better way for the decision authorities. The results were able to classify the WQ and examine the three-dimensional and temporal changes in it.
113.	(Scannapieco et al., 2012)	Sele River, Italy	Fuzzy Analysis	DO, NH_4, NO_3, $P_{Tot.}$, BOD, COD, E.coli / WQI	NA	Monthly, Semester (2002–2007)	The study developed binary and fuzzy logic methods to assess the water quality and classify them as per their quality level. It was able better understanding the fuzziness and thus, had higher accuracy.
114.	(Xu et al., 2012)	Zhangweinan River, China	FCA	EC, DO, COD, BOD, TSS, H, Cl, SO_4, N, NH_4-N, NO3-N, P, f, V-ArOH, CN, Hg, As, Cr / WQ	NA	Seasonal (2001–2009)	The research applied FCA for WQ analysis and pollutant identification integrated with cluster analysis and seasonal Kendall test. The model was able to classify as per the pollutant and identify seasonal variations.
115.	(Liu and Zou, 2012)	Luokou stream, China	IFMEM	pH, DO, COD, NH_3-N/ WQI	NA	Weekly (2011)	The author used IFMEM and the conventional FMEM for WQ monitoring and found that an improved version was able to produce better results with efficient data comprehension and gives a promise of a new approach.
116.	(Angulo et al., 2012)	Guadiana River, Spain	Fuzzy logic Technique, FIS	pH, EC, Tu, WT, $P_{Tot.}$, NH_4, DO, WL/ WQI	NA	Monthly (March 2007-January 2009)	The study is based on the episodes of the discharge affecting the water quality, eutrophication and fishes. The results were able to interpret the data to a more understandable form and detect the water quality change.
117.	(Ocampo-Duque et al., 2013)	Cauca River, Colombia	FIS	DO, FC, BOD, WT, PO_4, NO_3, Tu, TS, pH/ WQI	NA	Seasonal (2002–2010)	Fuzzy WQI was calculated using FIS and MCS tools. The integrated model was able to estimate the uncertainty and randomness of WQ data.
118.	(Wang et al., 2014)	Tseng-Wen River, Taiwan	VFEM	DO, BOD, pH, NH_3-N, TSS/ WQI	NA	Seasonal (1997)	The study used two methods of fuzzy binary comparison method and theory of variable fuzzy sets to evaluate the river WQ. The new method was able to improve the assessment and classify quality with more accuracy.
119.	(Mourhir et al., 2014)	Bouregre- Chaouia, Morocco	FIS	DO, BOD, COD, NH_4, $P_{Tot.}$, FC, TSS / WQI	NA	Quarterly (2007–2009)	The authors used FIS for water quality indexing to help with the decision making. The results of the new classification method were better than the conservative methods and can be used for risk mitigation.
120.	(Singh et al., 2015)	Yamuna River, India	FAHP	Cl. EC, pH, SAR, TDS, BOD, COD, TC, NO_3, N, Al, H, $P_{Tot.}$, NH_4, WT / EC, TDS, SAR, Cl, pH	R	Not given (1999)	The efficiency of FAHP was evaluated for river WQ. The model was able to rank areas as per the priority of the WQ. The model was found easy and robust for such an assessment.
121.	(Deng et al., 2015)	Yangtze and Beas River, China	FTSM, NAR, SVM, OSM, ARIMA, RBFNN, ANN-GT	DO, COD, WT, EC/ DO, COD, WT, EC	MSE, MAPE, CE, R	Weekly (2004–2013), Monthly (1986–2014)	The cloud model theory was applied along with the FTSM to manage the vagueness and uncertainty in WQ and integrated with the adaptive expectation model to enhance performance. The results were superior to other models considered i.e. ARIMA, RBF-NN, NAR, SVM, ANN-GT, and OSM.

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Table 2B (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
122.	(Zhang et al., 2019)	Jialu River, China	IFMEM	Q, WQ, human activity, COD / WQI	NA	Not given (2008–2017)	The author's objective was to assess the river health based on pressure-state-element extension and incorporated IFME model. Out of 20 inputs, pesticides, water consumption by industries, COD, and sewage discharge was found to be affecting the river health the most. The new model was effective in the assessment.
123.	(Singh et al., 2019)	Yamuna River, India	FC-WQI	DO, FC, BOD, pH, WT, NO_3, TC, EC / WQI	NA	Monthly (1994)	The study used a fuzzy approach for water quality assessment so that the vagueness of the data can be handled. The results were helpful for capturing and interpreting the WQ data.

efficiently. Huang et al. (2010) classified river WQ according to the pollution zones and examined WQ features, pollution source and change according to the sources. These questions were solved efficiently using statistical analysis, such as fuzzy comprehensive analysis (FCA) and factor analysis (FA). Scannapieco et al. (2012) applied fuzzy logic for WQ assessment and added Mann–Kendall test to analyse the trends when sampling regularity was decreased. The classification was based on the pollution level of macro indicators and extended biotic index. The study also considers only 5 membership functions to void crossing of more than two classes. This model allows an improved understanding of the limits of the WQ variables and thus, increases the accuracy of the evaluation.

FIS model has been successfully integrated for river assessment for estimating the river health and evaluating the need for water treatment in four studies. The model has shown good accuracy, flexibility to develop an enhanced classification, ability to deal with the nonlinear input variables and classify river WQ in natural language. Thus, the model helps decision making more convenient (Babaei et al., 2011; Mourhir et al., 2014; Ocampo-Duque et al., 2006; Raman et al., 2009).

Liu and Zou (2012) examined the improved fuzzy matter–element evaluation method by using the coefficient of variation to determine the weight and normal membership degree. The new method is more scientific, inclusive and effective than traditional calculating super scale method by avoiding abnormal values during the WQ modelling. Angulo et al. (2012) used fuzzy logic to assess the anthropogenic effect on river WQ variables. FIS model can detect the surges of wastewater discharge, changes due to eutrophication in the river, risk on fish growth and develop a cause-and-effect relationship. Ocampo-Duque et al. (2013) hybridised FIS as a probabilistic fuzzy hybrid model and added MCS to handle imprecise and vague input data. The model can identify the water pollution source and flexibly classify them into natural language thus, can be used for WQ indexing. Singh et al. (2015) applied fuzzy analytical hierarchy process (F-AHP) to evaluate river health on the basis of Chang's extension analysis. The method is easy and effective and can be applied to obtain priority weight. The model is effective and improves water treatment plans and projects. In another study, the improved fuzzy matter–element evaluation method was used by Zhang et al. (2019) along with the construction of pressure–state–response model for river health, which is based on the relationship of the variables and urban river health index. The model can solve incompatible complex problems related to WQ and asses the multi-index as urban WQI.

Deng et al. (2015) extensively compared the hybrid fuzzy time series (FTS) model with ARIMA, RBFNN, NAR, SVM, ANN–GCT and OSM models. FTS used heuristic GCT algorithm and Gaussian cloud algorithm. The model can reduce noise by calculating the length of the periodicity and improve the prediction performance considerably when compared with other models. For WQ management, Singh et al. (2019) proposed a fuzzy-based model along with the GIS model, which is a geographical regression method. The model can manage the vagueness of the data and index the WQ variables is considerably comprehensive form and track the changes in WQ. This information can help decision-makers to develop improved water management and monitoring policies.

Lu et al. (2010) utilised FSE model for river WQ assessment by incorporating entropy method, and the results are comparable to those of the conventional FSE performance. The combined technique is effective as the entropy method can simplify the calculation process by determining the weight of the assessment constraints. Fuzzy WQI was generated in the study. Fuzzy WQI can interpret the uncertainty and vagueness of WQ better than the normal WQ model. This way allows decision-makers to interpret and manage water-associated projects effectively. Xu et al. (2012) examined FCA coupled with CA and seasonal Kendall test. FCA classifies the WQ data into various quality levels, whereas Kendall test classifies them into different periods and identifies the change in the WQ which makes it an excellent WQ classification

model. Wang et al. (2014) designed a fuzzy evaluation model by integrating fuzzy binary comparison method and variable fuzzy sets. This model uses eigenvector of level H and thus increases the WQ assessment performance to a higher level by using developing interval scale rather than point scale.

Liou et al. (2003) explored the use of fuzzy set theory for data condensation using WQ indicators. The authors also applied the fuzzy c-mean approach for assessment. The model enhances the freedom of the decision-makers. This method can evaluate different scenarios and can present an enhanced assessment. Lee and Chang (2005) implemented PCA and fuzzy PCA to manage, compress and extract a large amount of WQ data. They considerably reduced the shortcomings, such as sensitivity to outlines, missing data and managing poorly distributed data, by adding fuzzy with PCA. FPCA results are sharper than those of normal PCA. Wang et al. (2008) coupled FSE with PCA. This method helps in the correlation of variables and thus contributes to an improved assessment by eliminating the less sensitive variables. The results are satisfactory and more efficient and considerably easier to perform than those of the typical PCA. Karamouz et al. (2009) applied fuzzy clustering for classification, Kriging method for regionalising spatiotemporal changes, AHP for selection of suitable sampling station, entropy theory for sample frequency calculation and GA optimisation. Various combinations of each method were used for classification, river monitoring station selection, WQ variable selection and standardisation. The two models can achieve the desired objective of finding potential monitoring station and can, therefore, be helpful in river WQ monitoring. Mouton et al. (2009) compared an expert knowledge-based method (i.e. hill-climbing algorithm) and a data-driven model (i.e. fuzzy rule-based modelling) for WQ modelling. Fuzzy sets were optimised by nearest ascent hill-climbing algorithm. However, data-driven models outperform all others. They also provide a suitable habitat for modellers. In a similar study, Alias et al. (2009) applied fuzzy AHP and compared it with the conventional techniques. The combined multi-criteria technique was used by reason of its flexibility, applicability, simplified use and capability to create WQI. This model can manage the vagueness of the data and rank results as Fuzzy WQI compared with WQI models.

3.3. Model assessment

An assessment of the various studies on fuzzy-based modelling of river WQ shows the following. All the scrutinised papers based on fuzzy models take the second place, and ANFIS is the top-related model since it has the advantages of ANN and FIS models. ANFIS can manage nonlinearity, uncertainty and fuzziness related to river WQ. ANFIS has several advantages in signal identification, inverse coding and noise cancelling. ANFIS produces enhanced results when data are normalised and reduced (Ahmed et al., 2017). Most studies use either one or a combination of multidimensional data analysis method in ANFIS, namely, GP, SC and FCM clustering. However, SC is a better tool than the others in handling river WQ classification (Tiwari et al., 2018; Yaseen et al., 2018a,b,c,d).

ANFIS model has shown good performance in handling river quality data as it has two types of parameter: linear and non-linear which can be attuned accordingly through the training stage (Jang, 1993). Along with the member, function plays a vital role and right choice leads to the best optimization of parameters and hence results in higher accuracy (Jang, 1996). Another deciding factor for designing a good ANFIS model is the identification method applied for splitting the input space. Popular techniques applied are grid partition, subtractive clustering and fuzzy c-means clustering with various function such as genfis 1, 2 and 3 (Jang et al., 1997). Therefore, ANFIS can perform well with non-linear data along with proper selective parameters decided by the designer.

Few models, such as DENFIS and EFuNN models, are introduced recently for river water modelling. They perform well in managing river WQ. However, the considered comparative models are not against the basic benchmarked models and other popularly used models.

FTS with Gaussian transformation can decrease noise, construct enhanced training data and apply soft partitioning. This method is superior to ARIMA, RBFNN, NAR, SVM, ANN and OSM models (Deng et al., 2015). FIS, FSE, FMOE and FMOLP approaches can deal with WQ uncertainties and are flexible in creating fuzzy rules, classifying them into classes as bad, average or good according to criteria, interpreting the skewed information and understanding the relationship between input variables. FMOE model has been used to identify the monitoring station and increase the success of screening, selecting and sequencing the optimal expansion substitutes (Ning and Chang, 2004). All these tools can efficiently classify river WQ and help monitor and manage the river system to create improved monitoring projects.

Various modified and hybrid versions of fuzzy set rules are available, but each can handle river WQ and create fuzzy WQI. In addition to that, they coupled with PCA, they can extract large amounts of data, outgrow the problem of outliners, handle missing data and poorly scattered data, and remove less related data (Lee and Chang, 2005; Wang et al., 2008).

Fuzzy based model is able to produce a high-level result and easily incorporates natural language prior knowledge to the model. This allows the parameter estimation algorithm to allocate a practical preliminary value which decreases the chance of getting unnecessary local minima. Another advantage is that the ability of extrapolation which can be used for analysis and revealing new information from the data (Lindskog, 1997). The Fuzzy rule-based model describes the input–output relation more realistically and has physical interpretation which has allowed many researchers in WQ research to choose FIS model rather than ANN models (Ocampo-Duque et al., 2006; Raman et al., 2009).

Fuzzy models are able to analyse and conclude from the given data about the behaviour of a complex system without prior specification of a functional structure in comparison to other black-box models (Jacquin and Shamseldin, 2009).

Nevertheless, the ability to establish a relationship between input–output variable becomes its limitation when too many input variables are there resulting to a number of rules and a complex system which is also known as “The curse of dimensionality” introduced by (Bellman, 1966). Thus, calibration become difficult.

4. Kernel-based AI models

4.1. Conceptual background

The concept of non-separable data in support vector network was founded in 1995. This type of learning machine uses nonlinear input mapping to a very high-dimensional space and ensures high generalisation. SVM depends on the so-called large margin factor and built based on the structural risk minimization principle. SVM can produce excellent accuracy compared with linear, kNN and NN classifier because it does not include knowledge. SVM comprises regularisation, kernel function and regression model complexity. Thus, it is known as a kernel model (Cortes and Vapnik, 1995; Vapnik, 2000). Kernel functions and parameter are selected to reduce the bound on the Vapnik-Chervonenkis dimension. The extended SVM was able to solve function estimation problems by applying Vapnik’s epsilon insensitive loss function and Huber loss function as shown in Fig. 5 with the general architecture of SVM (Chen and Yu, 2007; Yaseen et al., 2019c). SVM can be designed as linear, polynomial, spline, RBF network and MLP (Suykens and Vandewalle, 1999; Vapnik, 1998). SVM has been prevalently used to build a classification model for trend prediction and solving regression and time series problems (Raghavendra and Deka, 2014). Various models developed using kernel architecture are listed in Table 3.

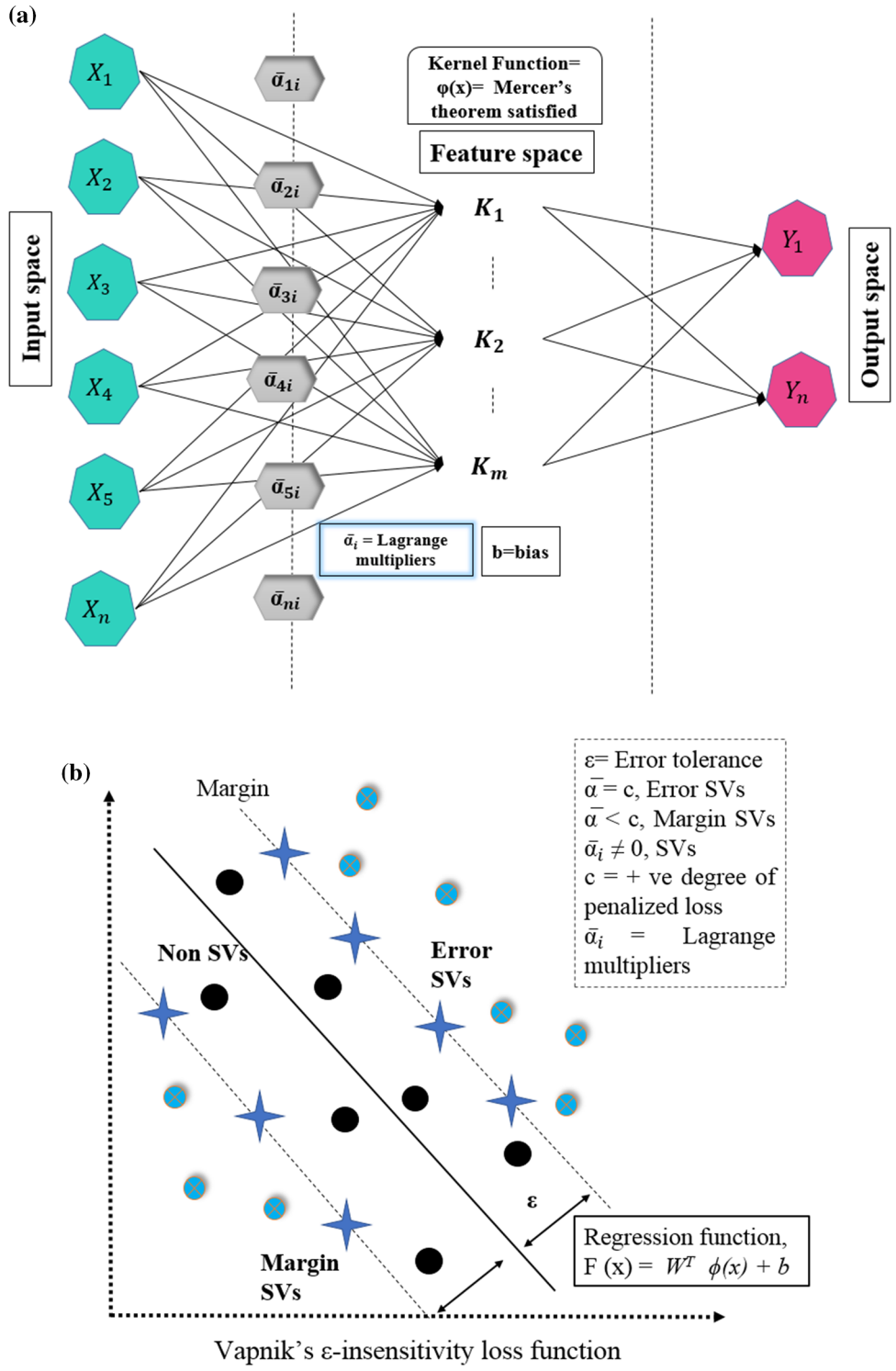


Fig. 5. a) Typical SVM architecture for river WQ modeling where $\bar{\alpha}$: weight and b: bias. Most common transfer functions are linear, polynomial, radial basis and sigmoidal, K = support vector, m: number of support vectors, φ(x) = non-linear function, b) Non-linear SVR vapnik's ε-insensitivity loss function, if the predicted value is within tube the loss is zero and outside the loss is the magnitude of the difference between the redicted value and the radius of the ε-tube.

4.2. Bibliographic review of SVM models

Wang et al. (2011) tested the use of semi-supervised regression model with SVM and compared it with MLR and SVM–GA models. The data in this project were retrieved using remote sensing and accuracy of the regression model is improved. Thus, the interpretation and monitoring of the WQ of the study area are also enhanced. In another study in the same year by El-Shafie (2011), SVM was compared with MLP and ENN (Ensemble) models to assess the level of performance in identifying a complex nonlinear relationship between WQ data to minimise expenses and labour charges. SVM performance is accurate, more robust and faster than other models with the need for minimum computational cost. Singh et al. (2011) applied support vector classification (SVC) for data reduction, used SVR as a predictive model and compared kernel-based discriminant analysis (DA), kernel-based partial least square (KPLS), ordinary DA and PLS techniques. SVM modelling results

Table 3
The summary of research conducted on Kernel-based models (SVM).

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
124.	(Wang et al., 2011)	Weihe River, China	Semi supervised-SVR, MLR, SVR-GA	COD, PI, DO, NH_3-N/ COD, PI DO, NH_3-N	MAD, MSE, DC	Not given (2006)	Two SVR models were modelled with one of them semi-supervised training setup and another with GA. The result was superior to multiple statistical regression methods. This model has the prospect of good and faster water quality management.
125.	(El-Shafie, 2011)	Johor River, Malaysia	MLP, ENN, SVM	DO, BOD, COD / DO, BOD, COD	R, MSE, CE	Monthly (1998–2007)	In this study, three models MLP, ENN and SVM were used to get the accuracy level with the least input variables to reduce time and need of a huge amount of data. The results show that SVM produced good results than MLP and ENN.
126.	(Singh et al., 2011)	Gomti River, India	SVC, SVR, KDA, KPLS, PLS, DA	pH, Al, Cl, PO_4, COD, BOD, DO / BOD	NSE, MSE, RMSE, R, A_f	Monthly (Site 1–3 1995–2005, Site 4–6 2005–2010)	The authors wanted to design a classification and regression problem-solving model for BOD using SVM models and linear models. Since WQ changes over a large geographical area and also influenced by seasonal variation in a larger area, it was found that SVMs can successfully classify them. SVC and SVR were more accurate and helped to optimize the monitoring program.
127.	(Noori et al., 2012)	Sefidrood River, Iran	SVM, ROSVM	Lat., Long., DO, EC, NO_3, $P_{Tot.}$, EC/ BOD	R, RMSE	Monthly (Not given)	The authors used SVM and ROSVM founded on orthogonal decomposition method for predicting BOD which usually needs a lot of time. The performances of both were up to the mark but six basis mode SVM was superior has it has more tendency to the centerline.
128.	(Gamble and Babbar-Sebens, 2012)	White River, Indiana, USA	SVM, LDA, SOM	Al, TOC, Cl, COD, DO, H, Fe, N, pH, NO_2, NO_3, $P_{Tot.}$, TSS, EC, SO_4, WT, Tu/ WQI	Cross-Validation	Monthly, seasonal (1991–2008)	The study executed LDA, SOM and SVM for linear and non-linear variable modelling and classification. PCA and K-mean Cluster analysis was used for correlating the WQ variables and grouping data. SVM classification was found to be more accurate than others.
129.	(Chen et al., 2012)	Jiajiang River, China	SVM	DO, COD, NH_3-N, $P_{Tot.}$, Hg, Cd, V-ArOH, CN, Fe, Mn / WQ	RE	Not given (May 2009-September 2010)	The research used factor analysis non-negative constraints for reducing the number of WQ variables along with SVM to identify source and WQ characteristics. The combined model could identify the source of pollution in the river so that management can be improved.
130.	(Jiake et al., 2013)	Heishui River, China	Hybrid SVM, ARIMA, BPNN	WT, Pb, DO / WT, Pb, DO	RMSE, MAPE	Monthly (2003–2008)	The author targets to solve classification problems, difficulty in predicting non-linear statistics and small data set using hybrid SVM and compare the result against ARIMA and BPNN. The hybrid model was more capable and accurate in water quality prediction.
131.	(Li et al., 2013)	Tiaoxi River, China	SVM, RBFNN	pH, DO, PI, NH_4-N, N, $P_{Tot.}$, / WQI	MSE, SC	Monthly (June 2010-May 2011)	The authors executed cross-validation and grid search to design better RBFNN and SVM models. The models improved the accuracy and performance for WQ assessment.
132.	(Nikoo and Mahjouri, 2013)	Jajrood River and Tehran Aquifer, Iran	PSVMs, FIS, FCT, SOM	TDS, EC, pH, HCO_3, Cl, SO_4, Ca, Mg, Na, K, HM, FC / WQI	MARE	Seasonal (2006–2009)	The study proved that PSVM can provide the mass function of WQ variables and well predict the water quality zoning than FIS, FCT, and SOM. This model can manage water pollution when appropriate data is available.
133.	(Modaresi and Araghinejad, 2014)	Kan River, Iran	SVM, PNN, kNN	NO_3, Cl / WQI	ER, EV	Not given (2003–2004)	SVM, PNN, and kNN models were used for WQI prediction. PNN and kNN were found to be a superior model than SVM because the determination of parameter C and optimization, as well as training, was also easier but SVM is a more useful tool when the aim is classification.
134.	(Liu and Lu, 2014)	Changle River, China	ANN, SVM	DO, WT, EC, pH, COD, NO_3-N, NH_4-N / $N_{Tot.}$, $P_{Tot.}$	RMSE, MRE	Monthly, Bimonthly, Trimonthly (January 2005-December 2010)	The author applied ANN and SVM to predict total nitrogen and phosphorus as pollution indicators. To reduce the stress on model trial-and-error analysis and GA plus structural risk minimization principle was employed. Due to these methods, SVM performed very well and accomplished better forecasting precision. WT was affected by $N_{Tot.}$ and $P_{Tot.}$ the most.

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Table 3 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
135.	(Kar et al., 2016)	Yamuna River, India	SVM	H, pH, TDS, Cd, Cu, Pb, Zn, Mn / Sa	DC, RMSE	Not given (2013)	The authors used a supervised spectral angle mapper for classification of data and clustering using Band Max, ISODATA and SVM. Because of the use of such superior methods, the SVM model was able to identify the pollutant with higher accuracy.
136.	(Olyaie et al., 2017)	Delaware River, USA	ANN, MLP, RBF, LGP, SVM	Q, pH, EC, WT, pH, DO/ DO	R, RMSE, MARE, NSE	Daily (July 2007- June 2012)	The authors implemented various machine learning methods to predict DO. Among all considered, SVM established to be a superior model. GP evolution was used to reduce the parameter, which intern decreased the modelling time.
137.	(Ji et al., 2017)	Wen-Rui Tang River, China	SVM, BPNN, GRNN, MLR	WT, pH, COD, BOD, NH_4-N, Pet, $P_{Tot.}$, N_{Tot}, COD_{cr}, f, EC, DO/ DO	MSE, DC, NSE	Bi-monthly (2004–2008)	This study aims to test the performance of AI models in a hypoxic water system which has complexity, high non-linearity and vigorous properties. The results showed that the most efficient predictive model was SVM then GRNN, BPNN and the least MLR, respectively. SVM was able to perform well with insufficient available data.
138.	(Bozorg-Haddad et al., 2017)	Sefidrood River, Iran	LSSVR, GP, GP-LSSVR	Na, K, Mg, SO_4, Cl, pH, EC, TDS / WQ	NSE, DC, RMSE	Monthly (1985–2005)	The study examined GA for optimization along with LSSVR modelling and sensitivity analysis to select the correct variables. GA-LSSVR was found to be a superior model than single models.
139.	(Xuejiao Li et al., 2017a,b)	Jialing River, China	Modified SVR, LSVR, BPNN	pH, PI, NH_3-N, $P_{Tot.}$, AT/ DO	MAE, MAPE, RMSE, DC	Monthly (2014–2015)	To enhance the model performance, authors used EEMD which can decompose data directly for non-linear and non-stationary data for modelling with LSSVM and modified SVR for DO prediction. Modified SVR was proved to be a superior model than other models considered.
140.	(Haghiabi et al., 2018)	Tireh River, Iran	ANN, SVM, MLP, GMDH	WT, EC, HCO_3, SO_4, Cl, TDS, Ca, Mg, Na, SAR, Q / WQ	DC, RMSE	Monthly (1960–2015)	The study suggested that all the models performed well in predicting river WQ but SVM overtook others. SVM results were most accurate when an RBF kernel was used. The data dispersion related value showed that all the models faintly over-estimated the values.
141.	(Fan et al., 2018)	Taizi River, China	SWAT, SVM	Geological, Climate, agricultural and biological data, EC, DO, BOD, COD, NH_3-N, $P_{Tot.}$, Q / WQ	DC, NSE	Monthly (1990–2002, 2007–2009)	The author's aim was to identify the area of restoration using geological, climate, agricultural, physiochemical and biological data. The study has used SWAT for identifying environmental indicators and SVM to understand the relationship between the data. The coupled model could establish and identify the relation and need for restoration.
142.	(Shan et al., 2018)	Pearl River, China	FSVM, SVM	pH, DO, COD, NH_3-N / WQ	SFE, F1-Score	Monthly (2012–2018)	SVM and fuzzy model hybrid were designed to overcome previous problems by removing the noise using the new two membership functions from the fuzzy system. The FSVM proved to outperform the SVM model and was able to solve the WQ evaluation.
143.	(Rehana, 2019)	Tungabhadra River, India	SVM, MLR	Q, WT, T_{mx}, MSp, Hs, W / WT	NSE, RMSE	Daily (1988–2005)	The author used SVM and MLR to design a predictive model for WT to estimate the risk of river water pollution and measures need to be taken for management. It was found that SVM outpaced the MLR model but the projection was more distinct in the case of MLR.
144.	(Kamyab-Talesh et al., 2019)	Sefidrud river, Iran	SVM	pH, DO, TDS, WT, NO_3, PO_4, BOD, Tu, FC/ WQI	DC, RMSE	Monthly (December 2007-November 2008)	The authors aimed to identify the most affected WQ parameter with SVM. Nitrate was found to influence the WQI the most. The model could perform up to the satisfaction which will help in WQ management in the study area.
145.	(Shen et al., 2019)	James River, USA	SVM, LS-SVM	Chl-a, Q, WT, DO, TSS, N, $P_{Tot.}$, DIN, PO_4 / Chl-a	RMSE, ME, AE, DC	Monthly (1990–2013)	LS-SVM was modelled for assessing algal bloom by considering seasonal variation. The empirical orthogonal function was applied for data dimension reduction. The study revealed to better use of this model can be done after logarithmic variable transformation.

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Table 3 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
146.	(Leong et al., 2019)	Perak River, Malaysia	SVM, LS-SVM	DO, BOD, COD, TSS, pH, NH_3-N/ WQI	SSE, MSSE, DC	Monthly (2009–2014)	SVM and LS-SVM were applied to predict WQI to find the best model. Linear, RBF and polynomial kernel function were tested and polynomial kernel performed the best with LS-SVM
147.	(Abobakr Yahya et al., 2019)	Langat River, Malaysia	SVM, ANN	pH, TSS, DO, NH_3-N, COD, BOD /DO	MSE, DC, CE	Not given (2006–2016)	SVM model was applied to better handling of non-linear, high-dimensional and local minima issue often found in the ANN model. SVM was able to predict DO and create an early warning system for river management.

are promising in handling optimisation, classification and prediction of a large amount of WQ data. Noori et al. (2012) developed reduced-order SVM (RO-SVM) by using orthogonal decomposition to reduce time consumption and compared it with the conventional SVM model. The obtained results are acceptable for BOD prediction. Modaresi and Araghinejad (2014) evaluated classification models, such as SVM, PNN and kNN. SVM performance is superior to others because it produces no errors during calibration and validation as it uses kernel function and only depends on support vector, which makes it a better model than ANN models for WQI modelling. Kar et al. (2016) used spectral angle mapper (SAM) and SVM to classify and identify metal contamination in river water. SVM performance is superior as it provides classes based on a priori spectral knowledge selected by the modeller. Spectral absorption depth is a good source of information regarding metals and can be utilised for other remote sensing techniques. Bozorg-Haddad et al. (2017) applied LSSVR, GP with PCA input optimiser and hybrid GA–LSSVR model. Since LSSVR doesn't have a mechanism for selecting coefficient and yet its precision is dependent on it, GA optimises it by finding the neighbouring global optima and selects the correct coefficient which modifies coefficient of the SVR producing better result than other models. Similarly, two other studies evaluated SVR performance. It is able to identify the most important WQ variables. Generalized SVM and canonical correlation analysis-based multivariable data downscaling method, when applied the accuracy and efficiency of the model, increases considerably (Kamyab-Talesh et al., 2019; Shan et al., 2018). Shen et al. (2019) addressed the problem of the complex interaction of WQ variables and inter-annual variation of eutrophication by using LS–SVM modelling and empirical orthogonal function (EOF) as a dimension-reducing tool to better comprehend the spatiotemporal patterns. The proposed AI model can handle the uncertain and multi-faceted data and identify the point of nutrient loading, which concurrently encourages algal bloom thus, this identification can help to control and manage algal bloom. Another study compared LS-SVM and SVM model performance for WQI prediction. Polynomial kernel function outperformed linear and RBF thus, chosen for final prediction. LS-SVM is a superior model than SVM as it can reduce the complexity of the problem by using linear equation rather than using quadratic programming like the conventional SVM model (Leong et al., 2019).

Gamble and Babbar-Sebens (2012) analysed the performance of predictive models by applying PCA method for the linear variable dimensional reduction and Kohomen SOM for nonlinear WQ variable reduction. The results from dimensional reduction were assembled using k-NN clustering analysis to reduce computational cost. Also, the authors applied linear DA to establish a relation between clusters as a linear classification method i.e., SVM, to deal with nonlinear variables. The results are promising; SVM is superior, but both models perform well. Chen et al. (2012) coupled SVM model with factor analysis and non-negative constraints which together are responsible for the variable reduction and classification to identify the source of water pollution. The results are satisfactory and provide an opportunity to understand the environmental status and decide on management strategies. Jiake et al. (2013) assessed the capability of hybrid SVM in handling non-linearity and nonstationary of WQ variables against ARIMA and BPNN models. PSO was applied to determine the variables for the SVM model. This way improves the accuracy of the results by avoiding overfitting or under-fitting. The hybrid SVM model outperforms ARIMA and BPNN models. Li et al. (2013) used SVM, Linear discriminant (LD) analysis and quadratic discriminant (QD) analysis models for WQ variables estimation. They found that SVM is better than other considered models. SVM is an effective classifier and predictive model but lacks the analytic capability in terms of spatiotemporal distribution.

In another study, Nikoo and Mahjouri (2013) developed a probabilistic SVM (PSVM) model; WQ data were fetched from FIS in a Monte Carlo analysis and were compared with SOM and fuzzy clustering technique (FCT). The PSVM model performance is better than the others. Two studies found that SVM is a good alternative to ANN. The

study used trial and error method and GA optimisation for both models. The authors found that SVM generalisation capacity is high. As a result, over-training is decreased and WQ can be predicted for non-point source with good accuracy and low risk (Abobakr Yahya et al., 2019; Liu and Lu, 2014). In another comparative study, Olyaie et al. (2017) assessed MLP, RBF, linear genetic programming (LGP) and SVM models together for DO prediction. The efficiency and accuracy of the model follow the order SVM the highest, then LGP, RBF and least MLP model for DO estimation. Though, LGP becomes highly applicable when it can present the empirical formula in a simple form. In the same year, Ji et al. (2017) applied SVM for hypoxic river systems and compared its results with those of MLR, BPNN and GRNN. BPNN is a traditional nonlinear regression model. GRNN is similar to BPNN but has no fixed architecture and needs an optimum number of hidden layers to deal with large datasets. SVM is superior with good performance and excellent DO prediction accuracy. Li et al. (2017a,b) coupled ensemble empirical mode decomposition (EEMD) with SVR and compared it with standard SVR and BPNN models. The hybrid model data were used for multimodal decomposition with EEMD since it is practical, sensitive, direct and self-adaptive. Regression analysis using SVR was performed, and the hybrid model was assembled to predict DO which showed that SVR is a better model than BPNN. Haghiabi et al. (2018) scrutinised the performance of ANN and group method of data handling (GMDH), using FFNN model, and used a network of simplified processes and SVM, which utilised transign RBF as the transfer and kernel function. SVM outperforms the others. GMDH and ANN models produce satisfactory results in WQ prediction. Fan et al. (2018) explored the effectiveness of the results of the coupling of SWAT (Soil and water assessment tool) and SVM model to assess the relationship between aquatic organisms. The results showed significant spatiotemporal deviation between variables and promising efficiency of the coupled model. Shan et al. (2018) hybridised SVM with fuzzy logic. The performance of the new FSVM is superior to that of the conventional model. The new FSVM can deal with noise by giving importance to different values and varying samples, and manage high-dimensional WQ data.

4.3. Model assessment

SVM is a statistical learning technique that can be applied for classification and regression problems. This method has good generalisation, high flexibility, global optimisation and statistical analysis capabilities. SVM model is accurate, robust and fast. The approach can identify complex nonlinear relationship better than other models, such as MLP and ENN, due to the capacity of the kernel model to construct expert knowledge (Raghavendra and Deka, 2014). Thus, SVM is known as a kernel model, and its selection is a priority to achieve excellent performance. The kernel can operate in input space and form nonlinear boundaries. Many functions, such as linear, polynomial, sigmoid and radial basis function, however, RBF is popular owing to its less or no error advantage during testing and validation and reliable performance compared with other models, such as PNN and kNN (Haghiabi et al., 2018; Modaresi and Araghinejad, 2014).

The principle of structural risk minimization makes SVM superior model than others since it reduces the upper bound of the expected risk whereas the traditional empirical risk minimization theory which is applied in ANN reduces error in the training data set (Wankhede and Doye, 2005). This risk management process helps SVM models in better generalization capability.

SVM performance is better when used as a classification tool. SVM training process is difficult than others since the parameter C of the algorithm has to be selected by the user and optimized for each class (Modaresi and Araghinejad, 2014).

SVM has been used as a regression model in many river WQ studies. This method improves the regression accuracy and interpretation and monitoring of nonlinear data as river WQ. SVR needs pre-processing of the training variables by mapping the input space to manage nonlinear data. SVR can be simultaneously used in data optimisation, classification and forecasting (Singh et al., 2011). A variant of SVM, such as LS–SVM, can be a good tool for regression and classification. This approach uses Karush–Kuhn–Tucker equation from the traditional equation. Dimensionality reduction tools, such as empirical orthogonal function, can be incorporated to comprehend the spatiotemporal patterns (Shen et al., 2019). This way makes the model flexible to the nonlinear input variables of river WQ.

Clustering techniques can be used with SVM to manage large datasets and decrease computational cost and data attribute for separating objects in a group; accordingly, the speed of the training process increases by utilising distribution properties (Bhagat et al., 2019; Gamble and Babbar-Sebens, 2012; Nikoo and Mahjouri, 2013). For similar results, dimensionality reduction tools, such as FA, PCA and SVC, can be used to deal with a large amount of data (Chen et al., 2012). SVM can efficiently predict the river WQ and simultaneously classify them. Thus, monitoring and managing by policymakers become easy to understand as river WQ is divided into standardised and defined WQ classes.

5. Complementary models

5.1. Conceptual background

Data-driven models of AI can be coupled with wavelet transformation to enhance their ability and deal with their limitations. This way greatly helps towards taking advantage of both techniques because these models are individually incapable of handling the difficulties related to understanding the complex and nonlinear WQ variable interaction. In 1984, wavelet was first presented as dilation of a single function and its transformation (Grossmann and Morlet, 1984). Wavelet algorithm consists of a dilation factor, a temporal translation function and a mother wavelet. Wavelet algorithm can overcome the shortcoming of other models with its capacity to extract important information from a large dataset, obtain data in a readable form by decomposing nonstationary signals into sub-signals at different levels and solving diagnostic, classification and forecasting problems when combined with various AI models. Wavelet algorithm can present signals in the time scale and analyse the relationship (Nourani et al., 2014; Rao, 2002). Various models developed using wavelet algorithm are listed in Table 4. The general process diagram of the wavelet-AI model has been shown in Fig. 6 (Yaseen et al., 2018a).

5.2. Bibliographic review of complementary models

Wavelet de-noising technique (WDT) was coupled with ANFIS model, its performance was compared in two studies with RBFNN and normal ANFIS. WDT– ANFIS model is accurate and fast with improved properties of de-noising and successfully predicts WQ variables as well as found better than the other two models. For optimum performance of the model, the selection of various variables is important; for instance, orthogonal wavelet was selected in the study because it is concise, easy to calculate and satisfactorily restores the original signal. A similar level of decomposition and threshold methods was considered, and the soft threshold technique is preferred since it produces an enhanced outcome (Najah et al., 2012; Najah Ahmed et al., 2019). Parmar and Bhardwaj (2015) developed ANN, neuro-fuzzy logic, wavelet and ANFIS models with wavelet transformation. Neuro-fuzzy wavelet was used with Daubechies wavelet to deal with random and spike series. The result showed that the wavelet couple model produced better results than of single models considered in the study. Barzegar et al. (2016) explored ANN, ANFIS, WNN and WANFIS models for salinity estimation. Discrete wavelet (DW) was used for pre-processing of the data to improve accuracy. Thus, WANFIS and WNN models outperformed other conventional models. Barzegar et al. (2018) explored the performance accuracy of ELM, wavelet ELM (WELM) and WANFIS models. The hybrid

Table 4
The summary of research conducted on complementary models (Wavelet-AI).

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
148.	(Najah et al., 2012)	Johor River, Malaysia	ANFIS, WDT-ANFIS	TDS, EC, Tu / TDS, EC, Tu	MAE	Monthly (1998–2007)	The aim was to reduce the noisy signal often found in WQ data making the prediction difficult. Thus, authors used normal ANFIS and Wavelet de-noising techniques with ANFIS to solve this problem. The new model proved to be superior and faster than ANFIS.
149.	(Y. Wang et al., 2013)	Songhua River, China	BWNN, ANN, WNN, ARIMA	NH_4-N, DO/ NH_4-N, DO	NSE, RMSEP, MPI, PICP	Monthly (2005–2011)	A predictive model was designed for NH_4-N and DO using BWNN and compared the performance with ANN, WNN, and ARIMA. BWNN was found to outperform all other models under the study and proved to handle better severe fluctuating time series data of WQ. The noise due to uncertainty was much lesser in the case of NH_4-N than DO and did well with structural missing data.
150.	(Ravansalar and Rajaee, 2015)	Asi River, Turkey	ANN, WNN	EC, Q / EC	DC, NSE, RMSE, MAE	Monthly (1984–2008)	Discrete WNN and ANN were applied for EC prediction. The results showed that WNN outperformed ANN and can be used for EC prediction 1 month ahead of time.
151.	(Parmar and Bhardwaj, 2015)	Yamuna River, India	ANFIS, ANN, RM, CW-CNF	COD/ COD	MAE	Monthly (1999–2009)	The study used five AI models to predict COD. Wavelet transformation (Daubechies wavelet) was used for selection and correlation. The result showed that the wavelet couple model produced better results than other single models considered.
152.	(Ravansalar et al., 2016)	Calder River, UK	ANN, WNN	DO, WT/ DO. WT	NSE, RMSE, MAE	Hourly (1 October 1994–30 September 1995)	The study focused on ANN and wavelet-ANN as well as reducing noise reduction for non-linear signals by discrete wavelet transforms (DWT). WNN performed very well and was suggested by the author that can be used for DO prediction in 30 min ahead.
153.	(Barzegar et al., 2016)	Aji-Chay River, Iran	ANN, ANFIS, WNN, WANFIS	Q, WT, Na, Mg, Ca, SO_4, Cl^- K, HCO_3, HCO_3/ Sa	DC, Normalized RMSE, NSE,	Monthly (Oct 1983-sep 2011)	Pre-processed data was done using Daubechies, Symlet and Haar wavelets to design hybrid models of ANN and ANFIS and then compared with the single model. The hybrid models showed the best performance when used with the db4 mother wavelet. In addition, to this, the wavelet with high vanishing moments has more reliability and performance.
154.	(Khani and Rajaee, 2017)	Clackamas River, USA	WR, WNN, ANN, MLR	DO, WT, Q/ DO	MAE, MAPE, NSE, R, APE	Daily (25 September 2009–4 March 2014)	Regression and ANN models were implemented along with wavelet transformation. WR was proved to better predictive model than others considered with high accuracy and simple when used in any scenario.
155.	(Solgi et al., 2017)	Karun River, Iran	WSVM, ANFIS	WT, DO, Q/ BOD	RMSE, DC, AIC	Monthly (2002–2014)	Two models SVM and ANFIS were studied which were used after decomposing data by wavelet transform and then PCA for co-relation important variables. WSVM provided better performance than the conventional counterpart with higher accuracy.
156.	(Huang et al., 2018)	Pearl River, China	FWNN, WNN, ANN, FCM, GA	COD, NH_4-N, DO, EC, WT, pH, Tu / WQ	R, DC, MAPE, RMSE, MSE	Daily (2013–2014)	Various predictive models were used for WQ variables prediction in which self-adaptive fuzzy c-mean was used to find fuzzy rules where GA was used as an optimization tool. FWNN was proved to be the most effective model of all.
157.	(Montaseri et al., 2018)	Nazuluchay, Tajan, Zayandehrud, and Helleh River, Iran	WGEP, ANFIS-GP, WANFIS, ANN, ANFIS–GP2	EC, Na, Cl / TDS	R, RMSE, MAE	Not given (1993–2012)	The study compared the performance of ANN with Grid partition ANFIS, subtractive-ANFIS, wavelet-GEP, wavelet-ANFIS for predicting TDS. A sensitivity test was performed to find the correlation. Wavelet-GEP was found to outperform all other AI models considered.
158.	(Rajaee and Jafari, 2018)	Blue River, Kansas	WNN, WGEP, WDT	DO, Tu, pH, Q, WT / DO Tu, pH	MAE, RMSE	Daily (1st January 2008-31st December 2011)	The research integrated discrete wavelet transformation with ANN, GEP and DT models which improved the overall performance. WGEP performance was better than others and able to handle extreme values.
159.	(Rajaee et al., 2018)	Chattahoochee River, USA	MLR, ANN, WNN, WMLR	pH, Q / pH	DC, RMSE, MAE	Daily (1 August 2012–31 January 2014)	The capability of MLR, ANN, WNN and WMLR models were evaluated to predict pH ahead of 3 days. WNN performance was better since it removed the noise caused by the pH shift.

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Table 4 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
160.	(Barzegar et al., 2018)	Aji-Chay River, Iran	ANFIS, WELM, ELM, WANFIS	EC/ EC	DC, RMSE, NSE	Monthly (1984–2011)	The authors have used a hybrid model of ELM and ANFIS. They have used autocorrelation function and partial autocorrelation function to find potential input. WELM and WANFIS proved to be a better model than the counterpart as they can deal with multi-step-ahead values of EC.
161.	(Zhu et al., 2019a)	Drava River, Croatia	WMLP, WANFIS, MLR	WT, AT/ WT	R, d, MAE, RMSE	Daily, Seasonal (July-August 2003,2006, 2015)	Wavelet transformation was coupled with MLP and ANFIS model and compared it with MLR for predicting water and air temperature. The result proved to be good when AI models were coupled with wavelet transformation.
162.	(Antanasijević et al., 2019)	Danube River, Serbia	SOM-LSI, WNN	WT, pH, EC, Al, COD, BOD, H, TSS, CO_2, HCO_3, NO_3, SO_4, PO_4, P_{Tot}, Ca, Mg, DO / DO	DC, R, RMSE, MAE, Bias, MAPE	Monthly (2002–2013)	The research aims to create a more accurate and less complicated predictive model for DO using the multilevel splitting approach with SOM-LSI and WNN. The results were satisfactory and better than published models In this case the data input size was comparatively low yet produced a similar or better result.
163.	(Parmar et al., 2019)	Yamuna, India	ANN, ANFIS, WANFIS, MLR	BOD / BOD	MAE, RE	Monthly (10 years)	The study found the superiority of WANFIS over ANN and ANFIS with the simultaneous prediction of future 9 months. Autocorrelation function and partial autocorrelation function was used to analyse stationary and non-stationary time-series data and Daubechies wavelet were used for decomposing random and spike series data.
164.	(Najah Ahmed et al., 2019)	Johor River, Malaysia	ANFIS, RBFNN, MLP, WDT-ANFIS	WT, Sa, Tu, EC, Cl, NO_3-N, PO_4, Fe, K, Mg, Na, E.coli / pH, TSS, NO_3-N	CE, MSE, DC	Monthly (2009–2010)	Model couple with de-noising technique shows better performance as per the authors, specially WDT-ANFIS when dealing with complex issues and dynamic processes. Three assessment techniques were used to find the most important input variable.

model outperforms the conventional models as a consequence of the enhancement made by the application of wavelet transformation, which can deal with non-stationary WQ data. In addition to that, the application of boosting ensemble method helps to reduce error rates which allow improvement in overall model performance. Two studies coupled wavelet algorithm with MLP and ANFIS and compared them with traditional MLP, ANFIS and MLR models. DW was applied because of its high efficiency. WMLP and WANFIS exhibit improved accuracy and performance during extreme weather conditions affecting WT and BOD values (Parmar et al., 2019; Zhu et al., 2019a). Elkiran et al. (2019) implemented BPNN, SVM, ARIMA and ANFIS model along with simple averaged ensemble (SAE), weighted average ensemble (WAE) and ensemble neural network (ENN) models for WQ modelling. Single models were tested against the models which were coupled with ensemble technique, among which ensemble technique coupled with ANFIS model performance was superior to others. Ensemble technique was able to improve the performance of the models.

Wang et al. (2013) applied bootstrapped wavelet (BW) with ANN (BWNN) and compared it with ANN, WNN and ARIMA models. BWNN model used Morlet wavelet basis function (WBF) with ANN and handled severely unpredictable and non-seasonal time series and missing WQ data. BWNN model shows a good result but must be improved in terms of cost and prediction. Three studies applied wavelet transformation (DW and wavelet-based regression -WR) with the ANN model to decompose WQ data into a multi-frequency time series against the conventional ANN. DW helped in noise reduction and manage nonstationary signals. This method produces an accurate and superior result than others. However, WR produces a better result with higher accuracy than WNN models (Khani and Rajaee, 2017; Ravansalar et al., 2016; Ravansalar and Rajaee, 2015). Similar work was presented by (Solgi et al., 2017) with SVR and ANFIS models. The data were decomposed using wavelet transformation, and PCA was used for WQ data relation identification and to determine the main sub-signal to increase the modelling speed. WSVR model with RBF kernel performance is found superior to all considered models in this study. Montaseri et al. (2018) applied ANFIS–GP2, ANFIS–SC_1, gene expression programming (GEP), WANFIS and WGEP models for TDS prediction. The hybrid models perform well, but the results from WGEP are exceptional because wavelet transformation is used to decompose the data into sub time series. Rajaee and Jafari (2018) integrated DWT with ANN, GEP and DT models to evaluate their performance for WQ prediction. Mother wavelets applied in the study were db2, db5, Haar, Bior1.1 and Meyer; amongst them, db2 and Meyer help the most in improving all the models' performance, decrease noise by separating the signals and deal with the extreme data. However, WGEP performs slightly better than other models in prediction accuracy and error, although all successfully produce better results than their conventional counterparts.

Huang et al. (2018) explored the use of fuzzy wavelet neural network (FWNN), ANN and fuzzy logic models and included self-adapted FCM clustering for determination of fuzzy rules. GA and gradient descent algorithm were incorporated as the optimiser. FWNN outperforms other models because of its capacity to manage uncertainty, fluctuation and non-seasonal time series in WQ data and produce high accuracy and feasibility. Rajaee et al. (2018) compared ANN, WNN, MLR and WMLR models. The pH prediction performance of the WNN model is better than those of the other models and can predict 2–3 days ahead due to the application of 'A Trous' wavelet transformation algorithm. This algorithm dilates the mother wavelet and maintains the same length. WNN performance was superior to others by removing the noise instigated by the pH shift.

5.3. Model assessment

Wavelet has been popularly used for time-series analysis in river WQ studies. This method tries to present localised and momentary change mapping in the system. Between two types of wavelet, namely,

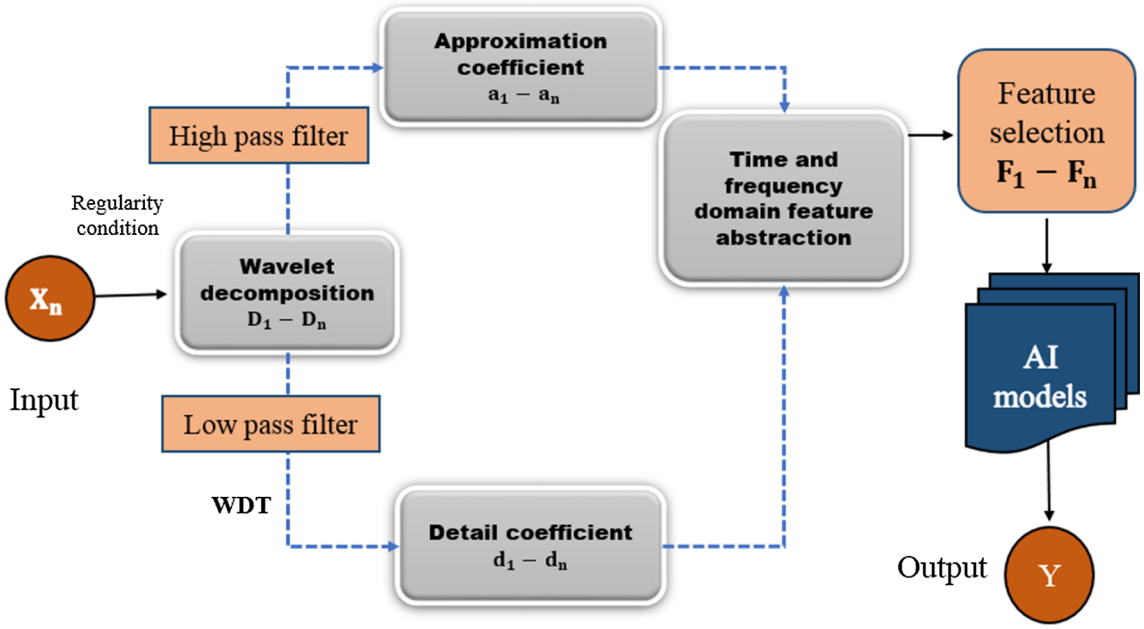


Fig. 6. Schematic diagram of hybrid wavelet-AI model: regularity condition with order N decides the choice of wavelet. Wavelet de-noising (WDT) is applied to adjust detail coefficient. X_n = Input variables, D_1-D_n = decomposition level, Y = Output variable.

continuous (CW) and discrete (DW), DW is more frequently used in river WQ studies as river WQ signals are mostly discrete and noisy; thus, they need decomposition and de-noising. DW in combination with de-noising increases the accuracy of the model, such as ANFIS and ANN, and decreases computational cost (Barzegar et al., 2018). Wavelet input time-series analysis is crucial to river WQ model because of the combined benefit of the wavelet algorithm and model can improve accuracy and performance. Wavelet analysis can be used for periodic river contamination identification and monitoring as it can identify the trend and the periodic changes (Olyaie et al., 2015).

A study by (Solgi et al., 2017) reveals that the modelling speed of a hybrid model can be considerably increased if PCA is used to select the main sub-signals among the numerous signals decomposed by W, then these sub-signals can be used as input in the model. PCA applies Kaiser-Meyer-Olkin test to assess the correlation between the variables. The test value less than 0.5 is considered inadequate for analysis whereas value more than 0.7 is allowed for further analysis (Hutcheson and Sofroniou, 1999). The combined form of the hybrid model and pre-processing analysis can be very helpful to deal with multi-variable WQ data.

Wavelet can considerably increase the overall performance of a model and can be combined with various models such as ANN. They have the ability to explicate spectral and temporal information. They are able to analyse time-series data and establish a relationship within non-stationary data allowing the couple AI model to perform better. Wavelet transform analysis has proved to more efficient than Fourier transformation and the energy variation plotted in time–frequency allows assessing the hydrological variation within the data at different scale and location (Partal and Kişi, 2007).

The selected mother wavelet is important to the performance of the model. Orthogonal wavelet can be a good choice because it is concise, easy to calculate and satisfactorily restores original signal and has a similar level of decomposition and threshold methods. Thus, this wavelet produces an enhanced outcome. Daubechies wavelet can also be used with increased random and spike series. Morlet wavelet can be applied in unpredictable and non-seasonal scenarios and in case of lack of data (Najah et al., 2012; Parmar and Bhardwaj, 2015; Wang et al., 2013). Many mother wavelets can be used to deal with multifaceted WQ data and must be select depending on the scenario and data availability. Thus, a mother wavelet with flexible and wide features is suitable to study time-series trends in river WQ research.

The simplicity of the wavelet coefficient and sub-signal endows the wavelet several specific characteristics, such as hidden period, dependence and easy diagnosis of the jump (Nourani et al., 2014).

Wavelet combined with ANFIS model provides the hybrid model using fuzzy rules with the advantage of a reliable result since it can handle uncertain and spatiotemporal seasonal WQ data. The hybrid model can handle extreme data fluctuation accurately under structural missing data (Barzegar et al., 2016; Wang et al., 2013).

6. Hybrid AI models

6.1. Conceptual background

AI models perform better with optimiser than without them because they have few limitations, such as high computation cost, inability to deal with NP-hard problems and difficulty with complex problems (Ghorbani et al., 2018). Optimisers, such as nature-inspired (NI) optimisation algorithms, have been applied to address these issues. Many NI algorithms mimic the natural behaviour of humans, animals and insects. ML is a full-fledged system, with its own algorithm, data structuring and learning process, but it must learn, adapt and evolve as nature does to create a robust system. NI algorithm has gained popularity owing to its advantages. (i) It is inspired by simple concept mimicked from natural examples, such as animal, insect or genetic behaviour. (ii) It is flexible and does not need special changes in the algorithmic structure; thus, it is easily incorporated as an optimiser. (iii) It only needs the representation of input and output variables. (iv) It applies deviation-free mechanism and random approach. (v) It can avoid local optima and deal with the challenges of real problems. Thus, it is more applied in many research areas than conventional algorithms. The general architecture of optimisers coupled with AI model such as ANN has been presented in Fig. 7.

GA was introduced in 1962 and improved considerably as time progresses. It works on selection, crossover and mutation operators (Goldberg and Holland, 1988; Holland, 1962). GA uses binary strings for coding; thus, it uses real variable vectors as chromosomes, real variables as genes and real numbers as alleles. The vector is decoded

and undergoes fitness evaluation, which helps decide whether the vector undergoes mutation or crossover. The evolution process repeats until the best performing model is found (Maier et al., 2014; Wright, 1991a,b).

'GA search by allocating efforts to regions of the search space based on an estimate of the relative performance of competing regions. In complex domain one expects perpetual novelty to be a characteristic feature… In this case, traditional search techniques… are likely to be misled and GA may be the search technique of the choice for machine learning system…' (Fitzpatrick and Grefenstette, 1988).

GP models, which are a class of EA that combine GA and GP, are based on the Darwinian principle of evolution theory proposed by Koza, 1997). It is a systematic, domain-independent method that spontaneously solves the problem without requiring the designers to have in-depth knowledge of the solution structure. GP functions as an optimiser and classifier. This method includes a syntax tree with branches as functions with arguments and leaves of the tree as variables and constant. Since GP uses GA, the primary operators are crossover and mutation (Poli et al., 2008). Compared with GA, GP models are more flexible and can evolve variable length. GP has various versions, such as linear GP, grammar-based GP, graph-based GP and tree-based GP, each with different presentations. This approach applies stochastic search properties, acts as a global search, is least trapped in local optima, varies in solution architecture, has freedom of expression to obtain an improved solution by searching relationships, has automatic removal of unwanted variables and finds meaningful associations from the given data (Jabeen and Baig, 2010).

Various algorithms are based on the behaviours of organisms. For example, particle swarm optimisation (PSO) algorithm uses the concept of bird flocking, fish schooling and swarming theory. PSO is based on a simple concept with the least coding. Its advantages over GA include enhanced performance due to group interaction and memory retention (Eberhart and Kennedy, 2002).

Another NI algorithm is the ant colony optimisation (ACO) algorithm, which is based on ant system, as the name suggests. This algorithm has positive feedback that rapidly discovers the appropriate answer; its distributed computation evades early convergence and uses constructive greedy heuristic for fast results. This approach is flexible, tough and population-based (Dorigo et al., 1996).

Shuffled frog leaping algorithm (SFLA) was introduced to solve discrete optimisation problems. SFLA transforms frogs into a memetic evolution chain, that is, a system of behaviour that must be passed from one frog to another similar to an infection. No physical change occurs, but the idea held by each frog improves. The difference from GA is that it only allows genetically alike individuals to interact, whereas SFLA spreads the idea to all characters (Eusuff and Lansey, 2003).

Similar to the natural behaviour of insects, such as fireflies that produce light to communicate, the hunt has been used to develop an algorithm. On the basis of this theory, the firefly algorithm (FFA) has been designed. FFA has several benefits over PSO and GA. (i) FFA can estimate better and converge faster towards optimality and (ii) is more efficient in finding the global optima (Yang, 2009). Consistent with the previous theory of FFA, bat algorithm (BA) has been designed based on bats' echolocation behaviour. BA consists of a good combination of PSO and GA (Yang, 2010). Various models developed using hybrid nature-based algorithm are summarised in Table 5.

6.2. Bibliographic review of hybrid AI models

GA is the earliest of the natural algorithms applied in river WQ studies. Ng and Perera (2003) applied GA, which works on the principle of natural selection where parent population is generated and improved by selection, crossover and mutation and used as an optimiser for the WQ model. GA operators play an important role along with sensitivity analysis to obtain the convergence point, which makes the model robust and effective. In another study, Icaga (2005) used GA as an optimisation tool to find and apply the best possible input combination. GA performance is better than dynamic programming. Burchard-Levine et al. (2014) evaluated the performance of GA and ANN models to increase the response time of the early warning system. GA–ANN model successfully provides a response 8 h ahead of time and can effectively understand nonstationary and vague data. In another study, Chatterjee et al. (2018) studied non-dominated sorting (NS-GA-II) modelling, which is similar to GA but optimises multiple objectives with ANN model, and compared it with NN-GA and NN-PSO models. NN-NSGA-II is superior to and more stable than other models. Jin et al. (2019) developed a model by integrating improved GA which uses multiple data sets of the sliding window for choosing optimum initial parameters for BPNN which was able to adjust the proper architecture to recognize the WQ deviation. The result from the integrated model and the normal BPNN were compared and shown significant improvement in predictability, effectiveness and reliability.

Zhou et al. (2006) enhanced ANN modelling result by integrating PSO optimisation technique, which is based on the cognitive and social interaction of flocking birds where particles are induced to their most promising solution. The solutions were compared with GA and BPNN algorithm and result show that PSO is robust and good tool for predictive models. Piotrowski et al. (2014) explored LM–ANN model as the main model and compared it with PSO, differential evolution (DE) with global and local neighbourhood-based mutational operators (DEGL), DEGL proxin, differential evolution–separate groups (DE–SG), JA-DE, self-adaptive DE and other adaptive methods. A total of 23 different training methods were used, and most were selected from DE, evolution strategies, direct search and PSO. Most of them are inferior to LM–NN performance in WT modelling. Li et al. (2016a,b) demonstrated the implementation of MLR, BPNN and SVR coupled with PSO as an optimiser. The authors prevented overfitting and bias by applying the leave-one-out cross-validation method during the training process. Performance capability is highest for PSO–SVM, then PSO–BPNN and lowest for MLR since among all SVM is more capable to handle the nonlinear relationship between variables. Keshtegar et al. (2019) evaluated the performance between polynomial chaos expansion (PCE) (a model that predicts practical problems using a stochastic model), MLP–PSO, MLP and MLR. The performance of PCE and MLP–PSO is similar and has higher predictive accuracy than others considered. In the latest research article by Azad et al. (2019), ANFIS-PSO, ANFIS–ACO_R, ant colony optimisation based technique used by ants to find food, and ANN were equated. GP, SC and FCM were used for FIS generation. ANFIS-ACO_R provides good results during testing, and PSO performs well during testing. Therefore, EA is a good optimising tool for modelling EC, TDS and sodium absorption ratio (SAR).

Mahmoudi et al. (2016) explored SFLA optimizer which is based on the social behaviours of frogs, which are sorted based on reducing or arising of the problem and GP was applied with the SVR model. SFLA–SVR is better than GP-SVR in terms of accuracy. In another research by Raheli et al. (2017), FFA, which is a swarm intelligence optimisation method based on the movement of the firefly, was introduced and coupled with MLP. The hybrid model accuracy in DO and BOD prediction exceeds that of conventional MLP, and the method can be used in other scenarios with available data. In another study, Li et al. (2017a,b) integrated FFA optimisation technique with SVR. The hybrid model is better and robust than the traditional SVM model in WQ indexing. FFA–SVM saves money and time as it is a direct and rapid testing technique. Yaseen et al. (2018a,b,c,d) verified the improved version of LSSVM with BA which uses the principle of echolocation capacity of the bat for locating its prey; Model Tree (M5) and multivariate adaptive regression splines (MARS) model were used for comparison. LSSVM-BA outperforms all other models considered. Chen et al. (2018) optimised BPNN model with an improved artificial bee colony (ABC) algorithm were applied for DO estimation. The results stipulated that improved ABC–BPNN is superior to traditional BPNN due to the better predictive capability and higher accuracy due to the

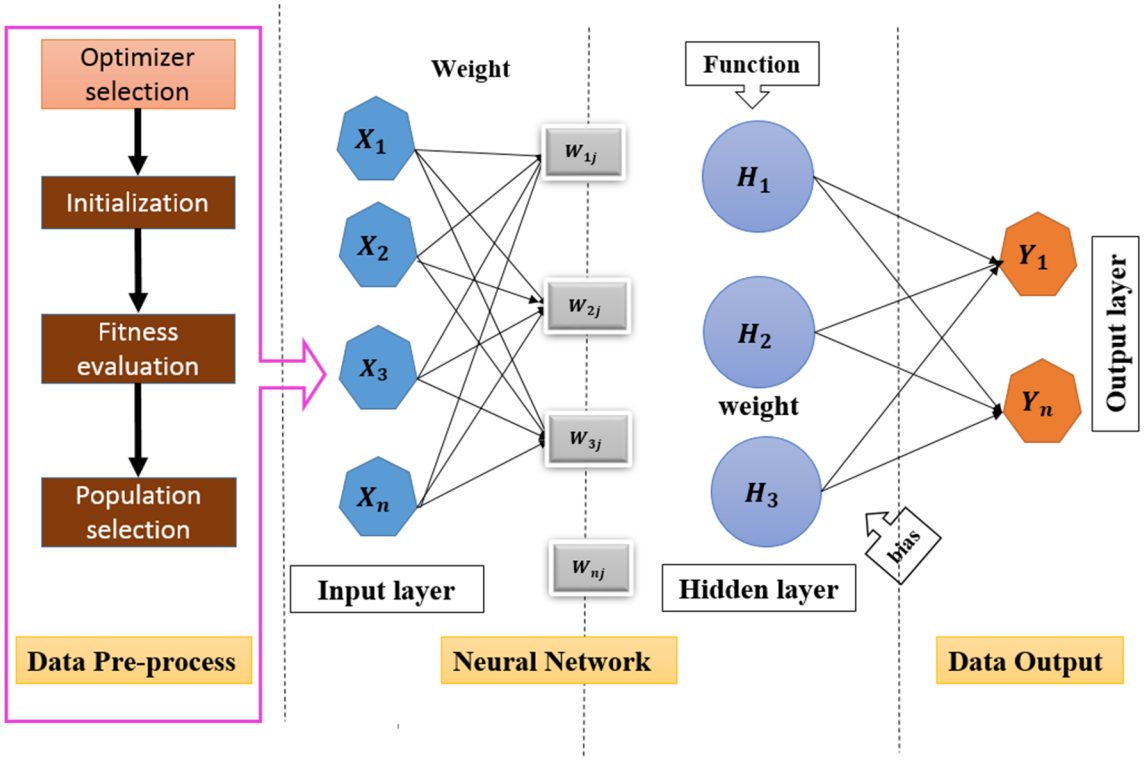


Fig. 7. General architecture of optimisers with neural network three-layer design (hybrid-model).

improvement in the speed of convergence and ability for local searching and generalisation.

Azad et al. (2017) investigated the performance of GA, ACO and DE coupled with ANFIS model. ANFIS–DE is the best optimiser for its high velocity, simple operation and capacity to detect the best response and the ability to improve uncertainty in WQ data. ANFIS–DE outperforms others, but all NI algorithms have the aptitude to optimise ANFIS as a predictive model.

Ahmadi et al. (2018) evaluated CA for classifying WQ and FA to determine the important variables. Five input selection methods were applied which are, correlation model, PCA and gamma test with backward regression and GA with gamma test. For BOD modelling LR, ANN and GP models were used. The results showed that the best input method is the gamma test with backward regression integrated with LR; gamma test with GA integrated with ANN is the best model.

6.3. Model assessment

NI algorithms are also known as EAs since they solve the problems inspired by biological mechanisms that evolved by a long, extended time of learning. The concept makes this group of algorithms unique as the algorithms are based on real-life system and easy to grasp. They are easy to couple with other models as optimisers, which helps deal with nonlinear and uncertain data (Ashrafzadeh et al., 2019; Maroufpoor et al., 2019). Their ability to deal with the complex problem in addition to their capability to perform a global search and local search, and find the nearest optima popularise their use amongst optimisation techniques.

The hybrid models that are used in river WQ assessment are population-based algorithms, such as GA, PSO, ACO_R, FFA and BA.

a) PSO can be more effectively implemented for hydrological data since it has low computational volume, easy to operate, not dependent on the problem, high convergence rate, global search capability, can deal with complex problems and can escape local minima (Azad et al., 2019; Taormina and Chau, 2015).

b) ACO_R uses a probabilistic approach that can find the best path through graphs. Although it has high affinity solve discrete problems, it can solve continuous WQ problems by applying probability density function. It applies global search techniques, can escape from local minima and can solve most of the complex issues but suffers from low convergence speed (Socha and Dorigo, 2008).

c) GA is a population-based stochastic optimization tool which can effectively deal with non-linear and non-differential WQ data issues by achieving optimum solution by the repetitive process. The operators can leap to different spaces with the look space to find the new best solution. It is a non-exhaustive process until the desired solution is achieved (Jakubcova et al., 2015; Yaseen et al., 2019b). These advantages allow them to solve highly complex and multi-parameter problems.

d) Another swarm optimisation technique is SFLA, which has been a least explored optimiser in river WQ studies. This method uses a set of frogs as a representation of solution and spreads similar to an infectious disease from one frog to another without discriminating; similar to GA, only blood relatives are accounted for (Mahmoudi et al., 2016).

e) Similar algorithms, such as FFA, BA, ABC and ACO_R have been explained in recent years in river WQ monitoring studies as an optimisation tool. They have high convergence speed, propensity for local and global searching, good generalisation and direct and rapid testing method (Naganna et al., 2019). These qualities enable the model integrated with them to achieve superior accuracy and great efficiency.

GA is an established and robust nondeterministic optimisation tool. GA generates parent population, is enhanced with selection, crossover and mutation, utilises probabilistic transition rule and easily converges. GA provides stable and efficient results (Ng and Perera, 2003). However, limited research has considered GP models. GP perform well when data are pre-processed using wavelet transformation. The linear and explicit representations of such model depend on convenience, efficiency and genetic operators. Additional research is needed for better utilisation of these models considering they can be easily manipulated, evolve during training and solve complex problems in an unexpected manner.

GP is a powerful tool to establish the relationship among input and

Table 5
The summary of research conducted on hybrid models where the artificial intelligence models integrated with nature-inspired algorithms (PSO, GA, ACO, etc.).

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
165.	(Ng and Perera, 2003)	Yarra River, Australia	GA	BOD, SOD / DO	CR, MR	Not given	The author's target was to use GA for calibration. As concluded GA operators play a vital role in model development and their selection depends on the sensitivity of model parameter and output. Uncertainty and sensitivity analysis was done by MCS. GA was robust and highly effective in water quality assessment.
166.	(Icaga, 2005)	Gediz River, West Anatolia	GA	As, BOD, Cd, COD, Cr, Cu, DO, E.coli, f, Strep., Mn, NH_3-N, NO_2-N, NO_3-N, Pb, pH, TSS, Tu, EC, Ca, Cl, K, Mg, Na, SO_4, TDS / WQI	Mean of CR and MR	Not given	The study aims to test the ability of GA in the field of water quality modelling using dynamic programming as optimization. The authors concluded that considering CR and MR together results in faster outcomes and GA have the potential used in such studies.
167.	(Zhou et al., 2006)	Yangtze River, China	ANN-PSO, BPNN, GA	DO, PI, pH, NH_3-N, Q (water), Q (wastewater), L (River)/ WQ	NA	Monthly (June 2003 to September 2005), Water flow and Volume of wastewater (1995–2004)	The authors predicted the water quality and future water pollution problem using ANN-PSO and simultaneously comparing with BPNN and GA performance. The results comply with model effectiveness and continuous increase in river pollution.
168.	(Piotrowski et al., 2014)	BialaTranowska River, South Poland	MLP-LM, MLP-PSO, MLP-DE	S, T_{av} and T_{mx} (city of Tarnow), N_{av} and N_{mx} for the town of NowySacz, Q and WT in koszyceWielkie / WT	MSE	Daily (1 November 1990–31 October 1995)	WT has influence over many biological and chemical processes and activities. The authors did an extensive comparative study using 23 different training methods and applied the LM algorithm to the MLP model to predict the WT using different population sizes. LM was found to be superior to most of them but few differential evolution algorithms were at par with LM. The study also concludes that the model performed well and the error was due to natural abnormalities like river freezing and melting process.
169.	(Burchard-Levine et al., 2014)	Songhua River, China	GA-ANN	pH, Tu, EC, DO, f, WT, PI, TOC, N-NH_4, $P_{Tot.}$ / TOC, PI, N-NH_4	MSE, MPE, R	Hourly ()28th March 2009 to 26th February 2012	The authors designed an early warning system using GA and ANN to increase the response time, if the water quality at the intake point shows abnormality. TOC, PI and N-NH_4 were successfully predicted ahead of time for 2 h using the GA-ANN model and performance may increase for larger time delays. This model demonstrated that with available data it can be a great tool for WQ assessment.
170..	(Mahmoudi et al., 2016)	Sefidrood River, Iran	SFLA- SVR, GP	Na, Ca, Mg, SO_4, Cl, EC, TDS / WQI	RMSE, DC, CE	Monthly-14 years (Not given)	The study objective was to find a more accurate model of WQ prediction by integrating SFLA-SVR together. The model was compared to GP which in the past study already shown to be an effective water quality modelling technique. All the eight WQ data shown improvement more than GP in SPLA-SVR. Its performance was found to be higher for simulating and predicting WQ time series
171.	(Li et al., 2016a,b)	Rivers, China	PSO-MLR, PSO-BPNN, PSO-SVM	WT, pH, COD, BOD, NH_3-N, N, $P_{Tot.}$, Pet, VP, COD, Hg, Pb, Cu, f, Zn, As, Cd, Cr, CN, LAS / DO	DC, MSE,	Not given (2009–2010)	The study used PSO optimization with MLR, BPNN, and SVM models for DO prediction. PSO-SVM was found to be the best model when used with RBF Kernel. BPNN showed more fluctuation resulting in unstable predictability however, results were good when data were close to average.

(*continued on next page*)

Table 5 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
172.	(Li et al., 2017a,b)	Euphrates River, Iraq	SVR, FFA-SVR	TDS, BOD, Cl, K, DO, EC, Na, Mg, Al, pH, Ca, PO_4, NO_3, H, SO_4 / WQI	DC, d, RMSE, MAE, RED	Monthly (2004–2013)	Hybrid FFA with SVR efficiency was tested for the assessment of WQI and compared it with the classical SVR. FFA-SVR doesn't use sub-indices and which leads to faster, expedient and direct prediction results with high predictive accuracy. It was suggested that this hybrid model is good for WQ prediction especially, in the semiarid area.
173.	(Azad et al., 2017)	Gorganrood River, Iran	ANFIS-GA, ACO_R, ANFIS, DE	EC, SAR, H, Cl, HCO_3, Ca, SO_4, Mg, Na, K/ EC, SAR, H	DC, RMSE, MAPE	Monthly (1996–2014)	The study predicts hardness and salinity using GA, DE, and ACO_R in combination with ANFIS. All the models produced a satisfactory result, but ANFIS-DE proved to be the most compatible one. The model was able to reduce uncertainty, increase performance and keep the overfitting under adequate range.
174.	(Raheli et al., 2017)	Langat River, Malaysia	MLP-FFA, MLP	COD, PO_4, K, Na, Cl, EC, pH, NH_4-N/ BOD, DO	R, RMSE, %RMSE, WI	Monthly (2001–2010)	The research used a hybrid predictive model using FFA along with MLP to do a comparative analysis to estimate BOD and DO. Uncertainty analysis was to assess the range of prediction output. MLP-FFA was found to be the superior model than the traditional MLP model.
175.	(Chen et al., 2018)	Yangtze River, China	ABC-BPNN, Improved ABC-BPNN	WT, pH, DO, PI, BOD, NH_3-N, Pet, VP/ DO	RE, DC, NSE	Daily (1–28 January, 2–15 March, 27 March-9 April, 21 April-10 May)	The author evaluated the performance of improved ABC-BP in respect to the ABC-BP model for WQ variables. Improved ABC proved to produce better stability, speed and accuracy because of the generalization, mapping, global iteration and local search capability. It was shown to produce efficient prediction results.
176.	(Yaseen et al., 2018a,b,c,d)	River in Washington, summit and New Jersey County, USA	LSSVM-BA, MARS, M5	WT, Q, pH, EC, DO / DO	MAE, RMSE, R	Monthly (2 January 2002-December 2016)	Various river station data were modelled with LSSVM which was incorporated with the Bat algorithm and was compared with MARS, M5. LSSVM-BA was found to be the superior model due to the coupled optimizer which resulted in the high correlation, better accuracy and improved performance.
177.	(Chatterjee et al., 2018)	Hooghly River, India	ANN-NSGA-II, ANN-PSO	pH, CL, H, Al, Tu, Cl_R /WQI	RMSE, ME, F-measures, MCC	Monthly (Not given)	Two training methods NSGA-II and PSO were combined with ANN modelling to predict WQ variables. ANN-NSGA-II proved to outperform the other two neural network models. It produced stable and highly accurate results.
178.	(Ahmadi et al., 2018)	Karkheh River, Iran	LR, ANN, GP	SAR, Cl, K, Na, Ca, SO_4, HCO_3, pH, EC, TDS, WT, Tu, PO_4, AT, COD, BOD, DO, TC, EC, NH_4, NO_2, NO_3, Mg /BOD	RMSE, MAE, DC	Monthly (1992–1996)	The investigation aims to select the best model from LR, ANN, and GP. Various input selection method was also used to find the best to predict BOD like correlation model, PCA, gamma test, backward regression and GA. ANN with a gamma test based on GA was the best model of all.
179.	(Keshtegar et al., 2019)	Tualatin River USA	MLP, MLP-PSO, MLR, PCE	WT, AT, W_1, EC, pH, Tu, RH / DO	R, NSE, RMSE, MAE	Daily (2004–2012)	PCE was tested against MLP, MLP-PSO, and MLR. The other model accuracy was found to be high when all inputs were used whereas PCE produced good results with only three. All three models were good as a predictive model.
180.	(Azad et al., 2019)	Zayandehrood River, Iran	ANN, ANFIS-ACO_R, ANFIS-PSO	EC, DO, TDS, SAR, H, CH, pH, Cl, Ca, Mg, SO_4, HCO_3, HCO_3 / EC, TDS, SAR	DC, RMSE, MAPE, δ %	Monthly (2001–2015)	This study designed three models i.e. ANFIS, Hybrid ANFIS-ACO_R and Hybrid ANFIS-PSO in which SAR and EC had the highest prediction accuracy than CH prediction values. The hybrid model produced better results especially ANFIS-ACO_R.

out of non-linear system thus, a prevailing tool for environmentalists and hydrologists (Olyaie et al., 2017). GP is a systematic domain-independent, a self-parameterizing tool which doesn't require pre-defined structure knowledge and tuning (Mehr et al., 2014).

Few optimisation algorithms use behaviour patterns. For example, GA goes to the in-depth knowledge and concept of genetics. PSO has no crossover and mutation transformation. Similar to GA, PSO is combined with different models, such as ANN, ANFIS and MLR, to increase predictive and classification potential. In each case, this approach is proven to be a good optimiser because it can enhance performance and has a retentive memory; particles are persuaded to achieve the most promising result in a short period, which also works as an advantage of decreased computational cost (Zhou et al., 2006).

7. Other AI models

7.1. Conceptual background

Various models are least explored than others in the field of river WQ assessment, prediction and classification. Thus, all the other models that do not belong to the other classes discussed in the previous section are collated here.

Model tree (M5P) uses a binary decision tree (DT) along with multilinear regression to predict continuous numerical attributes. The first step consists of splitting criteria to generate a DT. The second step consists of a pruning method to eliminate overfitting and the construction of a linear regression function. Compared with ANN and SVR, M5P model reveals the pattern and relationship in the data. This model is simple yet efficient and accurate for pattern recognition and relation establishment for large-scale data (Etemad-Shahidi and Bonakdar, 2009; Quinlan, 1992).

Random forest (RF) also uses the regression approach with a combination of tree predictors. This model uses a randomly selected section of training dataset for each tree. The method uses only one-third of the data, and the remaining is called out of bag. RF consists of several input variables and requires a large number of trees (Breiman, 1996; Breiman et al., 1983)

Another regression model developed during the 1990s by Friedman (1991) known as Multivariate Adaptive regression Splines (MARS) which is a combination of recursive partitioning and regression. The approach is a hierarchical forward-backwards subset selection procedure. This model consists of product degree, knots, dependent variables, intercepts and basic function. The model is flexible as it allows recursive splitting and produces a continuous model by substituting the step function.

GEP is based on a similar concept as GP and GA, however it has the features of a model (Danandeh Mehr et al., 2018). Specifically, GEP utilises individual population, fitness test and genetic operators (Sanikhani et al., 2019). GEP encodes individual population as linear strings of fixed length and later expresses them as nonlinear entities of different sizes and shapes. During the operation, the first step of GEP is similar to that of GA, whereas the second step is similar to that of GP. GEP can adapt and evolve. This method can be easily manipulated as a consequence of its use of chromosomes that are linear, compact and small. The approach always produces valid expression trees, reproduces nominated chromosomes with alteration and expresses them as expression trees for the next generation (Ferreira, 2001).

ARIMA model process includes model identification, variable assessment and diagnostic scrutiny. Identification is done to obtain stationarity and normality. Temporal correlation is verified by autocorrelation and partial autocorrelation. By following these steps, ARIMA determines the minimum Akaike information criterion to obtain the best model (Shahwan and Odening, 2007). Table 6 reports the studies conducted on the other AI models.

7.2. Bibliographic review of other AI models

Džeroski et al. (2000) applied M5 and ordinary regression tree (RT) models to address the issue of selective chemical WQ monitoring and its effect on river health. RT model successfully produces a good-sized structured generalisation of the input that is easy to interpret. The model can establish the relationship between the variables, which are expressed in mini and max values. Therefore, bio-indicators can be influenced by seasonal deviations. The biological state of the river can also be influenced by the chemical state. Nikoo et al. (2013) applied M5P and SVR models for the valuation of integrated water quantity management to minimise load on water supply and targets during the planning of projects. M5P model is a combination of a DT and multiple linear regression model which is applied on input/output variables using deduction learning for the better understanding of the input–output relationship. The better management skill of M5P model in handling environmental data and its capability shows the pattern and relationships in the data by the regression equation. Thus, its performance is superior to SVR. Sepahvand et al. (2019) examined the performance of M5P, RF, and bagging M5P along with GMDH models. Bagging M5P model outperforms other models because of the least standard deviation from the predicted value and shows high accuracy and validity due to low values of the vagueness indices. Its flexible approach also decreases the computational cost.

Kisi and Parmar (2016) compared least square SVM (LSSVM), MARS and M5 tree models. MARS can manage the complex nonlinear relationship and perform on forward and backwards stepwise patterns. As a result, the model can exclude unnecessary variables and select suitable variables. Furthermore, LSSVM, which works in the principle of structural risk minimisation, can deal with estimation, classification and regression analysis with a simple approach; it is robust against noises and has lower computational cost than the conventional SVM models. The results showed that both models are more accurate and better than M5 tree in predicting river WQ. Heddam and Kisi (2018) presented the application of LSSVM, MARS and M5 tree models. The performance of the three models is excellent, but each performs better than others in a different scenario for DO prediction. Input-max scaling was used for dimensionality reduction. The result concluded the effect of discharge may vary by place, but pH and WT influence over DO are undeniable and essential for correct DO prediction. Maier and Keller (2018) tested kNN, RF, SVM, MARS and extreme gradient boosting (XGB) models. The models were coupled with PCA as a pre-processing tool for managing high-dimensional data and compared with the models without pre-processing. The regression models with PCA perform better than the other models due to their higher accuracy, lesser error and ability to handle high-dimensional regression problem. Di et al. (2019) applied hierarchical clustering, expectation–maximisation clustering algorithm and time-series analyses to classify pollutants, short-time pollution condition and sudden unusual events. The data were normalised using Ward's method. The models perform well and can classify the pollutants.

Rankinen et al. (2019) applied the generalised linear model (GLM), which can manage non-normal error distribution, and boosted regression tree (BRT) model, which can handle nonlinearity and missing data. Their study aimed to predict river WQ condition in various future scenarios considering indirect factors, such as climate change, agricultural measures and environment policies. The model predicts the increase in annual mean WT to 17.3 °C between (2025–2034) and approximately 19.3 °C by (2055–2064) with an increase in precipitation of 10%. In addition, summer is predicted to face low flow. The study can predict an increase in annual concentration nitrate, suspended sediments and phosphorus by 16%–63%, 45%–146% and 38%–100%, respectively, in 2025–2034. Geetha Jenifel and Jemila Rose (2019) applied a linear model, SVM and DT. Recursive partitioning technique was applied to identify the relationship between variables. DT shows

Table 6
The summary of research conducted on the other reported for the River water quality simulations. Decision Tree, Regression model, dimensionality reduction, instance-based models etc.

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
181.	(Džeroski et al., 2000)	Slovenian rivers	M5, RT, LR,	BOD, Cl, CO_2, EC, $K_2Cr_2O_7$, $KMnO_4$, NH_4, NO_2, NO_3, O_2, AL, pH, PO_4, DO, SiO_2, WT, H/ WQI	NA	Not given (1990–1995)	The interaction of chemical and biological variables in the river was studied by regress tree analysis. The average, max or min values were used to interpret the WQ variable relation. It was found that few bio-indicators can be affected by seasonal variation. RT performs better by producing reasonably-sized structural generalization of the input variables in compare to M5 and LR.
182.	(Ömer Faruk, 2010)	Buyuk Menderes River, Turkey	Hybrid ARIMA, ARIMA, ANN	B, DO, WT / B, DO, WT	R, RMSE, MAPE, NSE	Monthly (1996–2004)	The study object is to overcome the weakness of ARIMA and ANN when dealing with both linear and nonlinear time series patterns. The hybrid ARIMA used a feed-forward and backpropagation network which was able to recognize times series outline and quite accurately dealt with the nonlinearity of the data.
183.	(Fahmi et al., 2011)	Klang River, Malaysia	MLR, PCA	DO, BOD, COD, TSS, pH, NH_4-N, WT/ DO	DC, RMSE, MSE,	Monthly (2003–2007)	The study aims to identify the WQ changes through DO prediction using MLR and PCA to get a better understanding of pollutant sources. PCA was used to reduce the huge and least influenced data into the more relative variable set so that MLR accuracy can be increased. MLR provides excellent results with the selected input where DO was most influenced by NH_4-N and WT.
184.	(Su et al., 2011)	Qiantang River, China	APCS-MLR, PCA, SOM,	Pb, NH_3-N, COD, Cr, DO, V-ArOH CN, Petro, Cd, Hg, BOD, $P_{Tot.}$ / WQ	R, Wilks'lambda, Chi-square, p-level	Yearly (2001–2004)	The objective of the study was to analyze the temporal pattern of WQ and pollution sources in the Qiantang River using MLR and multivariate techniques. The highest concentration among all variables only four i.e. Pb, BOD, NH_3-N and DO was used in MLR modelling. SOM categories 41 sites into three groups as per the pollution sources. The assessment dictates that pollution sources have a direct impact on WQ so more emphasis should be given to such data procurement methods.
185.	(Grbić et al., 2013)	Drava River, Croatia	LR- GPR, MLR-GPR	WT, AT, Q / WT	RMSE, MAE, NSE	Daily, Seasonal (1993–1998)	Gaussian process regression was implemented to model water temperature on seasonal variation and non-periodic input relation. Mutual Information technique was used for input selection. The multiple GPR produced high accuracy results than the conventional one.
186.	(Nikoo et al., 2013)	Zayandehrood River system, Iran	SVR, M5P	R_1, E_{vap}, S / TDS	MSE, R	Monthly (2001–2015)	In this study, the river system was explored which included reservoir, river and groundwater interaction using SVR and M5P model. The water quality management model was developed to manage the hydrological condition and M5P was the higher-performing model than SVR.
187.	(Zaman et al., 2014)	Zarinehroud River, Iran	GEP, ANN, ANFIS, GP2-ANFIS, SC_1-ANFIS	HCO_3, Ca, Mg, Na, Q / TDS	DC, R, MAE, RMSE, RRSE	Not given (1993–2010)	The study applied various AI models in which GEP, GP2-ANFIS, SC_1-ANFIS was used to predict TDS and proved to be successful. GP2 helps to divide the input space into subspaces. SC_1 is applied to evaluate grid points equal to datapoint and removes the need for grid resolution. Yet GEP performance was superior to others considered in the study as it can yield clear formulation of relationship.

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Table 6 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
188.	(Cole et al., 2014)	Delaware River, USA	GLScos, ARIMA, ANN, HFM	Q (upstream and downstream), W_1 AT, / WT	RMSE, NSE, PBAIS, d	Monthly, Seasonally (2008–2012)	The importance of temperature changes on flora, fauna and other water parameters was examined by using AI models. The study assessed the precise effect of WT and its changes in the study area. All the model performance was reasonable and shown the high impact of upstream discharge, air temperature, upstream discharge on water temperature. However, ARIMA was found to work only within available data, ANN gave flexible and accurate data and HFM was able to give value for seasonal variation.
189.	(Guo et al., 2014)	Yangtze, Jialing, Fujiang and other Rivers, China	ARIMA, SVM	pH, DO / pH, DO	MAE, MSE, MAPE, RMSE	Monthly (2008–2011) (21 October 2011- June 2014)	The authors designed an improved hybrid model that can predict WQ time series problems more accurately using input as pH and DO. The improved hybrid model was able to perform during WQ emergencies and its management, but its shortcoming was in predicting trip points.
190.	(Tang et al., 2015)	Yellow River Estuary, China	NEM-SDM	WT, S, W_1, Z, K_b, K_1, K_2 / DO	RMSE	Hourly/ Daily/ Monthly (April- September 2009–2011)	The researchers tried to understand the effect of freshwater inflow on Tu, DO and other factors under artificial hydrological regulation. 24 variables were considered out of which WT, Sa, wind Speed, depth, BOD decay rate constant, the oxidation rate of nitrite, Nitrification Rate over DO. The model suggests that the ecosystem might change from auto-tropic to heterotrophic under the turbidity changes and temperature changes.
191.	(Sharif et al., 2015)	Klang River, Malaysia	SOM, kNN	DO, BOD, COD, TSS, NH_3-N, TDS, pH, WT, EC, Sa, Tu, Ca, Mg, Zn, COD, TDS Tu, Fe, Na, PO_4-P, Cl, As, Cr, K, E. coli, TC / WQ	NA	Bi-monthly (January 2006 to November 2009)	The investigation objective was to identify the source of the pollution using the SOM and k mean algorithm. These AI tools helped in river management and able to identify the point sources like industrial activities, construction sites, urban runoff.
192.	(Sakizadeh, 2015)	Karaj River, Iran	LD, NB	pH, COD, EC, TDS, Tu, NO_2, NO_3, SO_4, TC, FC/ pH, COD, NO_3, SO_4	CC, RE, Wilk's lambda, Chi-square	Not given (2013)	The study used linear discriminant and naïve Bayesian classification methods for water quality assessment. Three classification methods were applied: twofold, leave-one-out and holdout method These methods were highly accurate but LB outperformed NB.
193.	(Deng et al., 2015)	Yangtze and Beas River, China	GCT-FTS, ARMA, NAR, SVM, ANN-GT, OSM	WT, EC, DO, COD / DO	MSE, MAPE, CE, R	Weekly, Monthly (2004–2013, 1986–2014)	The study introduced the Gaussian cloud transformation and fuzzy method to deal with the water quality randomness and non-linearity. The model done soft partitioning for uncertain data, the noise was reduced by dataset construction by periodicity, the fuzzy rule was able to handle uncertain data and the final performance of the combination was promising.
194.	(Keshtegar and Heddam, 2018)	Rivers, USA	MRSM, MLP	Q, pH, EC, Tu, DO / DO	RMSE, MAE, MBE, NSE, DC, d	Daily (2009–2016)	Conventional RSM was modified and compared it with MLP to examined the performance of the model for DO prediction. The modified form of RSM performance slightly better and more accurate than the MLP model.
195.	(Najafzadeh et al., 2018)	Karoun River, Iran	GEP, EPR, MT	Ca, Mg, Na, NO_2, NO_3, PO_4, EC, pH, Tu / BOD, DO, COD	SSE	Monthly (1995–2010)	GEP, MT, and EPR were modelled to predict BOD, COD and DO. All the models were able to deal with linear and non-linear variables where EPR showed the best result. The authors suggested that these models may not be physically sound and need a great deal of financial support for data extraction.

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Table 6 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
196.	(Tao et al., 2018)	Euphrates River, Iraq	Hybrid RMS, SVM	WT, Tu, pH, EC, Al, Ca, COD, SO_4, TDS, TSS/ DO, BOD	MAPE, MAE, SI, RMSRE, MRE, DC, RMSRE	Monthly (2004–2013)	In this study hybrid response surface model and SVR were used to predict BOD and DO with various WQ input variables. Hybrid RSM was proved to be an efficient mathematical model and was able to produce results higher to SVR with less approximation.
197.	(Kisi and Parmar, 2016)	Yamuna River, India	LSSVM, MARS, M5 Tree	NH_4, N, WT, TC, FC, pH / COD	DC, RMSE, MAE	Monthly (January 1999-April 2002)	MARS, LSSVM and M5 tree models were used to build a predictive model for COD. M5 tree performance was found lesser than the other two models. The inputs like ammonia, nitrogen and water temperature were found the most influence on COD prediction.
198.	(Sattari et al., 2016)	Lighvan Chay River, Iran	kNN, SVM	P, Cl, Ca, Mg, Na, SAR, SO_4, HCO_3, TDS, EC/ TDS, EC	DC, RMSE, MAE, R	Not given (1967–2011)	The study aims to predict EC and TDS using kNN and SVR with various input scenarios. Both of the model performance was satisfactory, yet the authors recommended SVR because it's more efficient in the field.
199.	(Jiang et al., 2018)	Songhua River, China	GHSOM	pH, EC, DO, COD, BOD, NH_3-N, P, Pet, VP, AiS, f, CN, S, Hg, Se, As, Pb, Cu, Zn, Cd, Cr, FC / COD, FC, $P_{Tot.}$, NH_3-N	R	Monthly (2011–2015)	GHSOM was modelled which has the benefits of SOM and HCA and it proved to be superior to the originals since it can automatically generate neurons, establishes the relationship between data, better classifier and can handle larger data.
200.	(Heddam and Kisi, 2018)	Fanno creek, canyon creel and Delaware river, USA	M5 Tree, MARS, LSSVM	WT, pH, EC, Q / DO	R, RMSE, MAE	Daily (January 2002-December 2016)	The study designed three models on LSSVM, M5 tree and MARS to predict DO for three stations. The results showed that all models are capable to predict DO successful, yet each performed better at different scenarios. The authors conclude that DO can effectively be predicted simply using WT and pH.
201.	(Keller et al., 2018)	Elbe River, Germany	LR, PLS, RF, ET, AdaBoost, GB, k-NN, SVM, ANN, SOM	Chl-a, Ga, CDOM, Tu, Diatoms / Chl-a, Ga, CDOM, Tu, Diatoms	DC, RMSE	Daily (24 June − 12 July 2017)	The authors modelled ten regression models and used processing methods i.e. PCA and min–max scaling. All the machine learning models performed well and able to handle the high-dimensional data in the study.
202.	(Maier and Keller, 2018)	Elbe River, Germany	kNN, RF, SVM, XGB, MARS	CDOM, Tu, Chl-a, Diatoms/ CDOM, Tu, Chl-a, Diatoms	DC, RMSE	Seasonal (2017)	Hyper-spectral data and AI models were evaluated for their modelling efficiency. The data was obtained by sensors and pre-processed using PCA and mini-max scaling. All the models performed well and have the capability to be used in the future.
203.	(Di et al., 2019)	Yangtze River, China	HC, EMC	pH, DO, COD, NH_3-N / pH, DO, COD, NH_3-N	SC, CV	Weekly (2016)	The authors applied clustering methods to assess pollution variation. The results showed short term variation and changes in WQ variables. The model applied produced reliable and good results as well as identified spatial–temporal WQ characteristics and pollution causes.
204.	(Ho et al., 2019)	Klang River, Malaysia	DT	NH_3-N, BOD, COD, DO, pH, TSS / WQI	Ap, P_c	Monthly (2001–2010)	The study used a decision tree for water quality index prediction using different input scenarios. The decision tree model was used to simplify the complex data and analyze and reduce nonlinearity presented by water quality predictions.
205.	(Rankinen et al., 2019)	Lepsamanjoki Catchment, Finland	Generalized LM, BRT	$P_{Tot.}$, N, NO_3-N, TOC, Color, Tu, TSS, WT, Q, AT, $P_{Tot.}$, / Chl-a	DC, RMSE, PBIAS, NSE	Seasonal (1985–2014)	Generalized LM and BRT were analyzed to model Chl-a in the river and found it is most influenced by total phosphorus and water temperature among variables other considered. Both model performed good.
206.	(Sepahvand et al., 2019)	Karkheh River, Iran	M5, RF, bagging M5P, GMDH	TDS, Ca, EC, Na, Mg, pH, HCO_3, Cl, SO_4, K, SAR/ SAR	RMSE, MAE, DC	Monthly (May 1971-May 2017)	Four predictive models were built for SAR prediction in the study area. The results show the bagging M5P model results were more accurate than the M5 tree, RF and GMDH. During the uncertainty analysis, the bagging M5P model outperformed others.

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Table 6 (*continued*)

	Scholars	Case study region	Applied models	Input/output (variables)	Performance metrics	Data time scale	Research remark
207.	(Geetha Jenifel and Jemila Rose, 2019)	Various River, India	RT, DT SVM, LM,	WT, DO, pH, EC, BOD, Lat, Long. / WQI	RMSE	Not given	Four WQ variables were modelled using data set from all over the country and used liner, recursion partitioning, and SVM. RP proved to be more efficient than SVM and LR created less error and high-quality result.
208.	(Peterson et al., 2019)	Kaskaskia, Sangamon and Meramec Rivers, Illinois	MLR, PLSR, GPR, SVM, ELMR	BGA, Chl-a, TDS, Tu, TSS/ WQ	DC, RMSE	Seasonal (June 2016-November 2017)	Canonical correlation analysis fusion was introduced to reduce noise along with MLR, PLSR, GPR, SVM and ELMR models. The combination increased the accuracy and eliminated model bias which was proved to be an efficient tool for prediction of water quality variables
209.	(Salarijazi and Ghorbani, 2019)	Zarringol and Ramian river, Iran	LR	Q, EC/ EC	RMSE, MBE	Not given (1968–2015)	The study applied linear and non-linear regression along with statistical improvements like logarithmic, inverse transformation and exponential techniques. The performance of the model was good in predicting EC.

the least error rate, manages considerable numerical data, produces good data quality and handles dataset with errors and outliners. Ho et al. (2019) evaluated the productivity of DT model to simplify, analyse and classify the WQI for decreasing its complexity and the non-linearity of data. The study concluded that DT model can be a cost-effective and process-efficient modelling methodology.

Sharif et al. (2015) coupled SOM and kNN to examine the classification and monitoring capacity of the combined model for river WQ. The clustering technique was used to reduce the number of variables, and k-NN was used for classification. The model can identify the point of pollutant source and recognise patterns in the data. Sattari et al. (2016) applied the kNN algorithm for classification and regression along with traditional SVR model. Both models could successfully predict WQ, but the study suggested that SVR model with the RBF performs better and is more applicable than the other model due to its high-performance, easy execution and flexibility in case of sudden changes in WQ variables.

Ömer Faruk (2010) explored the possibility of hybridising ARIMA with neural network models because both are individually incapable of handling linear and nonlinear time series data efficiently; thus, both were used separately and combined afterwards for improved modelling and prediction performance. The combined advantage of both models produces good results in identifying time-series patterns and nonlinear characteristics. Cole et al. (2014) tested the performance of generalised least square model with a cosine trend (GLScos), ARIMA and ANN models for WT modelling. ARIMA model performance is least accurate in predicting ahead of time, yet it does not show over-estimation bias and under or over prediction. GLScos model is better and highly accurate even though all models perform well. Guo et al. (2014) applied hybrid ARIMA–SVM; ARIMA and SVM models were optimised with PSO for DO and pH prediction. The hybrid model uses both components of the models and can process linear and nonlinear data with overall enhanced forecasting accuracy.

Su et al. (2011) implemented rotated PCA and DA to identify the point source of water pollution and evaluate the temporal changes in WQ. SOM was applied for clustering and visualisation, and ANN was applied for partitioning. Absolute principle component score with MLR was applied to obtain the quantity information. The results showed that the model is best for identifying pollutant sources which help in better manage the river pollution. Grbić et al. (2013) applied a probabilistic nonparametric approach, that is, gaussian process regression (GPR) with LR, to resolve nonlinear regression water pollution issues. The inputs were fetched by least-square criterion, and MLR was used with mutual share information. The first model can understand the periodicity of WQ. The second one can handle short term non-periodic components. Fahmi et al. (2011) combined PCA with MLR to simplify and better understand the complicated and vague relationship between WQ variables. The new model is better than the conventional one, reduces sampling time and cost, and eliminates data collinearity. Keller et al. (2018) studied a large number of models, such as LR, partial LR, RT, extremely randomised trees (ET), GB, kNN, SVM, ANN and SOM models. ET, SVM and ANN models are the best amongst the models. ET has the superior performance after pre-processing with PCA, and ANN is the second-best model with mini–max scaled pre-processing of WQ data. Another study also considered the application of the LR model for the prediction of EC. Different combination of logarithmic and inverse transformation statistical technique was applied for model improvement. The ability of the LR model to fit on the non-monotonic behaviour of data resulted in better performance in handling WQ data (Salarijazi and Ghorbani, 2019). Jiang et al. (2018) applied growing hierarchical self-organising map (GHSOM) as a classification model. The model can establish the correlation between the WQ variables and detect the main pollutants. The model is fast because it automatically generates neurons to achieve the specific clustering accuracy, applies hierarchical inheritance approach for assessment and creates clear boundaries. Peterson et al. (2019) implemented various predictive

models, such as MLR, PLSR, GPR, SVM and ELMR models with CCA, for spectral analysis of river WQ variables. The study aimed to incorporate decision-level fusion method, which produces promising results. The fusion increases the overall accuracy and efficiency of the model and eliminates biases.

Zaman Zad Ghavidel and Montaseri (2014) predicted TDS as an indicator for WQ assessment using ANN, ANFIS–GP, ANFIS–SC and GEP models inspired by the genotype-phenotype system. The performance follows the order GEP > ANFIS–SC > ANFIS–GP models. Najafzadeh et al. (2018) examined GEP, evolutionary polynomial regression (EPR) and model tree (MT). Gamma test was coupled to better understand the relationship between the variables. EPR, which is a nonlinear global stepwise regression method, is a better model than others. MT performs well as a classifier.

7.3. Model assessment

The models that fall in the categories in the previous sections are accumulated here. The number of users of such models are few, yet their performance is promising. The literature shows that numerous models have gained popularity in recent years. One such model is ARIMA, which was reported in 2010 (Ömer Faruk, 2010). Its performance improves when it is coupled with another model for it cannot deal with nonlinear time-series data, which may be the reason why it has been less implemented in river WQ studies (Wang et al., 2015). Its ability in likelihood estimation and forecasting, score test for model checking and time series modelling (Box et al., 2015).

Similarly, kNN model can perform classification with good accuracy, but its performance is inferior to that of SVR, which may be why kNN has been used less in river WQ research (Sattari et al., 2016). However, this model needs exploration along with research possibility of hybrid k-NN models.

Regression models are popular models after conventional models. For example, M5 can yield good generalisation, better understand the variable relationship, avoid overfitting and achieve high accuracy (Sepahvand et al., 2019).

DT models have been successful in event prediction and can be applied for recognising WQ variation. They are able to solve classification and regression problems This model applies top-down, greedy search without backtracking and can be used for categorical and numerical data management (Brown and Myles, 2009). DT can decrease complexity, classify and handle nonlinearity with least error rate and handle a considerable amount of data (Geetha Jenifel and Jemila Rose, 2019). There are many variants of DT which are yet to be tested in the river WQ research such as logistic MT, reduced error pruning tree, naïve bayes trees, alternative DT (Khosravi et al., 2018). Due to model problem-solving technique (recursive dive and conquer) and simplification of the complex issue allows more applicability in complex WQ data modelling. Addition to that DT models are easy to interpret and can predict an unpredictable outcome (Ho et al., 2019).

MLR model in an older model, but recent studies have shown that the hybrid version of the MLR model has been mostly applied. The reason MLR is a mediocre model (as concluded from Section 2.2) compared with the ANN model. The studies have shown that MLR coupled with dimensionality reduction tools or optimiser can perform well in pollutant source identification and handle non-periodic variation. Moreover, it can understand complex, nonlinear and vague river WQ data when optimised (Fahmi et al., 2011; Grbić et al., 2013).

Models like MARS have been least implemented in river WQ as it easily gets trapped with overfitting and solving regression problems is narrow (Abdulelah Al-Sudani et al., 2019). However, pruning can help to increase the forecasting accuracy which can be achieved by the backward process using generalized cross-validation (Wang et al., 2010). MARS model flexibly and capability to manage high dimensional regression and classification problems makes it a good model to be tested in river WQ modelling. It can separately identify the additive contributions (Friedman, 1991).

8. A general appraisal of global literature review

8.1. Prerequisites of new model developments

The application of AI models has rapidly progressed over the last two decades along with various improvements. A continuous innovative attempt has been to develop the best model for river WQ. Fig. 8 shows the research development in river WQ field over the last two decades. From 2000 to 2007, the average research published was 2 per year. Steady growth was observed from 2009 to 2012, during when the average research published grew to 10 studies per year. River WQ modelling gained momentum during 2009–2012 due to increased awareness of the application of AI models and the advantages for WQ prediction and assessment for river health management. Furthermore, rapid advancement in AI model practices can be observed from 2013 to 2019; average reported research publication was 20 per year and still increasing. Additionally, the data from 2018 (31 papers) and 2019, 31 papers. In the coming years, the advancement of the application of AI models and optimising tools will grow more popular in the scientific community than before. In the last few years, synchronous growth of published research has indicated increased concern about the river water pollution and its effect on the study area, which is ubiquitous worldwide. Increased frequency of AI model research can realise its possible implementation in river WQ research and in real-life uses, such as river health monitoring, classification of a pollutant point source, customised policy-making and water treatment management (Bhagat and Tiyasha, 2018). This apprehension is affected by factors, such as increasing population, agriculture, industrialisation and urbanisation, which directly and indirectly affect river health. Thus, AI models are needed as a result of their superior understanding and applicability to handle the complex relationship between the river WQ and various factors responsible for the change in WQ.

In previous assessment sections, the individual AI models have been assessed on the basis of model strength to deal with complex and nonstationary river WQ data. Fig. 8 clearly presents the growth of AI models over the decades. Moreover, numerous architectural changes have been observed along with this exponential growth of application of AI model. The continuous progression implies the need for further exploration because each model has its own limitations and benefits. This section highlights a few weaknesses of the models for the need of designing new advanced models.

a. The 209 reviewed studies revealed that various models were used 311 times. ANN or any form of ANN was implemented successfully as a good predictive model for 133 times (43%), as shown in Fig. 9. Despite the success rate of ANN, it has disadvantages, such as the need for large data for training, overfitting problem, local minima issues and false-negative frequency issues. ANN also lacks the understanding of complex environmental data.

b. Fuzzy logic and fuzzy interference systems are the oldest and extensively applied algorithms in river WQ after ANN. Their application in 48 studies (15%) makes them the second most applied model after ANN models. However, they have limitations, such as needing rules for combining the appropriate membership function, which prevents the less experienced user from achieving the best performance out of the data. Moreover, the rule cannot differentiate amongst important input factors.

c. The next is SVM (32 studies), which was implemented in 10% of research and can solve regression, classification and prediction problems. This approach faces problems in the choice of kernel and high computational cost. SVM may suffer due to data inconsistency and inability to handle probabilistic forecasting. The nonlinear models cannot be easily deduced due to the complication in mapping nonlinear space

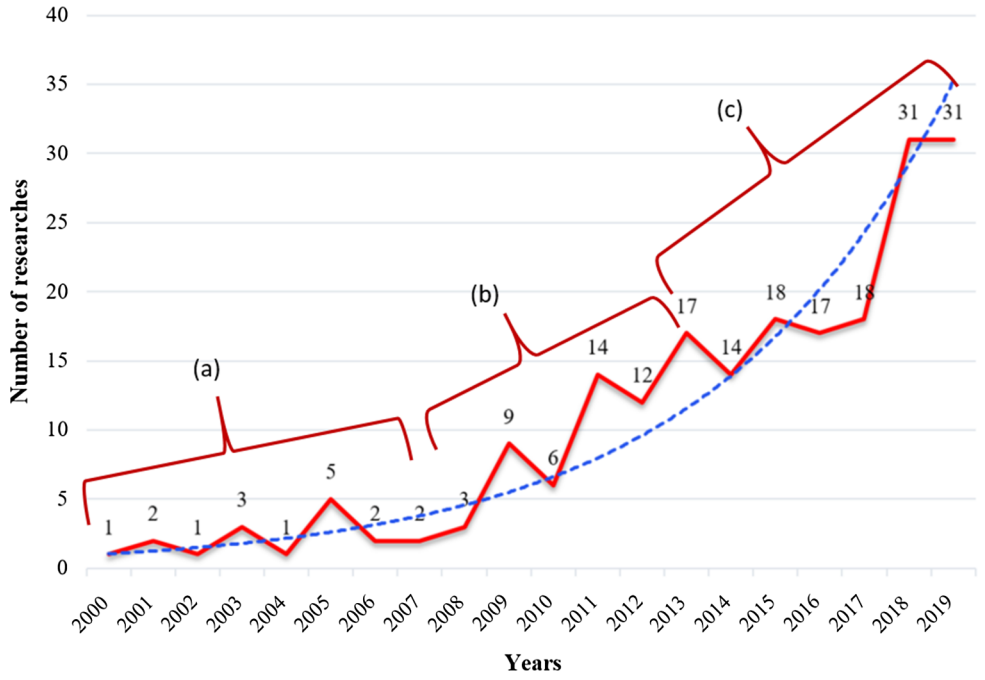


Fig. 8. Exponential Growth of AI model from 2000 to 2019: (a) 2000–2008 reveals the early phase for AI model in river WQ research (b) 2009–2012 was expansion phase with steady growth (c) 2013–2019 suggests rapid phase.

into high-dimensional space, which interferes with the computational speed.

d. Various optimisation techniques using NI (18 studies) were integrated with these models as support, but they also faced various issues. For instance, PSO faces the problem of unrestrained particles, which tends to leave the feasible space. As a result, its convergence speed decreases. SFLA needs prior input normalisation to achieve good accuracy. ACO_R is hindered by random initialisation and excess parameter requirement and needs tuning.

e. EAs, such as GA, suffer from early convergence issue, do not use local search information and take a long time to tune. GP is dictated by natural behaviour, which makes it random and unable to guarantee the required results. GP suffers from bloating, that is, excessive growth of the code without any substantial growth in the fitness (Silva and Costa, 2009). GA and GP designed with binary expression produce invalid expression and cause increased computational cost. GP, GA and GEP can be easily manipulated but lead to functional complexity. This complexity hinders the redesign of the same model and production of the same result (Ferreira, 2001).

f. The performance of the wavelet algorithm (16 studies) depends on the choice of the mother wavelet and the decomposition level. This condition results in its unpredictable accuracy. Despite a few issues, the wavelet algorithm complements other models very well and increases their performance level.

g. It was observed that river WQ modelling is associated with uncertainty feature and thus uncertainty analysis for the dataset, models and input variables should be investigated for better understanding of the predictability performance (Chen and Chau, 2019; Shamshirband et al., 2019; Yaseen et al., 2019a).

h. It has been observed that most of the research study did not give much thought for the data division techniques. During the study commonly employed data splitting is simple random sampling which is an unsupervised technique, it is efficient, simple in application and produces low bias (Lohr, 1999). However, it suffers from a high discrepancy of the model performance error estimation. Under this technique other methods are SOM, physical process-based, domain knowledge-based, trail-and-error and user rule-based methods are few which have been frequently applied. Among them, the trial-and-error method has less variance in the model performance and can be combined with more techniques but takes a lot of time and there is no proper theoretical background (Bowden et al., 2002). However, this minute consideration of data splitting ration can have a significant effect on the result (Dawson and Wilby, 1998). Thus, utmost precaution should be taken when considering WQ data and its statistical properties.

i. Throughout the literature, it was observed that several types of research were conducted on the baseline of comparative AI models capacity with respect to the predictability performance. This was seen in journals mainly related to water quality researches such as journal of hydrology, environmental monitoring and assessment, environmental science and pollution research, stochastic environmental research and risk assessment, water supply, water resources management, science of the total environment, etc. As a matter of fact, this kind of researches adds no value to the base knowledge of river water quality modelling. The establishment of water quality modelling using the feasibility of artificial intelligence models should be incorporated an innovative contribution where insightful aspects of water quality are discussed, debated and recognized. In addition, there must be an informative reason explained for the behaviour of the predictive model.

8.2. Most explored rivers

Fig. 10 shows the number of studies in each country along with a map that presents the most explored and modelled river streams worldwide in the last two decades. The leading AI research on river WQ area have been accomplished in China (42 out of 209). This result suggests that AI models are popular and gaining interest as time progresses. Most study areas are from Asian countries, such as China (42), Iran (31), India (22), Malaysia (23), Taiwan (8), Korea (4), Iraq (4), Bangladesh (2), Thailand (1) and Hong Kong (1), representing 138 out of 209 papers. This result can explain why AI models are continuously gaining popularity in developing countries. Moreover, Asian countries are endowed with perennial and non-perennial rivers, which demand much research on surface water resources, such as rivers. Rivers are the lifeline for agricultural and industrial growth, and agricultural countries that grow crops, such as rice and sugarcane, need a substantial

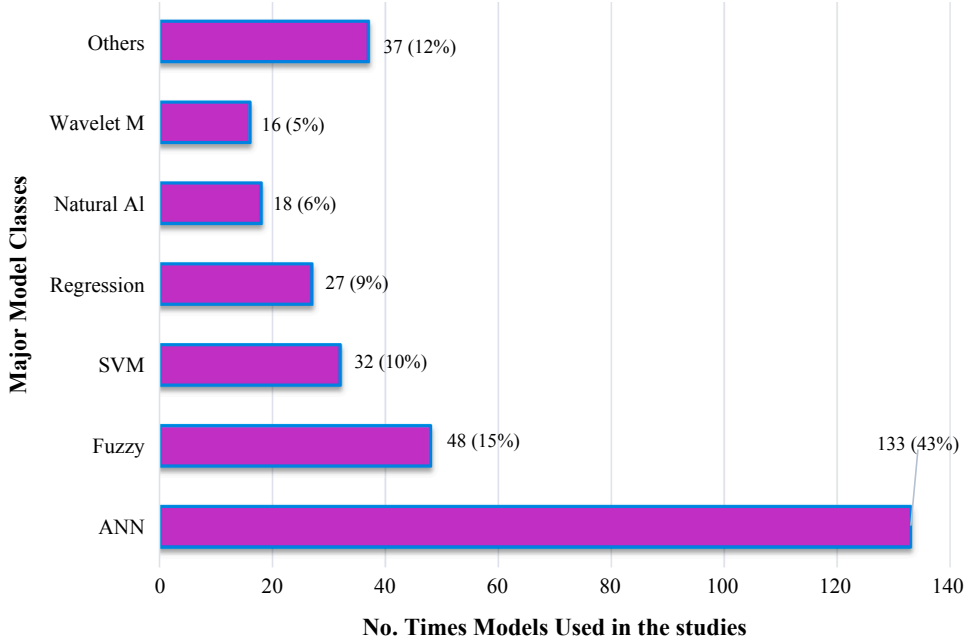


Fig. 9. AI models distribution among reviewed papers in last two decades: The presentation identified the number of papers which applied AI modeling and their percentage share.

amount of water for irrigation. Thus, increased research evidently manages and monitors river WQ to maintain healthy crops. After China, USA has the second majority of 31 papers, which can be justified by its large land area with rivers and advanced monitoring station where data are freely available for study worldwide. Thus, USA is a popular choice since modelling scientists need good quality data for testing new models.

Many countries and river sources are yet to be explored. A large number of studies should be done considering diverse conditions for each study area, which can help design models applicable to various situations. Although the African continent ranks third by land size and comprises 54 countries, only 2 studies have been reported. The reasons are (i) many of its countries are under-developed and developing thus, still working on their water-related policies and don't have comprehensive policies (ii) many countries are still in the initial stage of industrialisation and use natural methods of farming, which leads to minimal pollution concerns (iii) The mass population is less educated leading to less acceptability and addition to that, there is less or no teachers for AI-powered education to prepare the next generation (iv) fewer funds for AI research and education (v) concerns in the data collection transparency, ethics and proper utilization of the data (Pedró et al., 2019; Ukpong et al., 2019). However, additional studies must explore the unique diversity amongst various African river streams. Similarly, the number of river WQ research reported in Australia are only 2 possibly due to the least orientation of researchers towards AI techniques. The cause of such alignment is due to the acceptance of the AI technique since, it will cause a shift in the job, will need to upskill the workforce, needs implementation of safer and ethical policies, increased cybersecurity privacy and advanced data collection system (Hajkowicz et al., 2019).

The ecological quality of river water quality can be articulated as organic and non-organic variables. Regular monitoring the environmental and hydrological condition of a river zone can help to a better insight into the changes in the variables, even the macro or microhabitat. Regarding this, there is a challenge in dealing with the existing quality indicators and the specified reference value of them. It has been pointed out in the introduction that most of the changes in water resources are related to anthropogenic activities. The only salvation for humankind to solve the self-created problems in designing effective water-related policies which should be knowledge based on the catchment area, frequent intensive monitoring data, environmental indicators, effects of previous legislation and the social and economic activities in the area. To reduce the sudden change in WQ due to industrial or municipal wastewater discharge or an extreme condition like a flood, it is essential to rely on advanced water and wastewater treatment, integrated environmental monitoring and automatic water quality modelling system. Both last techniques can be equipped with remote sensing and sensor technology to obtain high-quality data. Nevertheless, this kind of technology implementation is the need of the hour but still, a challenge for the hydrologists, environmentalist and data scientists as of the WQ has a wide range of indicators and variables. Their measurement and understanding of inter-relationship and the influence of each other can be difficult to interpret.

8.3. *Water quality output consideration*

All previous review and research papers have mostly emphasised on input data, which are essential for modelling process, and various input optimisation techniques have been used to ensure the most potential input is used relative to the relationship established with the output variable. Although researchers have considered the input–output relationship for modelling, the least thought and analysis have been considered in the WQ review papers about the output variable selection. The reason is that the basic approach of these studies is to establish the relationship between the input and output. Another reason can be those input variables are used for learning, optimisation and dimensionality reduction, and a relation is established based on their sensitivity towards the chosen output variables; thus, they are the focus of the study. Furthermore, the output is commonly decided based the issue addressed by the researchers such as, if the issue in hand is algal bloom then output considered is Macrophytes whereas if the problem is salinity then TDS or EC is taken as output not by applying any predefined technique.

The major outputs highlighted in Fig. 11 represent the number of papers that have used each output variable out of all the considered studies. Thus, the current study emphasises the assessment of the output and why few selected outputs or a single, rather than multiple, output is selected in many cases. The figure presents that, out of 209

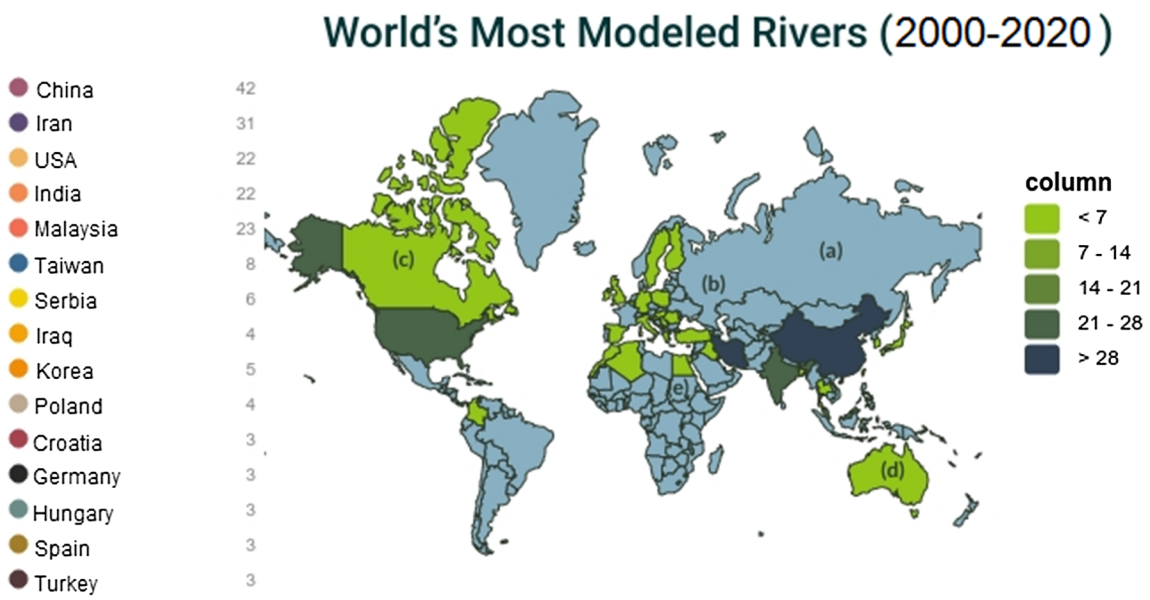


Fig. 10. Globally most studied river sources using AI techniques in last two decades: (a) Asian river system is the most studied area (130) (b) European river system (38) (c) American river system (30) (d) Australia (2) and (e) Africa is the least explored (2).

assessed papers, 55 papers used WQI, which indicates the multiple numbers of output variables; the remaining 154 used single or few outputs. WQI is mainly considered in the case of river WQ classification and trend determination where multiple variables are assessed against the government standards and classified in simplified categories.

The most popular WQ variable is DO. DO is a critical WQ and river health indicator. DO is part of various water-related reactions, such as aeration, underwater photosynthesis, chemical oxidation and decomposition. Sufficient amount of DO is essential for not only for humans but also aquatic plants and animals (Gray, 2017). DO holds the highest part in chemical WQ assessment and is very easily affected by other variables. Thus, finding the DO can provide a considerable amount of insight on river health, WQ changes and further treatment requirements.

Few variables may have an indirect and synergistic effect. WT affects the solubility of the water and works as a catalyst in various chemical and biological processes in the river. It also inversely affects the solubility of DO. Another essential WQ variable is pH, which is the potential of hydrogen ion and one of the largest initiators of all chemical and biological processes, such as corrosion, decomposition, solubility and toxicity. Similarly, BOD and COD, which are the fourth and ninth most used variables, respectively, are the WQ indicators that manifest microbial and chemical reactions occurring in the presence of oxygen in the water. BOD is one of the first WQ variables discovered for river health detection. BOD and COD signify the need for treatment or the extent of river pollution.

Few WQ variables are considered indicators of another major pollutant contaminating the river water. For instance, an increase in Chl-a signifies an elevated chance of algal bloom. Macrophytes indicate the growth of phytoplankton and eutrophication in the water, which kills aquatic plants and animals, such as fish. Nitrogen compounds, which are second to DO, also indicate and encourage algal bloom, eutrophication and dendrification (they create NO_x that leads to global warming). Along with nitrogen, phosphorous indicates the same but also informs about industrial wastewater contamination. Similarly, EC indicates mineralisation and salinity in the water (Al-Shujairi et al., 2015).

As per the reported survey, it is evident that the performance capability of AI is dependent on the water quality multifaceted nature. While designing a predictive or classification model, the complexity depends on the number of different inputs integrated i.e. fewer is the better whereas too many inputs increase the intricacy of the model. It is worth to highlight, the applicability of AI models with the various model version as classified in Fig. 1 can change depending upon the geographical location, government policies, local population behaviour, climate change and other unique scenarios. Thus, there is no issue in the application of AI model in simple cases, however, the ever-changing WQ due to the dependent variables can cause a serious impact on the modelling result. For instance, oxygen depletion or organic decomposition or biological activities (aquatic plants, animals or microbes) can alter the amount of DO, BOD and COD. Another example can be nitrogen related compounds which changes its structure quite often due to change of other factors (nitrogen fixation{ammonia}, nitrification{nitrate}, denitrification {nitrite to molecular nitrogen}).

8.4. WQ data variation time scales

The survey shows that most AI model papers have considered the use of monthly data for river WQ modelling, as shown in Fig. 12. A total of 92 papers provided information about the uses of monthly data out of 110 papers, which provided information on the data step. This percentage can be considered as 50% because 46 papers have not reported on the data collection interval amongst all reviewed papers, followed by weekly (9 papers), yearly, bi-weekly and bi-monthly (3 papers each). The main reason of the highly conducted research on monthly river WQ include unavailability of sufficient project budget, few analytical laboratories, lack of technical and skilled human resources, policies of decision-makers and meeting the basic need for modelling, monitoring and managing river health. In addition, monthly WQ monitoring can be highly essential for multiple water supply usage, agriculture and water treatment plant design. Thus, the prediction of the monthly time scale can contribute to several environmental engineering perspectives to obtain considerable insightful information.

A total of 12 research papers conducted time-series hourly data: 8 from the USA, 3 from China and 2 from UK. Accordingly, hourly data are from developed countries with sufficient funding and advanced monitoring stations that can obtain hourly data effectively with highly equipped laboratory facilities. Similar findings can be observed in case of daily data; out of 26 papers, 9 are from USA and 8 are from European countries, which may again be due to availability of funds and facilities. This finding signifies that these countries can manage and collect large amounts of hourly and daily data to produce good and effective models.

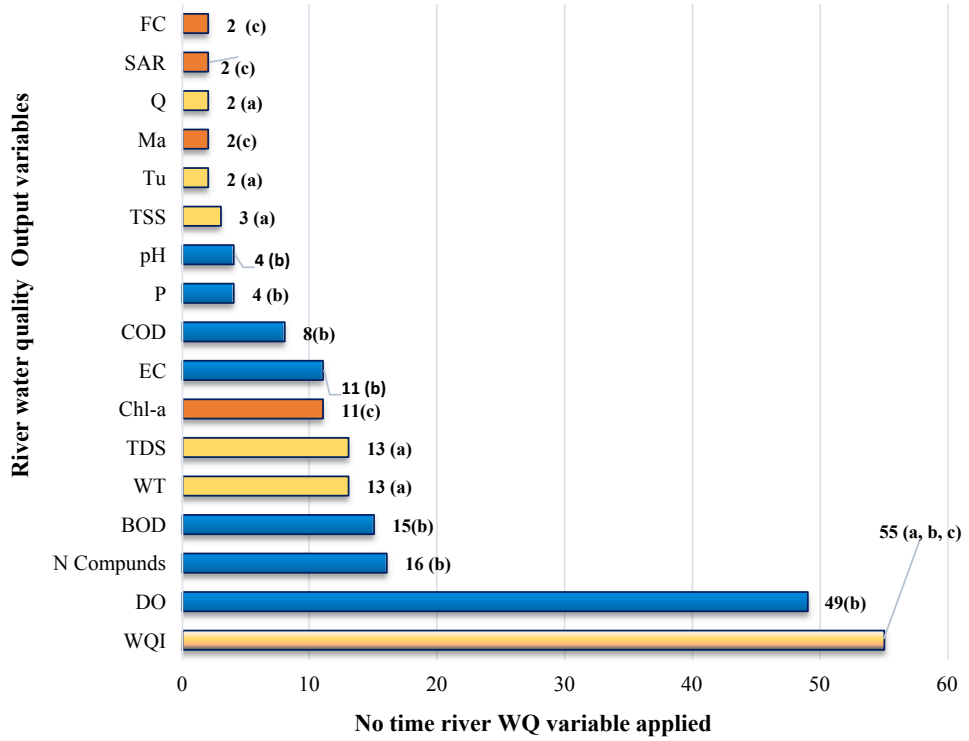


Fig. 11. Recurrently applied modelling Outputs for river WQ modeling: (a) These are the physical water quality parameters (b) This represents the chemical parameters and most preferred for modeling (c) The biological parameters are generally used as bio-indicators.

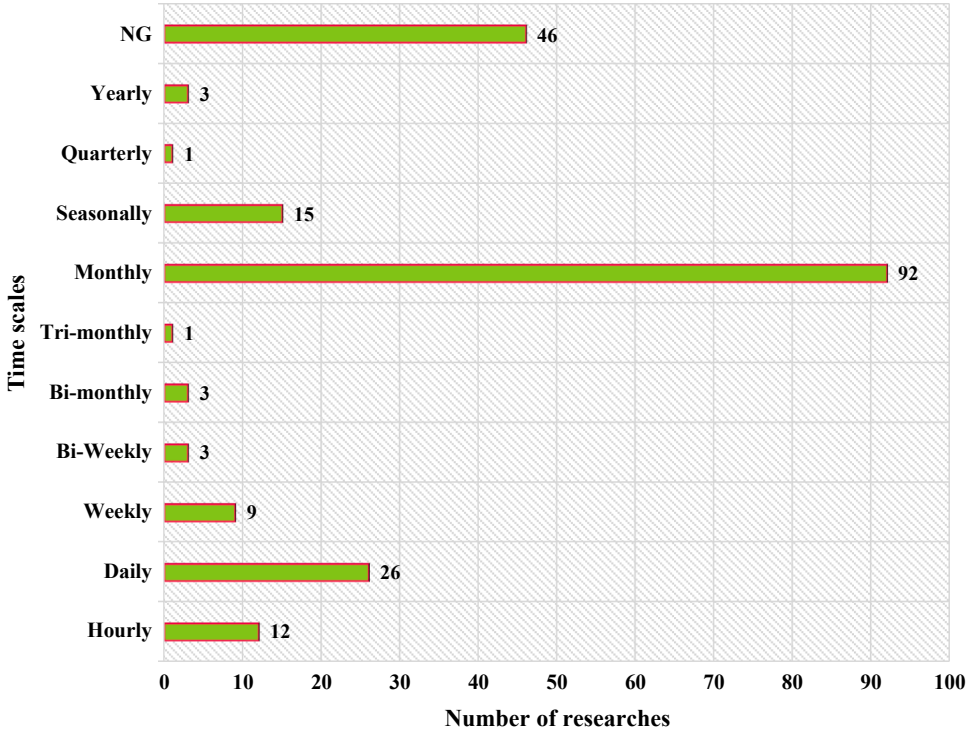


Fig. 12. Multiple time scales (i.e., Yearly, Quarterly, Monthly, Weekly, Daily, Hourly) of water quality variables predicted using the potential of AI models.

Consequently, better measures can be taken against river pollution. One of the unique time steps is seasonal where few specific months are targeted. A total of 15 papers, which used seasonal data collection to save funding by collecting data for a precise period of the year, which were already evaluated for changing trends in the specific time of the year. Thus, the resources must be concentrated on those specific time intervals. Another possible reason is that the rivers selected for the study are non-perennial (seasonal).

Furthermore, the decision of selecting the appropriate data time scale depends on the objective of the study. For instance, if the researcher aims to understand the point source of pollution, then daily time scale will most likely be used to provide improved insight into the

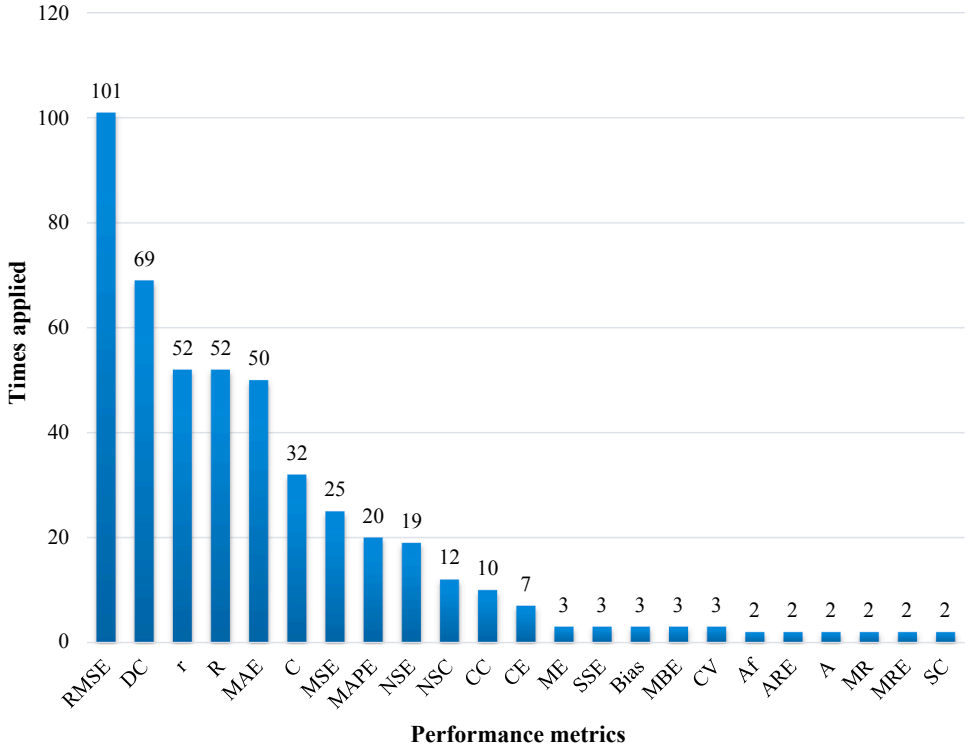


Fig. 13. The computed performance metrics for the applied predictive models evaluation and assessment.

WQ changes due to industrial effluent discharge, domestic wastewater discharge and agricultural runoff. Similarly, if the study targets seasonal variation, such as flood, drought or any other unique natural changes affecting river WQ, then specific months or weeks of the year and seasonal time steps are preferred. Using quarterly and yearly data collection may be inappropriate as it does not add any value to the research. Selecting the correct time interval of data collection is important to achieve the objectives because the river has the capacity for self-purification, which indicates the WQ can change naturally over time thus, it is important to select the correct time interval of data collection to achieve the objectives.

8.5. Performance metrics

Performance metrics are statistical criteria that help developers assess the calibrated modelling performance on various platforms. They also translate performance accuracy and efficiency into understandable and expressible form. Over 30 evaluation criteria have been used in river WQ modelling, and few are represented in Fig. 13.

AI models mostly use quantitative error measuring technique. The most frequently used square error measuring systems are root mean square error (RMSE 101 papers), Nash Sutcliffe efficiency (NSE 19), mean square Error (MSE 3) and sum of square error (SSE 3) based on the square of the differences between the actual and predicted values influenced by the error. RMSE is popular as of it can show several deviations, is reliable and provides high weight to error. However, outliers should be removed. Coefficient of determination (DC 69), which is the second most preferred matrix, evaluates the predicted value produced by the model in respect of the actual value, which provides the model performance and ranges from 0 to 1 (Nagelkerke, 1991).

Another group consists of absolute errors, namely, mean absolute error (MAE 50), mean absolute percentage error (MAPE 20) and absolute relative error (ARE 2). They are the absolute difference between the actual and predicted values. They only provide information about the extent of the error but not model validity; however, they have good sharpness (Willmott and Matsuura, 2005). Correlation of coefficient (CC 10), coefficient of efficiency (CE 7), coefficient of variance (CV 3) and Nash Sutcliffe coefficient (NSC 12) gauge the strong relation between data and relative movement of two datasets with different units (McCuen et al., 2006).

Many studies have reported classified river WQ into classes (C), namely, good, average and worst per the standard WQ guidelines of the study area. For better understanding, these studies have kept separate Cs, and 32 papers have reported this classification (Yaseen et al., 2018c).

Among all the computed performance metrics, although correlation coefficient (R) and DC: determination coefficient was widely used for models evaluation, they are oversensitive to high extreme values and insensitive to additive and proportional differences between model predictions and observed data (Legates and McCabe, 1999; Moriasi et al., 2007). On the other hand, the MAE was majorly used for evaluation owing to its less sensitivity to the extreme values in the predicted dataset in comparison with the RMSE (Fox, 1981). Studies reported Nash-Sutcliffe efficiency (NSE) is a normalized statistic that determines the relative magnitude of the residual variance compared to the measured data variance (Nash and Sutcliffe, 1970). According to the guidelines of hydrological model performance evaluation proposed by (Moriasi et al., 2007), the performance of a hydrological model is very good, if NSE is more than 0.75. Hence, this is another remarkable metric to be investigated for such type of modelling.

Other informative metrics can be examined for better modelling evaluation and assessment which have not applied are, volumetric efficiency (VE) metric can be explored to circumvent some problems associated with NSE. Unlike the usual correlation coefficients or the NSE, VE would be particularly helpful in comparing the performance of similarly scaled. It has the additional advantage of being easy to calculate. The modified index of agreement (m-IA) developed by (Willmott, 1981) as a standardized measure of the degree of model prediction error, is one of the essential metrics for evaluating predictive models and thus it is giving more insightful aspect on the modelling certification. Although it provides some improvement over DC, it is still sensitive to extreme values due to the squared differences in the mean

square error in the numerator (Legates and McCabe, 1999). In addition, the disadvantage of the results given by the previous DC and NSE arises from the fact that the differences of observed and predicted WQ use the squared values. Consequently, large errors in time series can be over-estimated whereas small values can be neglected. In order to overcome this limitation, Willmott (1984) introduced a generic form of index of agreement, Willmott's Index (WI). The advantage of the WI is that the errors and differences are given their appropriate weighting and are not inflated by their squared values. To reduce the effect of absolute differences during peak values of WQ and to improve the effect of absolute low values differences, the relative index of agreement (r-IA) is applied which is a substantial metric that measures the differences between the observed and simulated values as relative deviations (Chadalawada and Babovic, 2019). Another significant metric is Kling-Gupta efficiency (KGE), it provides a diagnostically interesting decomposition of the NSE, which facilitates the analysis of the relative importance of its different components (correlation, bias and variability) in the context of hydrological modelling (Gupta et al., 2009). Persistence Index (cp) based on the assumption that the process is a Wiener process that aims of comparing the performance of the models against a simple model using the observed value of the previous day as the prediction for the current day as opposed to a constant mean. The percent bias (PBIAS) measures the average tendency of the simulated data to be either higher or lower compared to the observed data. According to (Moriasi et al., 2007), the PBIAS $< \pm 10$ indicates very good performance model. The performance of the predictive models could be evaluated as well through visual inspection of graphical representations such as scatter plot, box plot, relative error, Taylor dia0gram and violin plot. Those plots are very useful to have a better understanding of the prediction capacity.

9. Conclusion

River water quality research interest has grown over the years and the amount of data analysis generated from such research needs efficient handling, assessing and predicting tools. Correspondingly, AI technology has proved to be a powerful tool which has been successfully applied in various field including hydrology and environmental engineering. Considering the sub-classes of the AI models most popular model is ANN models then fuzzy, kernel-based models, complementary (wavelet) and hybrid (nature-inspired) models. The review has gone through more than 200 papers which have addressed the river water quality modelling to better assess, predict and manage the current issue of surface water pollution. The assessment has been divided into two parts in this study. The first part is the individual assessment of each model groups which describes the unique assessment derived from the bibliographic review. However, the reviewed literature reveals that these traditional models fail to grasp the full aspect of uncertain, non-stationary, noisy and nonlinear data when compared to the complementary and hybrid model. Though, each of them has their own strength allowing them efficiently working in the WQ modelling and monitoring area. Thus, to better understand the AI models a general appraisal has been prepared. Thus, the second part of the valuation has been categorized into five sub-section: (i) It focuses on the disadvantages of each models leading to the next innovative modelling algorithm. It also explains why there was and still is need of more advanced models research which will allow further research to fill the gaps pointed out in the review. (ii) Since the review research focused on river WQ modelling; thus, all the rivers which have been studied around the world are mapped in a single figure which shows the most explored river. Even though the researches in river WQ modelling has significantly amplified although the literature survey found that many of the river water sources are still unexplored worldwide especially African and Australian continent. This assessment allows researchers to focus on the new study area with unique environmental, ecological, hydrological and geological signature. Furthermore, it gives the opportunity to test the new model in different scenarios and validates its performance (iii) The section pointed out the most considered output variables. While these models performance is based on the input-out selection, it is evident that result is more optimistic when the relation between the input–output is considered more minutely and tools like dimensionality reduction, sensitivity analysis, mutual information and extrapolation are employed to find the most sensitive input-out association. This allows the researchers working in the field of hydrology, data science and computer modelling to work more on the consideration of the relationship and factors influencing the variables. (iv) This section discusses the time step chosen by the reviewed papers, revealing that many of them opted for longer time steps which may affect the time series analysis of WQ. Larger time scale also limits the scope of visualization of seasonal and periodic variation which is may severely damn the decision making in case of emergency and disasters like flood and drought. (v) The last part of the general assessment deals with the choice of performance metrics. There are many performance metrics widely used however, the selection of the most appropriate statistical criteria is very essential for proper judgement of the model. Thus, this section gives information about the application of the most common techniques and the frequency of their preference.

In section 8.2 "most explored river" also points out another challenge the researchers are facing and continue to face. The tremendous expansion of AI application is affecting the work and everyday life. AI involvement in many industries including water and wastewater management and decision-making policies sometimes makes the users reject it as these systems are inscrutable and biased. Thus, there are countries which are still not implementing AI (Africa, Australia, Russia, North America. Refer to Fig. 10). It is true that the AI system is needed to solve the complex problem but gaining users and decision-maker trust is also essential to realise the full potential of AI technology. Providing more information on the system accuracy, comparative assessment of the performance indices, water quality indexing and system behaviour can increase model acceptability in real-life application. Such information will give better feedback from the user which can be used for model improvement through feature engineering, architectural adjustment, parameter tuning.

Research work on river WQ modelling with AI is distributed over a wide domain and it is very much difficult to collect the whole range of WQ which may also include heavy metals, agricultural waste and chemicals, immerging contaminants, sediments, industrial chemicals, land use etc. which is one of the limitations of the study. Thus, the survey has been limited to the research studies who considered the most common and easily identifiable WQ variables. Another limitation of the study is ever-changing river WQ and thus there will always need new models and testing them in a new environment.

There is much scope in the field of river water quality modelling and these challenges will keep the generation of new innovative ideas. The next section presents the potential research and challenges which need to be addressed by forthcoming researchers.

10. Potential future research directions

'To get a better system lot of application of an expert system and decision support system has to be coupled together to overcome the barrier to achieve more. Such the system will also have to deal complexity, uncertainty, inconsistency, imperfectness, incompatibility etc.' (Stephanou and Sage, 1987).

AI models have been continuously, successfully and effectively implemented in river WQ modelling studies, as concluded by 209 reviewed research papers published in high-index international journals from 2000 till 2020. AI models are combined tools of mathematics, statistics and biological science. They can overcome the individual weaknesses by behaving, thinking and learning. AI becomes machine learning when combined with learning.

'Human-like behaviour can be called as AI, but unless it automatically

learns from data and improves, it is not Machine Learning.' (Carbonell et al., 1984).

AI models have several advantages in solving complex problems of nonlinear river WQ data. They are reliable, lets easy decision for policy-makers, easy and effective in the implementation and cost-effective and can deal with large-scale data. However, they need large labelled training datasets and suffer from limited capability and functionality. Various stages, such as inputs, architecture and training, can be altered depending on the preferences of the researcher to achieve desired performance and outcome. AI models can still be improved in river WQ because WQ concerns are increasing worldwide. This paper is a glimpse of what has been done with one step forward and what may be done in the future. Here are several suggestions to achieve the goal:

a) AI models are progressing massively to provide reliable and robust intelligence methodologies for solving environmental engineering problem theoretically. Thus, the application must be practised as real-life problem solutions.
b) AI models are data extensive tools and can achieve high accuracy with a consistent, regular and adequate amount of data if proper and competent monitoring stations are established, which can provide continuous, stable and accurate data worldwide.
c) Many study areas have problems with obtaining the data, and several issues are difficult to overcome. Thus, additional AI models that can overcome the problem of missing data and lack of continuous data should be designed.
d) The issue of missing data and data unavailability, specifically stations unreachable due to climatic condition, can be overcome by utilising the recent advancement of technologies, such as GIS and remote sensing. These technologies can be used often for resource management and environmental modelling issues. The high-resolution data from the satellite can be used for improved spatial river WQ modelling.
e) With a similar problem about data, additional models should integrate the features of a black-box model and white-box models; in another term, the conjugal between the physiochemical mechanism happening in river systems must be considered, and numerical approach must be used to predict river WQ changes. Understanding the relationship between the WQ and the mechanisms that influence each other can produce an improved result with fewer data inputs.
f) Different inputs and outputs are used in river WQ analysis and few works with new variables. Given that every change around the study area may affect river WQ, future research studies must include additional variables, such as population change, seasonal runoff and amount of industrial influent and effluent. The influences of these factors on one another should be established to provide a broad spectrum for understanding and identifying river pollutant sources and predicting possible sudden changes. These studies will provide a good chance for water treatment plants to cope with emergencies.
g) A new model that is designed and tested should be tested against the benchmarked models, techniques and issues in river WQ. This kind of benchmarking technique will help researchers around the world find better solutions for real-time crises already identified by other researchers, such as data from arid areas, semi-arid areas, flood-prone zones and other diverse conditions where model competency can be checked for a scenario. This method will develop robust and field applicable models to help solve real-life river WQ management issues.
h) Research data, monitoring station data, modelling information and benchmarked model application should be easily and freely available online to broaden the horizon of environmental research. Freedom for data access will provide a broad aspect of model performance based on information of benchmarked models and available data. Moreover, researchers can evaluate their work easily and efficiently whilst producing models applicable to various environmental scenarios.
i) Few branch models are less explored in river WQ assessments and need additional research, such as ward NN, hopfield NN, Dempster–Shaft theory, kNN, DT and deep learning models. These models have not been implemented in this field. Deep learning uses multilayer architecture, can compute nonlinear input–output mapping and learn through a general-purpose learning technique (Sulaiman et al., 2018). This method can understand and extract natural language and complex information, achieve high accuracy, provide an effective optimisation approach for end-to-end performance, provide a fast and minimum need of tuning and use unsupervised learning (Lecun et al., 2015; Schmidhuber, 2015).
j) Attempts have been already made to create a hybrid model with high strength and power to deal with nonstationary, random, complex and vague data. These efforts should be encouraged for the proposal of superior models in the near future (Tao et al., 2018). Various optimisers have been developed over the years, and many remain unexplored in river WQ. Such optimisers include bacterial foraging optimisation (Das et al., 2009), amoeba-based algorithm (Zhang et al., 2013), artificial plant optimisation (Cui et al., 2012), flower pollination algorithms (Yang, 2012), grasshopper-insect-based algorithm (Saremi et al., 2017), wasp-insect-based algorithm (Theraulaz, 1991), fruitfly-insect-based algorithm (Xing and Gao, 2014), glow worm-insect-based algorithm (Krishnanand and Ghose, 2009), dragonfly-insect-based algorithm (Mirjalili, 2016), shark optimisation (Hersovici et al., 1998), whale optimisation (Mirjalili and Lewis, 2016), bean optimisation (Xiaoming Zhang et al., 2010a,b), dove-bird-based algorithm (Su et al., 2009), eagle-bird-based algorithm (Yang and Deb, 2010), cuckoo search (Yang and Suash Deb, 2009), bird mating (Askarzadeh and Rezazadeh, 2013), monkey-animal-based algorithm (Mucherino et al., 2007), wolf-animal-based algorithm (Liu et al., 2011), lion-animal-based algorithms (Yazdani and Jolai, 2016), artificial fish-swarm algorithm (Li, 2003). Termite (Roth, 2005), marriage in honey bees (Abbass, 2001), bee collecting pollen algorithm (Lu and Zhou, 2008), krill herd (Gandomi and Alavi, 2012), grey wolf optimiser (Mirjalili et al., 2014), Earthworm (Wang et al., 2018), salp swarm, (Mirjalili et al., 2017) nomadic people (Salih and Alsewari, 2019), sooty tern (Dhiman and Kaur, 2019), harris hawks (Heidari et al., 2019), side-blotched (Maciel et al., 2020) and color hormony (Zaeimi and Ghoddosian, 2020) can be implemented in future studies. For better visualisation, the timeline is presented in Fig. 14.
k) Many models, such as RBF, RNN, PNN, ELM, DENFIS, EFuNN, MARS and regression models, have been successfully applied for river WQ assessment but never coupled with any bioinspired optimisation technique. Fig. 14 presents the combined architecture of possible hybrid modelling using WQ input–output with AI models and NI algorithms for future research.
l) Regression models, such as GPR, DT, RT, MT, GLM and ET, are least explored in river WQ studies. They hold promise in classification and prediction competencies, and close attention is needed to explore and enhance their performance. Their performance can be increased using optimisers as recommended earlier.
m) Close attention should be paid towards the architecture of the model, such as the optimum number of layers, weight and bias values, data allocation for training, testing and validation, and method selection for model calibration.
n) The performance of wavelet transformation with various AI models in river WQ research has been considered for data pre-processing to extract the underlying features and de-noising time series. It has been used in WQ modelling but pre-processing of temporal and spatial data utilised in river WQ models need further research. Additionally, W can also be evaluated against other AI models that can deal with nonlinear data, such as GRNN, PNN, DENFIS, EFuNN, FIS, NARX, MARS, ENN, DT, RT and MT. Moreover, these models

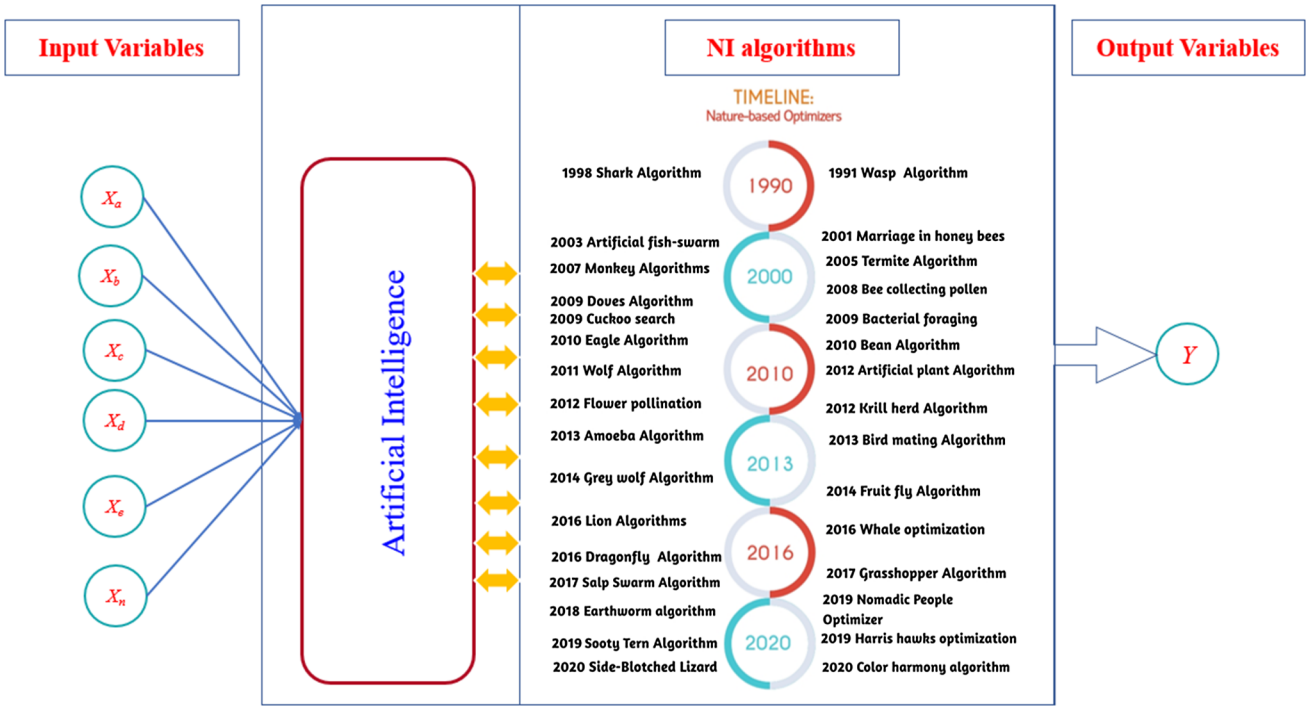


Fig. 14. AI model architecture with NI optimizers: The input variables (X) represents various water quality variables (such as TSS, pH, EC, DO, BOD, COD). AI models and Unmapped bio-inspired algorithm (animal/ insect/ microbe/ bird/plant based optimizers) are the combined hybrid model which have never been hybridized before in WQ research studies. Y represents the selected output variable which can help in understand the river health of the study.

must be comparatively studied to establish the extent of their success.

o) The computational explosion is an issue yet not fully addressed by AI researchers. The explosive increase in the knowledge growth leads to the need for the understanding of the interaction of the networks which are coordinating in a methodical fashion. To solve the problem by equational discovery and symbolic regression technique. This technique focuses on the structure identification, parameter estimation, system variables and decreases the discrepancies between simulated behaviour and observed system behaviour to find an optimal model. In contrast to that, AI models estimated the parameter value with fixed structure and losses the illustrate ability of the model thus, needs additional implications (Tanevski et al., 2020). Thus, with the increasing accessibility of data symbolic regression technique and artificial intelligence have to come together to better understand the internal networking of the model and on-field data interaction to design more applicable model. Quantifying investigation and getting structural knowledge is the only redemption for an effective solution to the complex real-life river pollution issues.

p) Increasing river pollution and its dangerous effect on life has ensured the need for environmentalist and hydrologist to work towards better solutions to overcome this life-threatening disaster and a political issue (Beven, 2016). They need to work on designing better assessing and predicting model. Additionally, by the study of 209 papers which contain WQ data ranges from 1968 to 2018, it is evident that river WQ is subjected to change-over-time, may get influence by various other variables, there is a chance of emergence of new contaminants and other uncertainty which may follow with time. This a limitation of river WQ modelling which the scientist has to face since there is always a probability that a superior performing model may not perform very well next few years and new conditions. Thus, continuous progress and testing have to be done to deal with river pollution and manage it well.

q) After reviewing all the research paper, it can be concluded that the process of contaminants mixing in rivers or natural streams is a complex and cause a synergistic reaction leading to change in the WQ composition. Thus, it should be considerated during the experimental process to include other variables such: irregular velocities bed configuration, sediment load and the dead zones (Kisi and Yaseen, 2019).

11. Abbreviation

Accuracy (Acc), Absolute mean prediction error (AMPE), Absolute average deviation (AAD), Absolute principle component score (APCS), Absolute error (AE), Absolute percentage error (APE), Accuracy factor (A_f), Adaptive boosting (AdaBoost), Adaptive neuro fuzzy interference system (ANFIS), Alkalinity (AL), Alkaike index (AIC), Aluminum (Al), Alternative fitness genetic Algorithm (AFGA), Ammonia (NH_3), Ammonical nitrogen (NH_3-N), Analytical hierarchy process (AHP), Anionic surfactant (AiS), Ant colony optimization (ACO_R), Arsenic (As), Artificial neural network (ANN), Artificial bee colony (ABC), Artificial Intelligence (AI), Auto associative neural network (AANN), Auto regressive integrated moving Average (ARIMA), Average absolute relative error (AARE), Average accuracy (AA), Average information theoretic criterion (Aic) Average relative error (ARE), Air temperature (AT); Back propagation neural network (BPNN), Bat algorithm (BA), Bayesian network (BN), Bayesian regulization (BR), Biochemical oxygen demand (BOD_5), bi-carbonate (HCO_3), Blue green algae (BGA), BOD decay rate constant (K_b), Boosted regression trees (BRT), Bootstrapped wavelet neural network (BWNN), Boron (B); Calcium (Ca), Cadmium (Cd), Calcium sulphate ($CaSO_4$), Carbon di-oxide (CO_2), Canadian council of ministers of the environment (CCME), Canonical correspondence analysis (CCA), Carbonate hardness (CH), Cascaded fuzzy system (CFS), Chaotic nonlinear dynamic algorithm (CNDA), Chemical oxygen demand (COD), Chemical oxygen demand digested by potassium dichromate (COD_{cr}) Chloride (Cl), Chloride sulphate ($ClSO_4$), Chlorophyll α (Chl- α), Chromium (Cr), Cladocera (Clad.), Class precision (Pc), Classical Neuro Fuzzy (CNF), Cluster analysis (CA), Co-active neuro-fuzzy inference system (CANFIS), Coefficient of efficiency (CE), Coefficient of variation (CV), Colored dissolved organic matter

(CDOM), Continuous wavelet (CW), Copepod (Cop.), Crossover rates (CR), Copper (Cu), Cyanide (CN); Day of the year (DOY), Decision tree (DT), Degree of agreement (d), Delta percentage (δ %), Depth (Z), Determination coefficient/ coefficient correlation (DC), Discrete wavelet (DW), Discriminant analysis (DA), Dissolved inorganic Nitrogen (DIN), differential evolution (DE), differential evolution–separate groups (DE–SG), differential evolution with global and local neighborhood-based mutational operators (DEGL), Dissolved oxygen (DO), Dynamic evolving neural fuzzy inference system (DENFIS), Dynamic factor analysis (DFA); Electrical conductivity (EC), Escherichia coli (E.coli), Empirical orthogonal function (EOF), Ensemble empirical mode decomposition (EEMD), Ensemble neural network (ENN), Evaporation (E_{vap}), Evolutionary Algorithm (EA), Evolutionary Polynomial Regression (EPR), Evolving clustering method (ECM), Evolving Fuzzy Neural Network (EFuNN), Expectation-maximization cluster (EMC), Extreme gradient boosting (XGB), Extreme learning Machine (ELM), Error rate (ER), Extreme randomized trees (ET), Error value (EV); Forest land use factor (F), Fluoride (f), Factor analysis (FA), Fat and Oil (F&O), Fecal coliform (FC), Feed forward back propagation (FFBP), Feed forward neural network (FFNN), Final predictor Error (FPE), Firefly algorithm (FFA), Full complexity prediction square error (FCPSE), Fussy interference system (FIS), Fuzzy analytic hierarchy process (FAHP), Fuzzy clustering technique (FCT), Fuzzy C-mean (FCM), Fuzzy comprehensive evaluation (FCA), Fuzzy contingent Evaluation (FCV), Fuzzy multi-attribute analysis (FMAA), Fuzzy multi-objective evaluation (FMOE), Fuzzy sets Theory (FST), Fuzzy similarity measures (FSM), Fuzzy synthetic evaluation (FSE), Fuzzy time series model (FTSM), Fuzzy water pollution index (FWPI); Gaussian cloud transformation (GCT), Gaussian process regression (GPR), Gene expression programming (GEP), General regression neural network (GRNN), Generalized Least square model with cosine trend (GLScos), Genetic algorithm (GA), Genetic programming (GP), Geographic information system (GIS), Gradient boosting (GB), Green algae (Ga), Grey relational method (GRM), Grid partition (GP2), Group method of data handling (GMDH), Group method of data handling (GMDH), Growing hierarchical (GH); Hardness (H), Habitat Quality assessment (HQA), Habitat modification score (HMS), Heat flux model (HFM), Heavy metal (HM), Hierarchical agglomerative cluster (HAC), Hierarchical clustering analysis (HCA), Hopfield Neural Network (HNN), Humidity (Hs), Hybrid evolutionary algorithm (HEA), Hydrazine (NH_4-N); Improved fuzzy matter-element evaluation method (IFMEM), Index of agreement (IA), Inflow (Q_{in}), Interactive fuzzy multi-objective linear programming (IFMOLP), Iterative self-organizing data analysis (ISODATA), Iron (Fe), Irradiance (Irad); Kling-Gupta efficiency (KGE), k-nearest neighbors (kNN), Kling-Gupta efficiency (KGE); Latitude, (Lat), length (L), Lead (Pb), Levenberg marquardt (LM), Linear discriminant (LD), Linear regression (LR), Linear alkylbenzene sulphonate (LAS), location similarity index (LSI), Longitude (Long); Macrophytes (Ma), Machine learning (ML), Macrophytes index of river (MIR), Magnesium (Mg), Manganese (Mn) Magnesium sulphate (MgSO4), Modified index of agreement (m-IA), Matthews correlation coefficient (MCC), Mean Absolute Error (MAE), Macrophytes biological index of river (IBMR), Mean error (ME), Mean absolute relative error (MARE), Mean absolute deviation (MAD), Mean absolute percentage error (MAPE), Mean bias error (MBE), Mean percentage error (MPE), Mean prediction interval (MPI), Mean relative error (MRE), Mean sea level pressure (MSp), Mean square relative error (MSRE), Mean square error (MSE), Mercury (Hg), Methylene blue active substance (MBAS), min/max autocorrelation factor analysis (MAFA), Mineral oil (M_{oil}), Model tree (MT), Modified response surface method (MRSM), Monte carlo simulation (MCS), Multi-layer perceptron (MLP), Multi linear regression (MLR), Multi non-linear regression (MNLR), Multiple regression analysis (MRA), Multivariate adaptive regression Spline (MARS), Multivariate linear regression model (MLRM), Mutation rate (MR); Naïve bayesian (NB), Nash sutcliffe efficiency (NSE), Nash-Sutcliffe efficiency (NSE), Nash criterion (NASH), Nature inspired (NI), Net ecosystem metabolism with system dynamic model (NEM-SDM), Nitrogen (N), Nitrate (NO_3), Nitrate Nitrogen (NO_3-N), Nitrification Rate (K_1), Nitrite (NO_2), Non-dominated sorting genetic algorithm-II (NSGA-II), Nonlinear auto regression with exogenous input (NARX), Not applicable (NA); Optimally pruned (OP), Order series Method (OSM), Organic nitrogen (N_{Org}), Output weight optimization-hidden weight optimization (OWO-HWO), Oxygen (O_2), Oxidation rate of nitrite (K_2); Partial least square (PLS), Partial least square regression (PLSR), Particle swarm optimization (PSO), Pearson correlation coefficient (R), Percentage of mean (M%), Persistence Index (cp), Permanganate index (PI), Percent bias (PBIAS), Persistence Index (cp), Petroleum (Pet), Phosphate (PO_4), Potassium dichromate ($K_2Cr_2O_7$), Potassium permanganate ($KMnO_4$), Phosphate phosphorus (PO_4-P), Phycocyanin (Pc_1), Physical process-based model (PPBM), Polynomial chaos expansions (PCE), Possibility measure based fuzzy support function machine (PMF-SFM), Potassium (K), Precipitation (P_{pre}), Prediction Accuracy (Ap), Prediction interval coverage probability (PICP), Principal factor analysis (PFA), Probabilistic neural network (PSVM), Probabilistic/ Probabilistic neural network (PNN); Quadratic discriminant (QD), Discharge (Q), Quantity of assimilated polluted water (Q1), Quantity of water evaporation (Q2), Quantity of water leaked to ground (Q3), Quantity of water precipitation from runoff (Q4); Radial basis function neural network (RBFNN), Rainfall ($R_{pre.}$), Reactive phosphorus (P_{React}), Recurrent neural network (RNN), Recursive partitioning (RP), Regression model (RM), Regression tree (RT), Regression value (r-value), relative index of agreement (r-IA), Relative error (RE), Relative humidity (RH), Relative error distribution (RED), Release from reservoir (R_1), Residual Chlorine (Cl_R), Re-substitution Error (RE), River pollution Index (RPI), River macrophytes nutrient index (RMNI), Root mean square error (RMSE), Root relative squared error (RRSE), Root mean square error proportional (RMSEP), Root mean square relative error (RMSRE), Rotated principal component analysis (PCA), Rotifera (Rot.), Reduced order support vector machine (ROSVM), Runoff (R_F); Salinity (Sa), Secchi depth (SD), Scatter index (SI), Sediment rating curve (SRC), Sediment oxygen demand (SOD), Self-organizing network based monitoring (SOM), Short radiation (Rs), Shuffle complex evolution (SCE), Shuffled frog leap algorithm (SFLA), Silica (Si), Silicon oxide (SO_2), Simulated annealing (SA), Single factor evaluation (SFE), Single averaging ensemble (SAE), Simpson diversity index (D), Sodium (Na), Sodium absorption ratio (SAR), Soil and water assessment tool (SWAT), Spearman correlation (SC), Spectral angle mapper (SAM), Spillage (S), Standard error of prediction (SEP), Streptococcus (Strep.) Subtractive Clustering (SC_1), Sulfide (S^2), Sulfate (SO_4), Selenium (Se), Sum of squared error (SSE), Sunshine hour (S_h), Support function machine (SFM), Support vectorclassification (SVC), Support vector machine (SVM), Support vector regression (SVR); The daily average (Tav), The daily max air temperature (Tmx), The daily run off / The declination of Sun (S), The mean bias (BIAS), Thermo tolerant coliform (TF), Time delay neural network (TDNN), Total dissolved solid (TDS), Total nitrogen ($N_{Tot.}$), Total coliform (TC), Total organic carbon (TOC), Total phosphorus ($P_{Tot.}$), Total solid (TS), Total suspended solid (TSS), Total volatile solid (TVS), Transparency (Tr), Travel time (Tt), Turbidity (Tu); Ultraviolet (UV), Urban land use factor (U); Variable fuzzy evaluation model (VFEM), Variance inflation factor (VIF), Volatile hydroxyl benzene (VHB),Volatile phenol (V-ArOH), Volatile phenol (VP), Volumetric efficiency (VE), Volume error (VE); Water quality (WQ), Water quality index (WQI), Water temperature (WT), Wavelet (W), Wavelet basis function (WBF), Wavelet de-noising technique (WDT), Wavelet neural network (WNN), Ward neural network (wNN), Weighted averaging ensemble (WAE), Willmott's index (WI), Wind speed (W_1); Zinc (Zn).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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